Linear mixed effects models for event-related potential research

ADVANTAGES, THEORY, AND A PRACTICAL WALKTHROUGH

MJ Heise, Ph.D. · mj.heise@ucsf.edu
Society for Psychophysiological Research · October 14, 2026



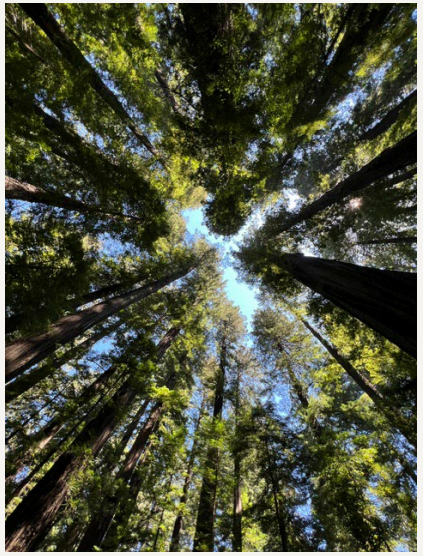
SPR SOCIETY FOR PSYCHOPHYSIOLOGICAL RESEARCH

About Me

- B.S. in Child Psychology from the Institute of Child Development, University of Minnesota
- Worked in Nathan Fox's lab at the University of Maryland and Angel Lillard's lab at the University of Virginia
- Ph.D. in Psychology with a minor in Quantitative Methods from the University of California, Davis
- Currently a statistician in the Division of HIV, Infectious Diseases, and Global Medicine at the University of California, San Francisco

- My hope for this workshop is that you leave feeling more confident about fitting LMEs in your own research.







Prior work on LMEs for ERPs

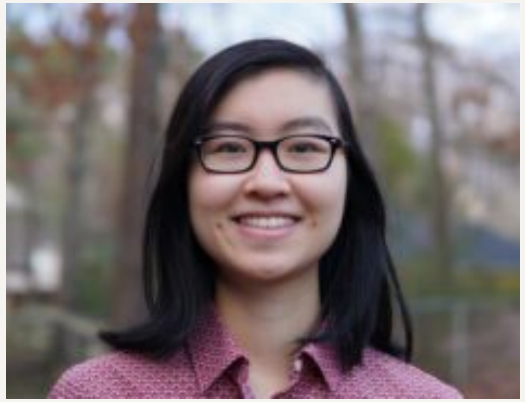
Contents lists available at ScienceDirect

Developmental Cognitive Neuroscience

journal homepage: www.elsevier.com/locate/dcn

Utility of linear mixed effects models for event-related potential research with infants and children

Megan J. Heise [a,b,*], Serena K. Mon [b], Lindsay C. Bowman [a,b]



Serena Mon

Looking for post-docs beginning summer 2027!

serenamon2027@u.northwestern.edu

Contents lists available at ScienceDirect

Developmental Cognitive Neuroscience

journal homepage: www.elsevier.com/locate/dcn

Next-gen tools

Analysis of event-related potential difference waves can benefit from linear mixed effects modeling: Recommendations for analyses and general model fitting

Megan J. Heise [a], Serena K. Mon [b], Lindsay C. Bowman [c,d,*]



Lindsay Bowman

What is your current role?

Faculty 0%

Postdoc 0%

Research staff (lab manager, research scientist, etc.) 0%

Graduate student 0%

Undergraduate student 0%

Other 0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Describe your experience with regression and/or mixed effects models.

New to statistical analysis generally

0%

New to regression

0%

Experienced with linear regression, but new to mixed effects models

0%

Have fit and published mixed effects models, but not for ERPs

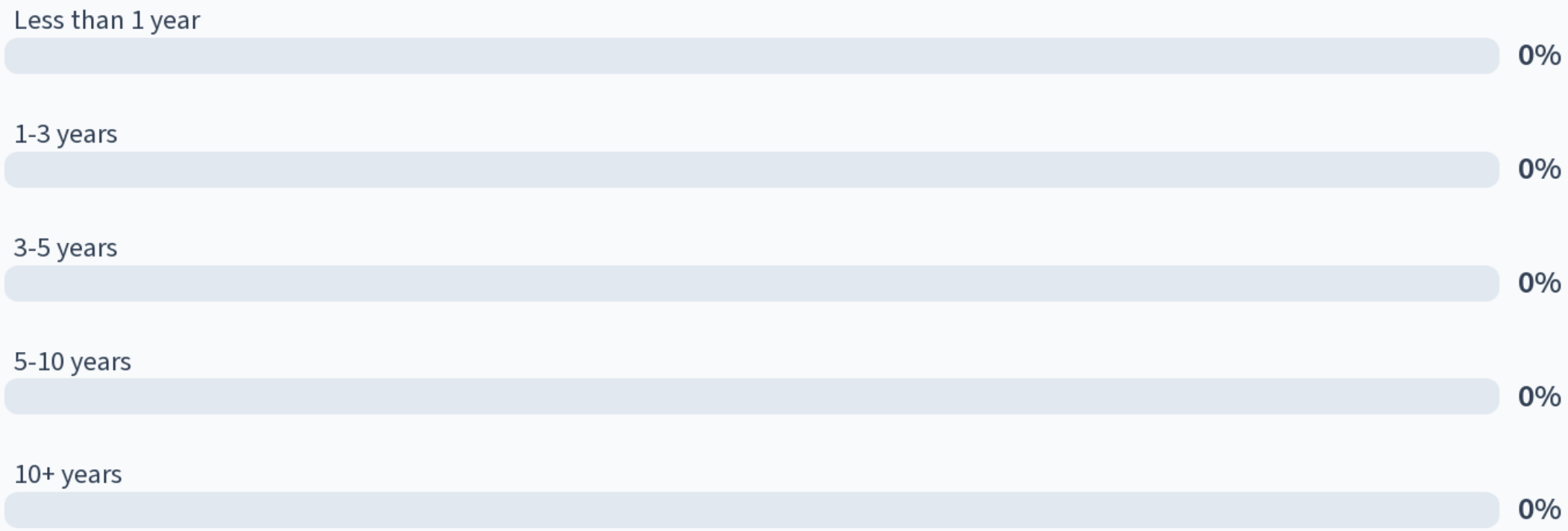
0%

Have fit and published mixed effects models with ERPs

0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

How long have you been conducting event-related potential (ERP) research?



Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Where are you coming from?



Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Do you have any burning questions (questions you are hoping will be covered today)?

Nobody has responded yet.

Hang tight! Responses are coming in.

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Roadmap for today

- Concepts & theory
  - Benefits of LMEs
  - LMEs as an extension of linear regression
  - Simulation demonstration evaluating LMEs vs. linear regression
- Special cases
  - Difference waves
  - Uncharted territory: Latency
- A practical, step-by-step walkthrough
  - Setting up your data, specifying your model, and troubleshooting
  - Power analyses the hard way & the easy way

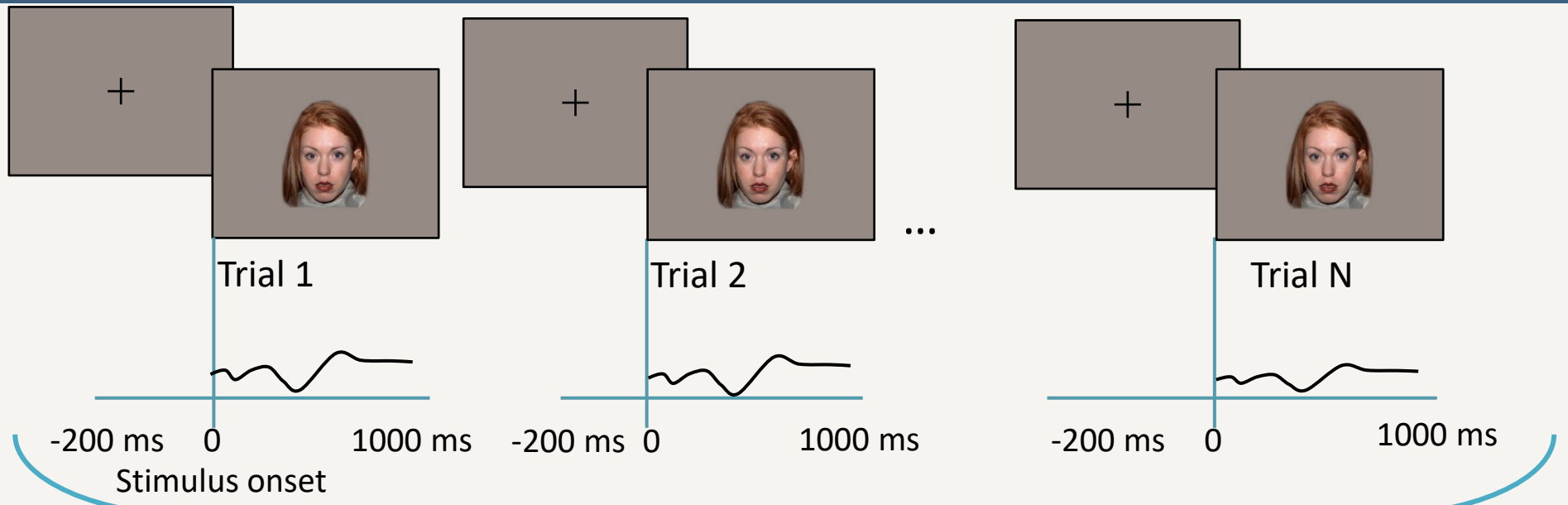
01

Challenges with ERP research

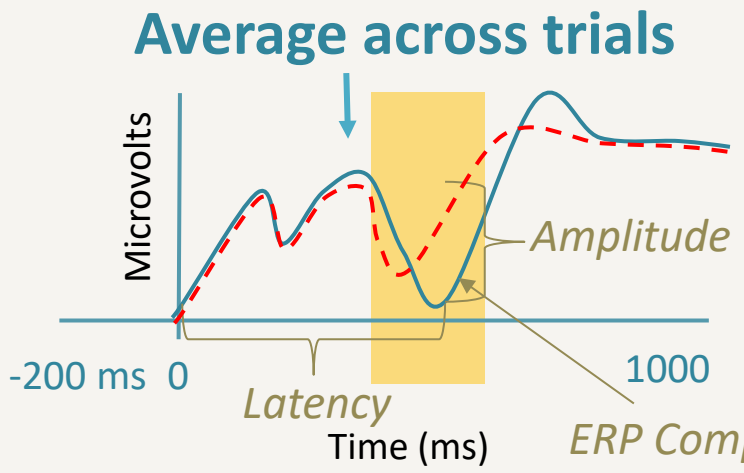
Why ERP research needs a better analytic toolkit

Event-related potentials (ERPs) reflect neural activity in response to a stimulus event





- ERP emerges from the average of voltage changes across many trials



- Examine ERP Component Amplitude & Latency to index neural processing across conditions of interest

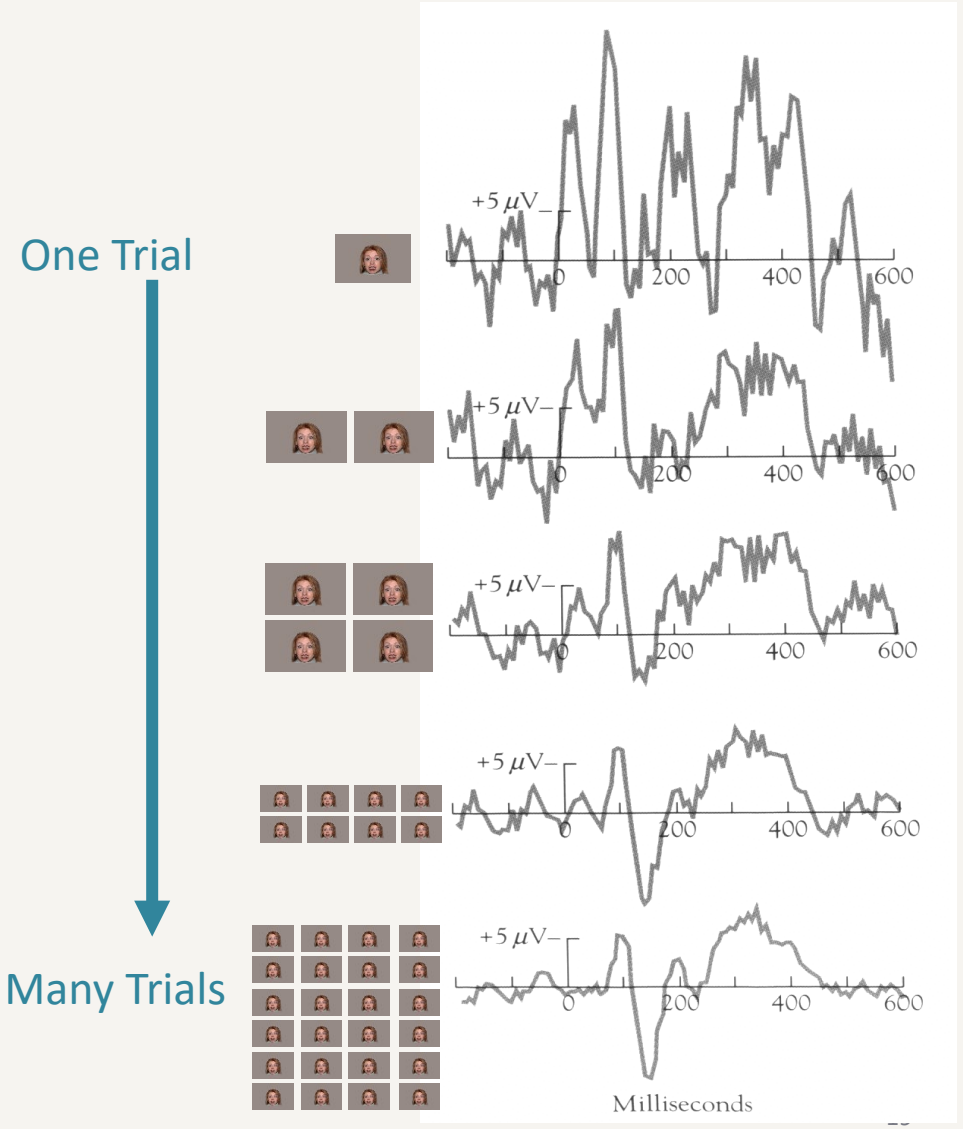
3-year-olds

6-year-olds

Unless otherwise stated, this workshop will focus on linear mixed effects models (LMEs) advantages for ***mean amplitude***

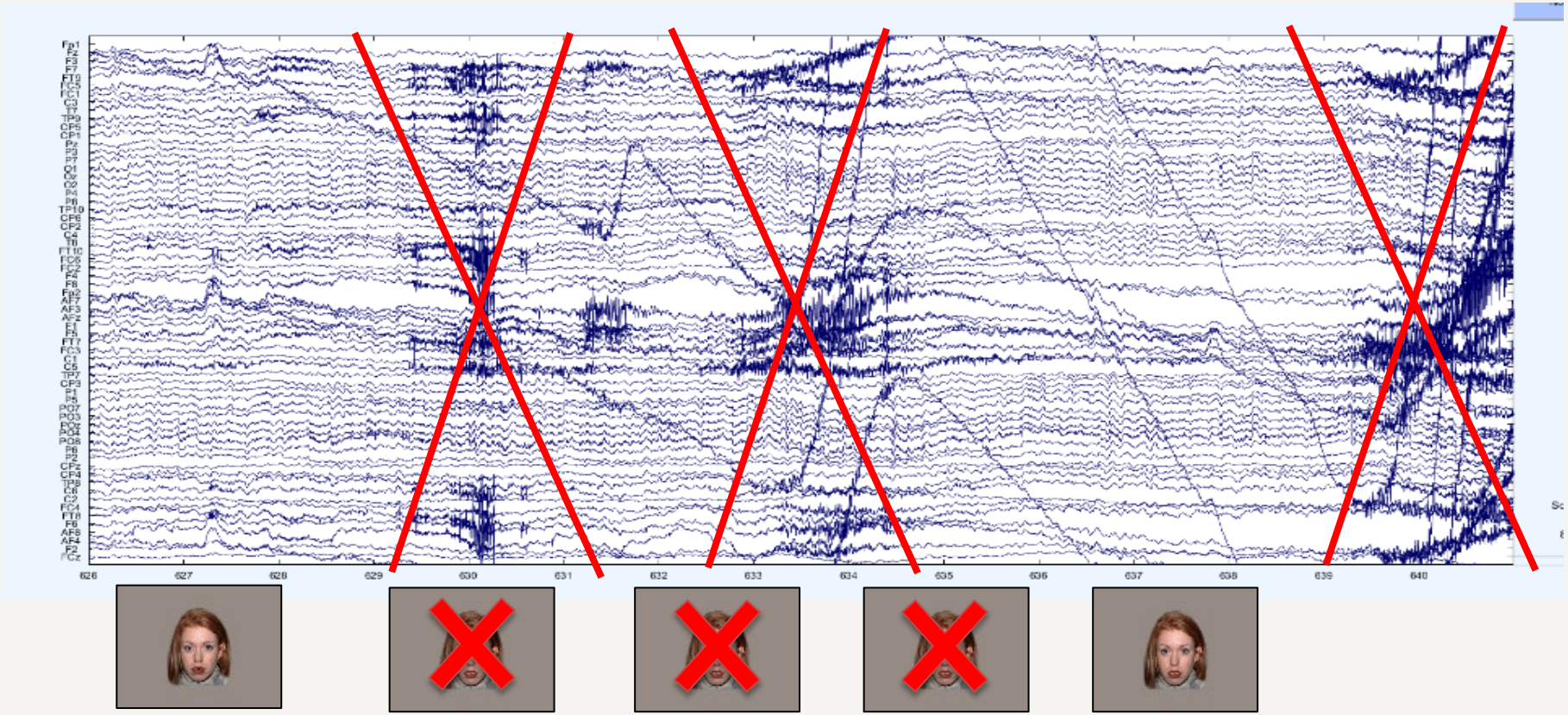
Challenges with ERP collection & analyses

- Many trials are required to reveal an interpretable ERP
  - More trials average together results in a larger signal relative to smaller noise, and a clearer ERP



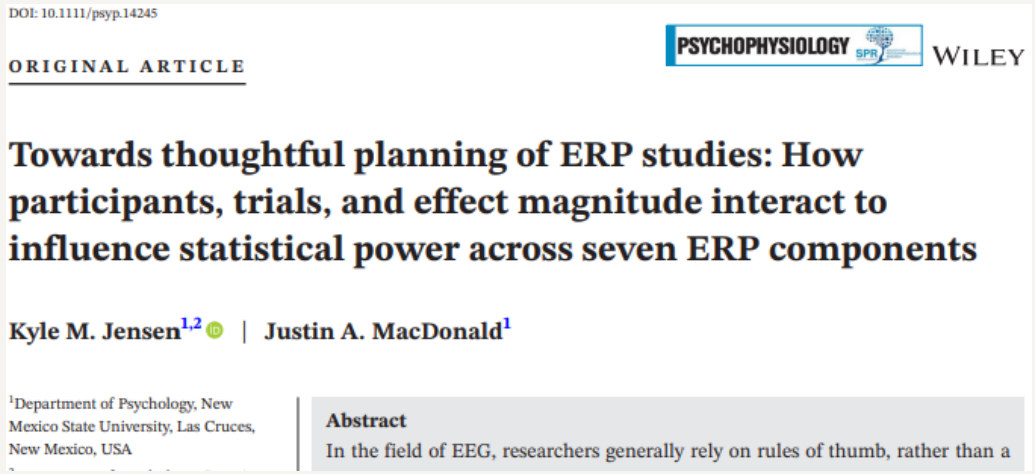
Challenges with ERP collection & analyses

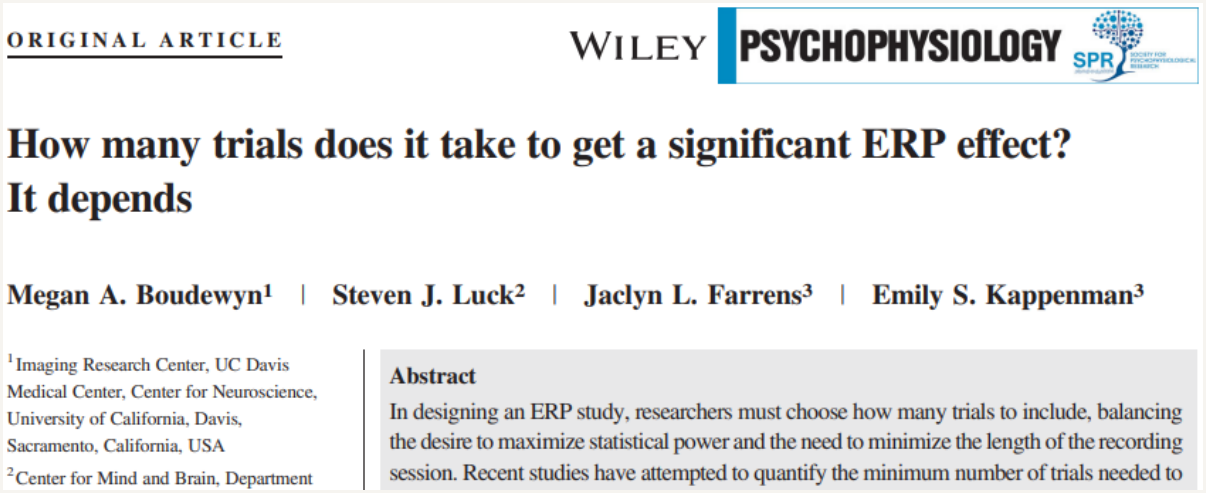
- Segments of ERP data are removed where there is excessive noise (e.g., from muscle tension, motion artifact, sweat, etc.)
- When too many individual trials are removed from a participant, researchers use "casewise deletion" to remove all data (to ensure high data quality across participants)



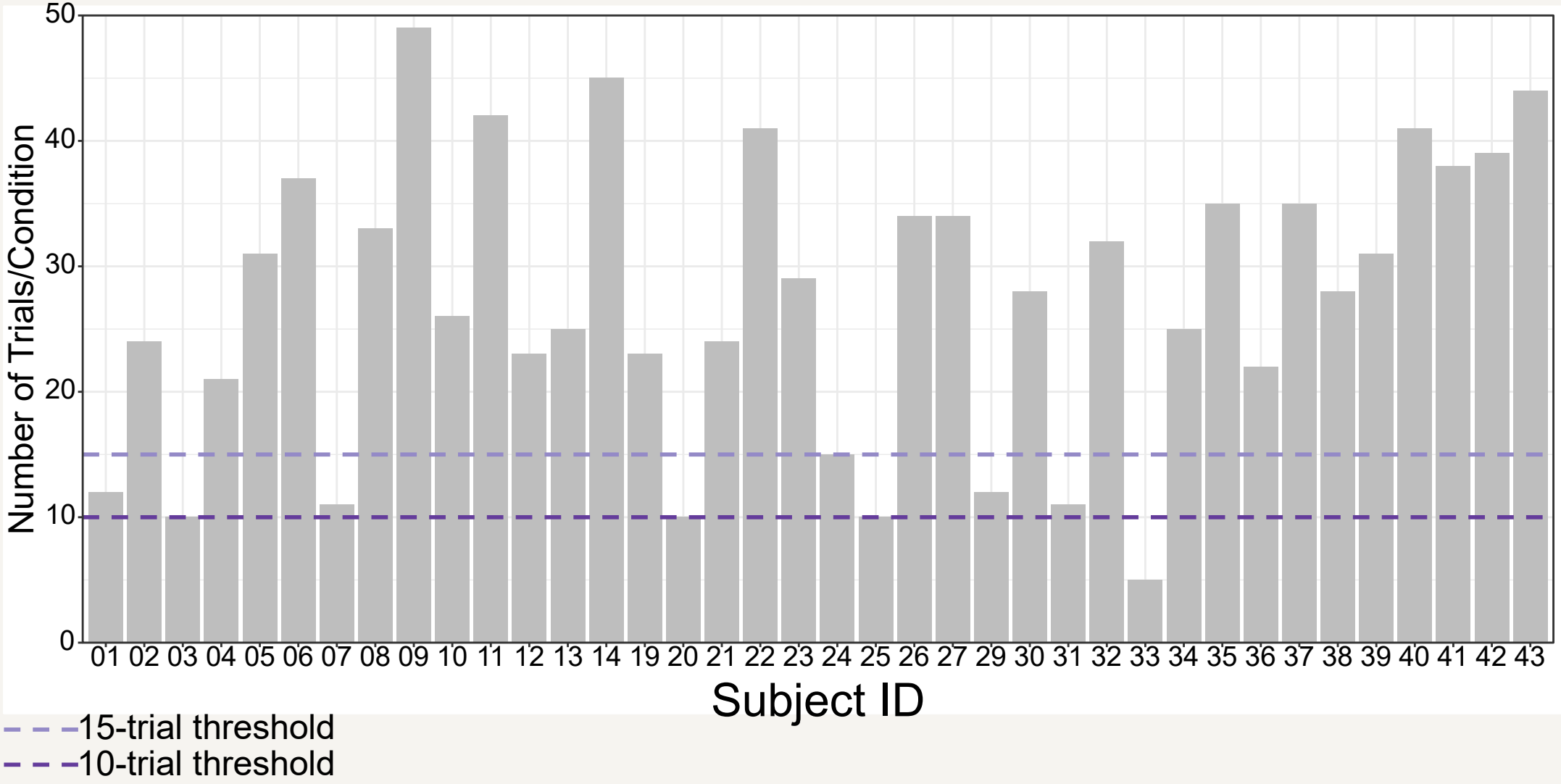
How many trials are required to analyze a “clean” ERP?

- It depends!
- Researchers have to make judgement calls about how many trials are required for inclusion in analysis.





Selecting trial-count thresholds



Heise et al., 2022

Real data: Preschoolers viewing emotional faces



Heise et al., 2022

How many trials are required to analyze a “clean” ERP?

- It depends!
- Researchers have to make judgement calls about how many trials are required for inclusion in the analysis.



Challenges with ERP collection & analyses

1. Decision-making on trial exclusion cutoffs
2. Too few trials → reduced participants → reduced power

Data loss in developmental ERP research

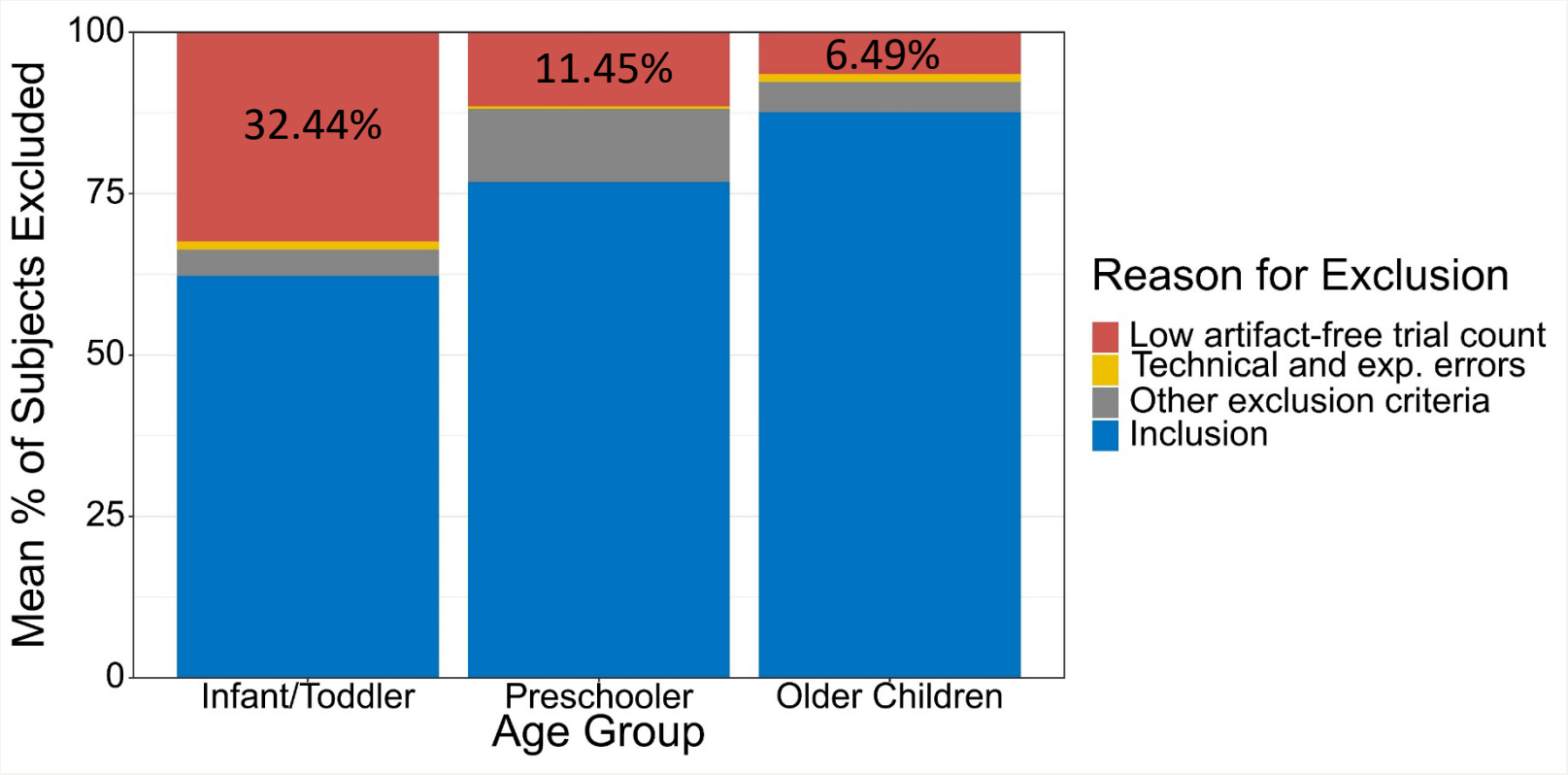


Figure 2. Stacked bar plot of the mean percent of excluded participants in studies requiring at least 10 trials/condition for ERP analysis (N=53)[1], representing the most common threshold used in our literature review (see Appendix A). All studies were published in the journal *Developmental Cognitive Neuroscience* from January 2011 to April 2021. Infant/Toddlers = 0- to 35-month-olds; Preschoolers = 3- to 5-year-olds; Older Children = 6- to 13-year-olds. (see Heise, Mon & Bowman, 2022)

Challenges with ERP collection & analyses

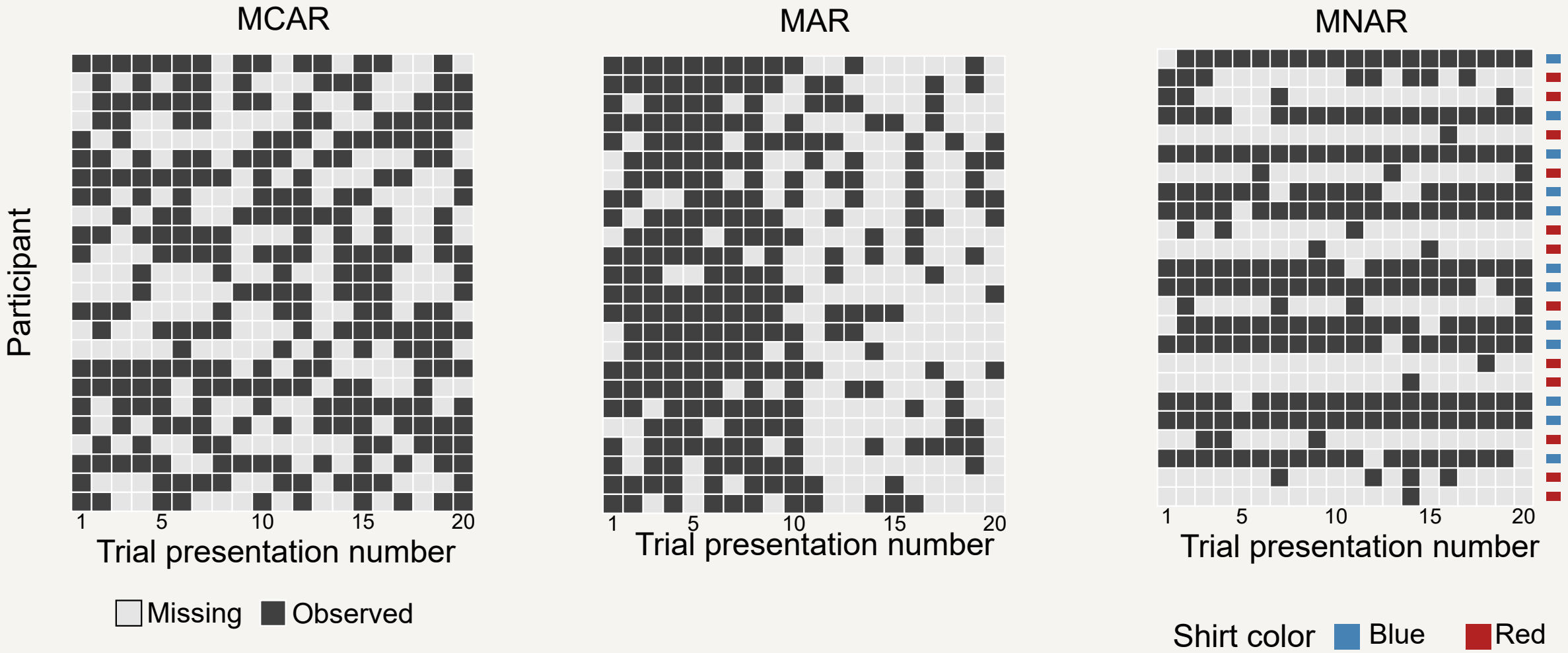
1. Decision-making on trial exclusion cutoffs
2. Too few trials → reduced participants → reduced power
3. Time and $ to collect data that isn't analyzed
4. Bias induced when data are not missing completely at random
   - Casewise deletion violates assumptions in statistical analyses unless data are missing completely at random

Types of missing data

Type	Definition
MCAR — Missing Completely at Random	Missingness unrelated to any variable, measured or unmeasured
MAR — Missing at Random	Missingness fully predictable from measured variables (i.e., missing patterns are fully explained by covariates)
MNAR — Missing Not at Random	Missingness related to unmeasured variables or the missing values themselves (e.g., participants with severe symptoms never participate)

Baraldi & Enders, 2010; Graham, 2009; Little & Rubin, 2002; Rubin, 1976

Types of missing data



Challenges with ERP collection & analyses

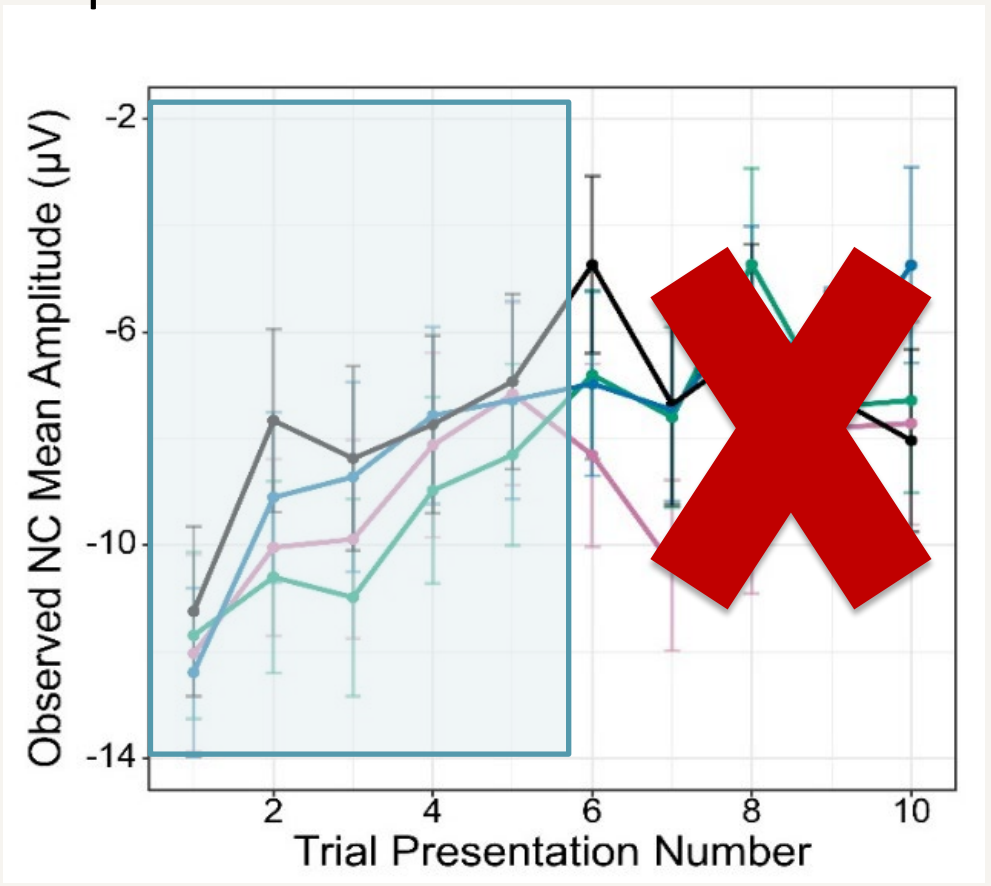
- Examples of data that are MAR
  - More missing data in younger children
  - More missing trials toward the end of the experiment



More missing data in younger children



Less missing data in older children



Challenges with ERP collection & analyses

- Trial-count thresholds (commonly 10 or 15 trials) are arbitrary — and the choice measurably changes your results
- Casewise deletion reduces statistical power to detect real condition or group differences
- Casewise deletion represents a large sunk cost — recruiting, scheduling, and running infants/children is expensive in time and money
- Casewise deletion violates missingness assumptions unless data are missing completely at random (MCAR) — rarely true

Why don't regression or ANOVA models account for missing data at the ERP trial-level?

- Because regression/ANOVA can model trial data directly
- Because trials are averaged into one mean per participant per condition before t...
- Because regression/ANOVA use casewise deletion automatically
- Because trial-level missingness is corrected by the Bonferroni procedure
- None of the above

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Why don't regression or ANOVA models account for missing data at the ERP trial-level?

Because regression/ANOVA can model trial data directly

0%

Because trials are averaged into one mean per participant per condition before the model "sees" them

0%

Because regression/ANOVA use casewise deletion automatically

0%

Because trial-level missingness is corrected by the Bonferroni procedure

0%

None of the above

0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Why don't regression or ANOVA models account for missing data at the ERP trial-level?

Because regression/ANOVA can model trial data directly

0%

Because trials are averaged into one mean per participant per condition before the model "sees" them

0%

Because regression/ANOVA use casewise deletion automatically

0%

Because trial-level missingness is corrected by the Bonferroni procedure

0%

None of the above

0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

LEARNING CHECK

Why don’t regression or ANOVA models account for missing data at the trial level?

A. Because regression/ANOVA operate on hierarchical (multilevel) trial data directly

B. Because trials are averaged into one mean per participant per condition before the model ever sees them

C. Because regression/ANOVA use casewise deletion automatically

D. Because trial-level missingness is corrected by the Bonferroni procedure

02

Concepts & theory

- ***Review of linear regression & its assumptions***
- *The problem: ERP trials aren't independent*
- *The fix: Overview of linear mixed effects models*
- *Fixed vs. random effects*
- *Partial pooling*
- *Maximum likelihood*

Linear mixed effects models (LMEs)

Also referred to as:

- multilevel models
- random-effects models
- hierarchical linear models

Why are they "linear"?

- Linear mixed effects models assume the outcome is continuous
- There are also generalized linear mixed models (GLMMs) that use a link function to connect a linear predictor to the mean of the outcome (e.g., logit link, log link) and can take different distributions (e.g., binomial, Poisson)

LMEs are an extension of regression

Linear regression: estimates the relation between a dependent variable/**outcome** and one or more independent/explanatory/**predictor** variables

- "As X changes, what happens to Y?"
  - e.g., as stimulus intensity changes, what happens to mean amplitude?
  - E.g., as age changes what happens to mean amplitude?

Linear regression (review)

$$y_j = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon_j$$

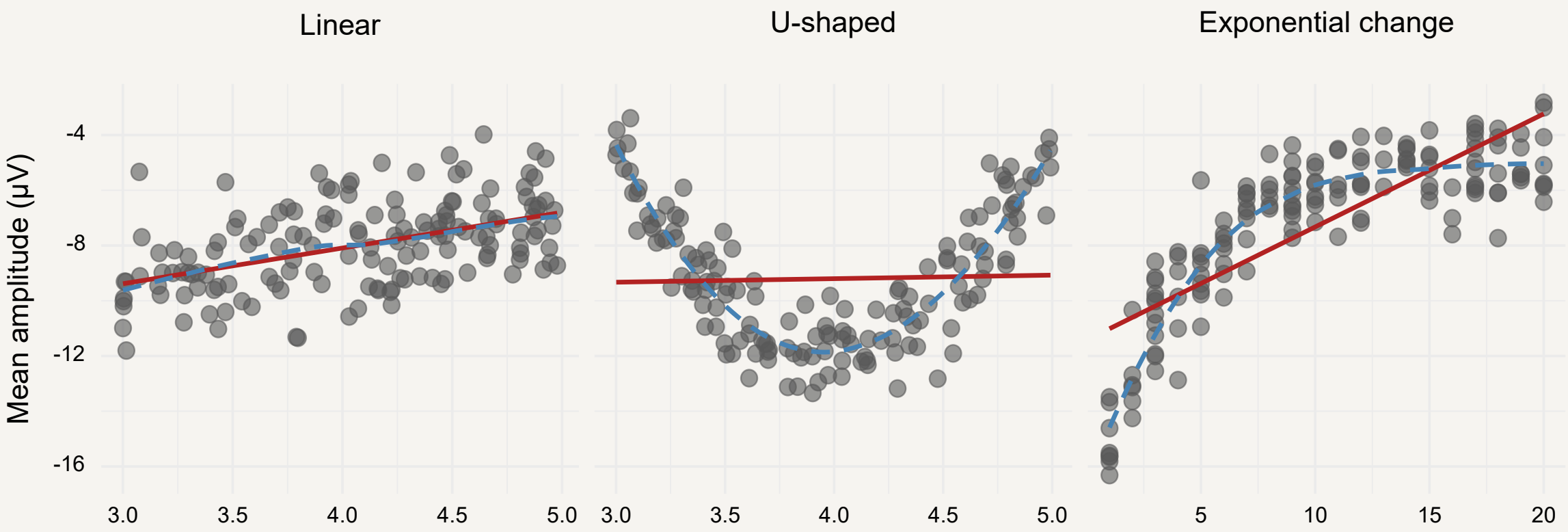
- β_0 = intercept;
- β_1, β_2 = coefficients of predictor variables
- $\varepsilon_j \sim N(0, \sigma^2)$ is residual error

Linear regression (review)

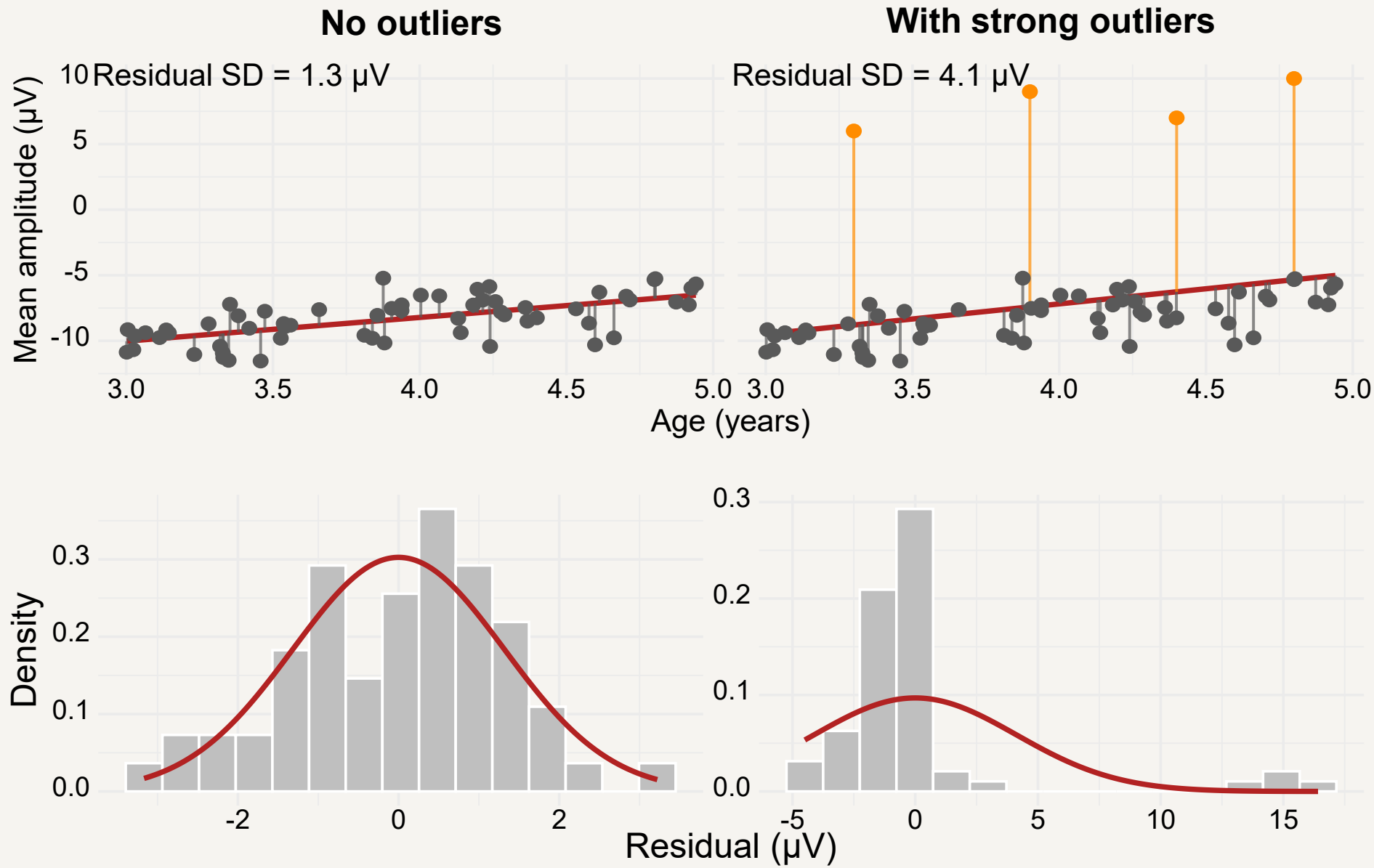
$$\text{ERP mean amplitude} = \text{intercept} + \text{Age} + \text{Condition} + \varepsilon$$

Every participant is assumed to share the exact same relation between predictors and outcome

Assumptions of regression: linearity



Assumptions of regression: normality



To meet the normality assumption of regression, the predictors must be normally distributed.

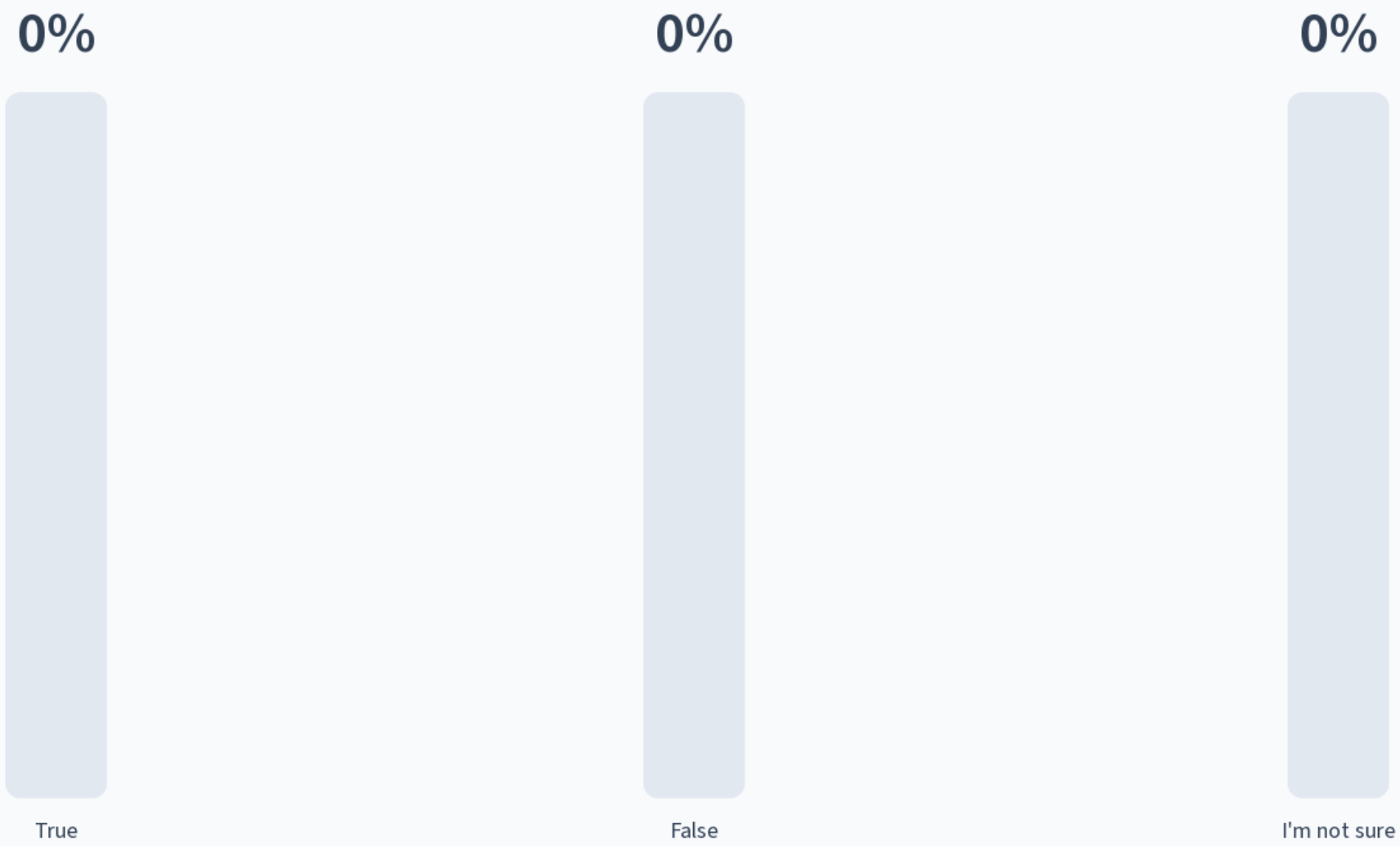
True

False

I'm not sure

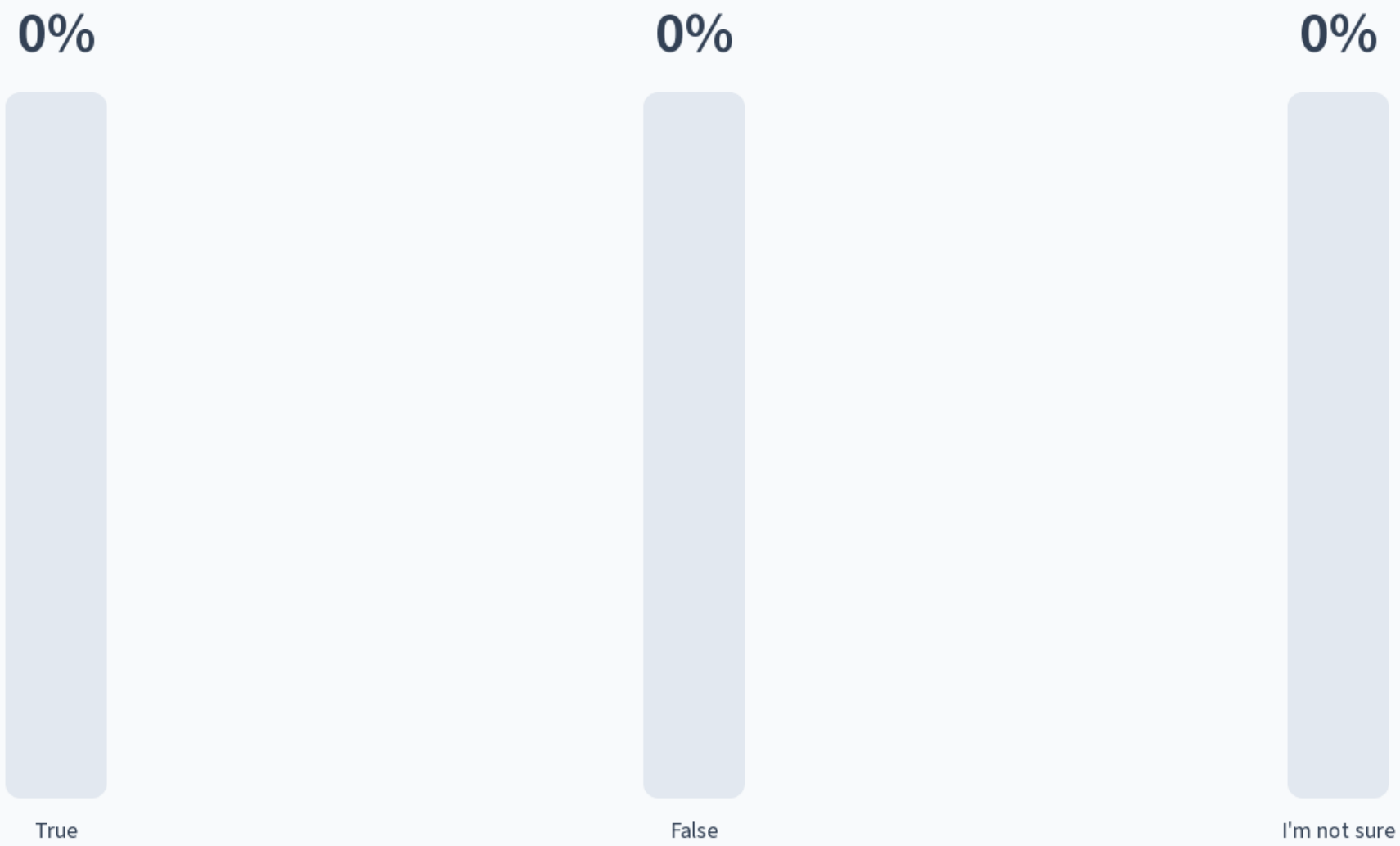
Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

To meet the normality assumption of regression, the predictors must be normally distributed.



Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

To meet the normality assumption of regression, the predictors must be normally distributed.



Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Normality of residuals

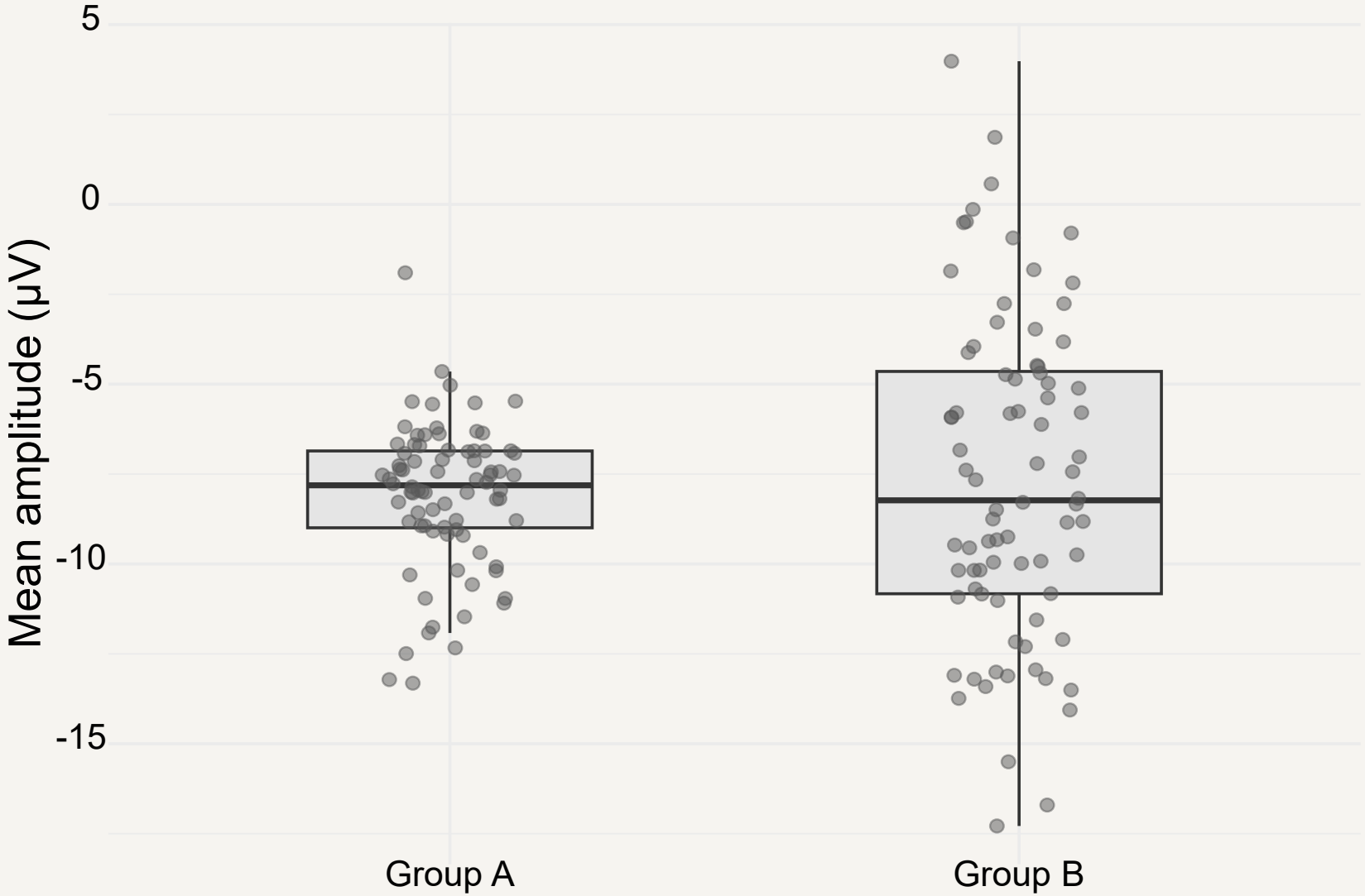
The assumption of normality is for residuals, but predictors don't necessarily have to be normally distributed.

- Predictors *can* be normally distributed, categorical, binary, etc.

A very skewed predictor or a predictor with extreme values can cause problems via:

- A non-linear relationship with the outcome (mean amplitude)
- Containing a strong outlier that pulls the line of best fit away from most of the data
- Leading to non-normal residuals if the model doesn't fit the data well

Assumptions of regression: homogeneity of variance



Assumptions of regression

Assumptions of regression:

- Linearity
- Normality of residuals
- Homogeneity of variance
- Independence of residuals

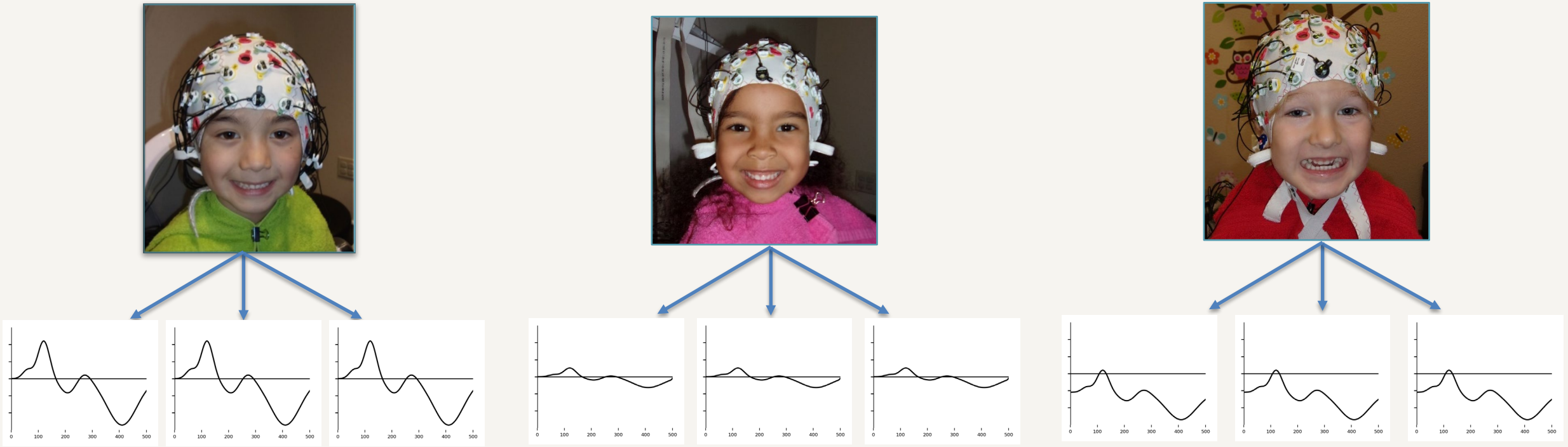
02

Concepts & theory

- *Review of linear regression & its assumptions*
- ***The problem: ERP trials aren't independent***
- *The fix: Overview of linear mixed effects models*
- *Fixed vs. random effects*
- *Partial pooling*
- *Maximum likelihood*

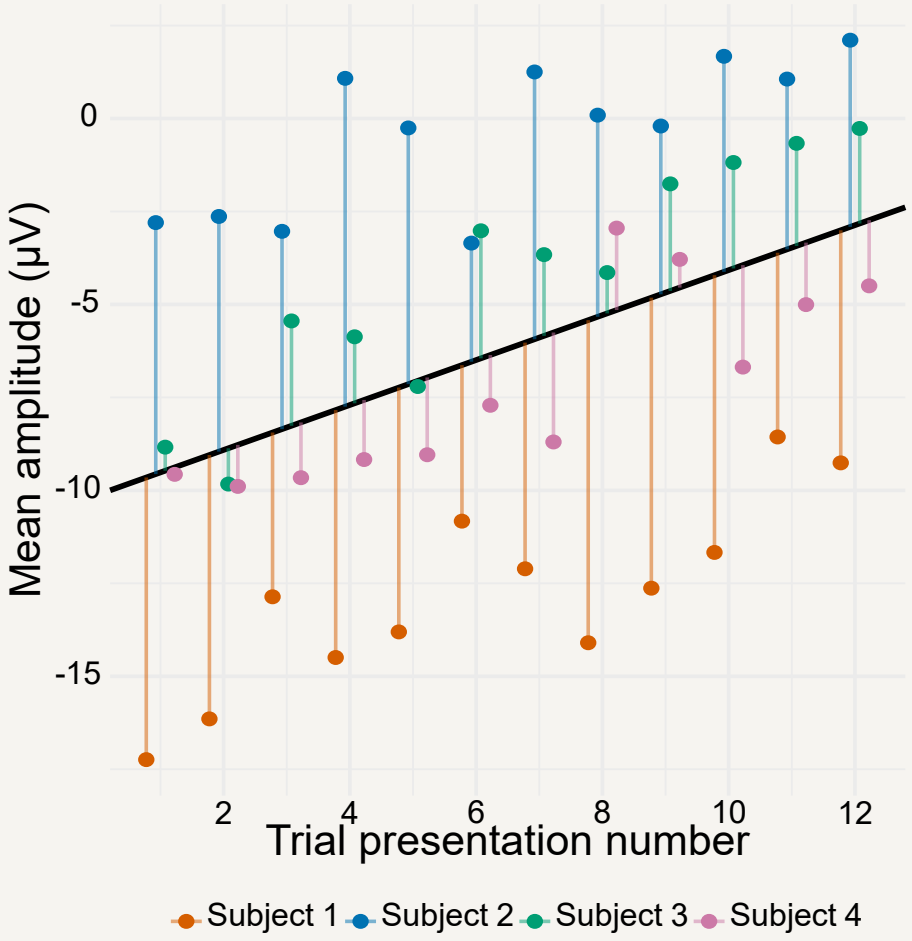
ERP trials are non-independent

ERP trials within a participant are correlated with each other



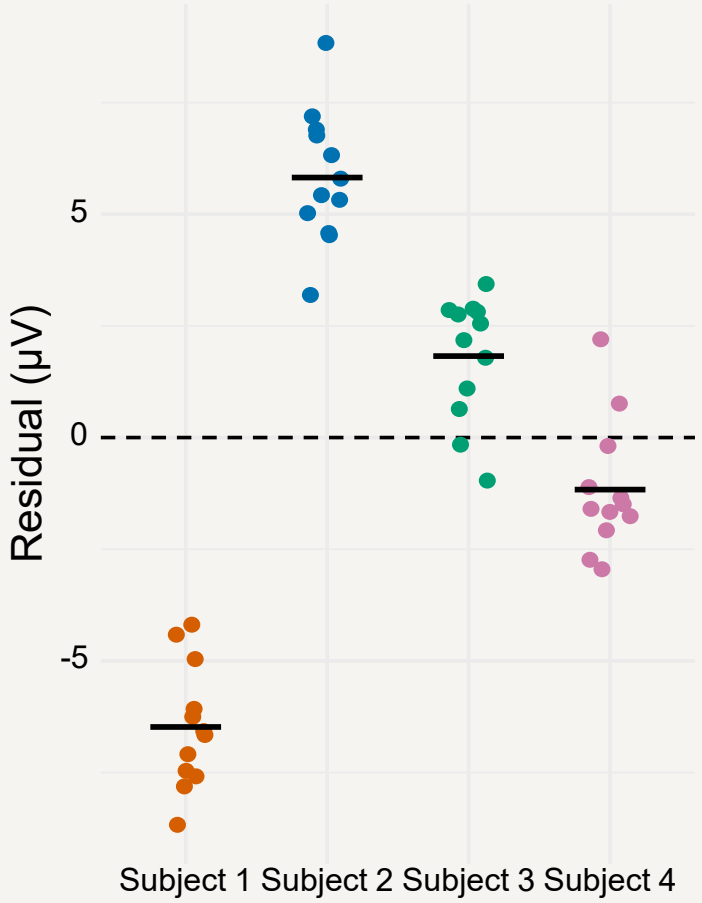
Non-independence of residuals

Residuals cluster within subject



Black line = line of best fit; segments = residuals

Same residuals, grouped by participant

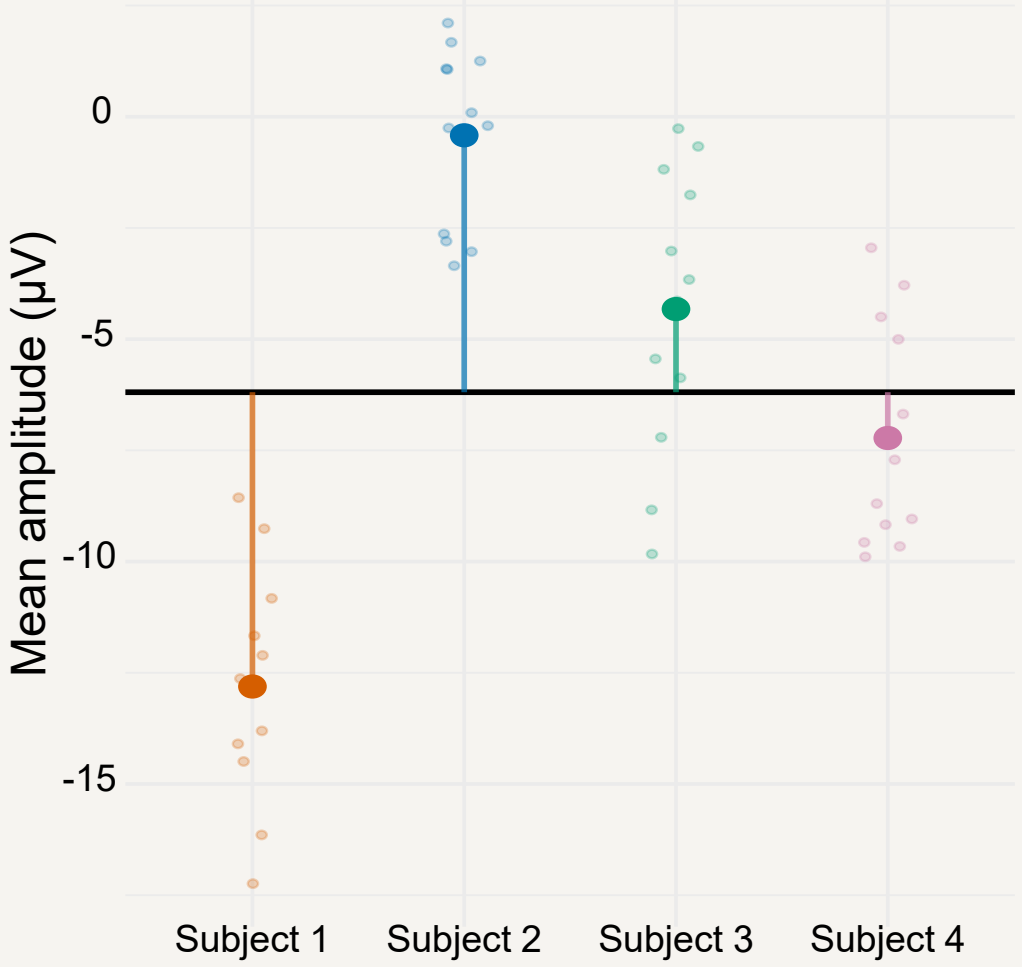


Black bar = participant's mean residual

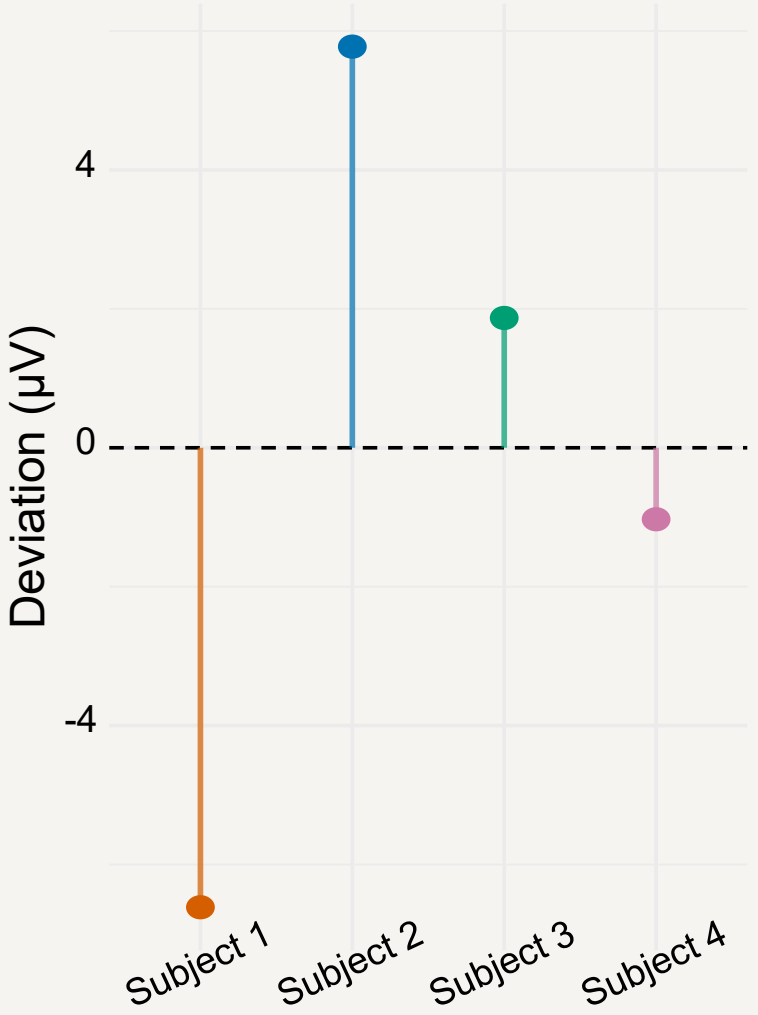
Regression solution to non-independence of residuals

Averaging: one mean per participant

Black line = grand mean;
large dots = participant means; faint dots = trials



Participant deviations from the grand mean



LMEs are an extension of regression

Assumptions of LMEs:

- Linearity
- Normality of residuals
- Homogeneity of variance
- ~~Independence of residuals~~

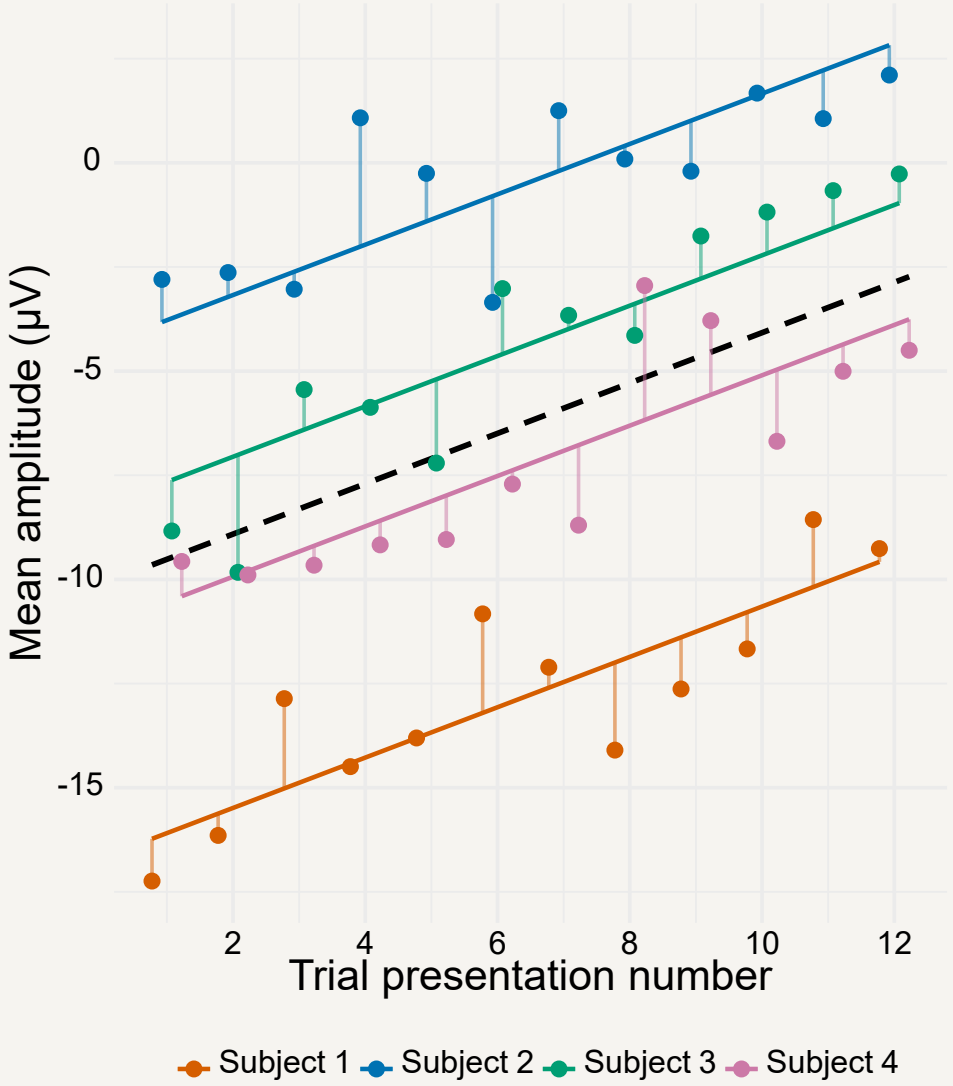
02

Concepts & theory

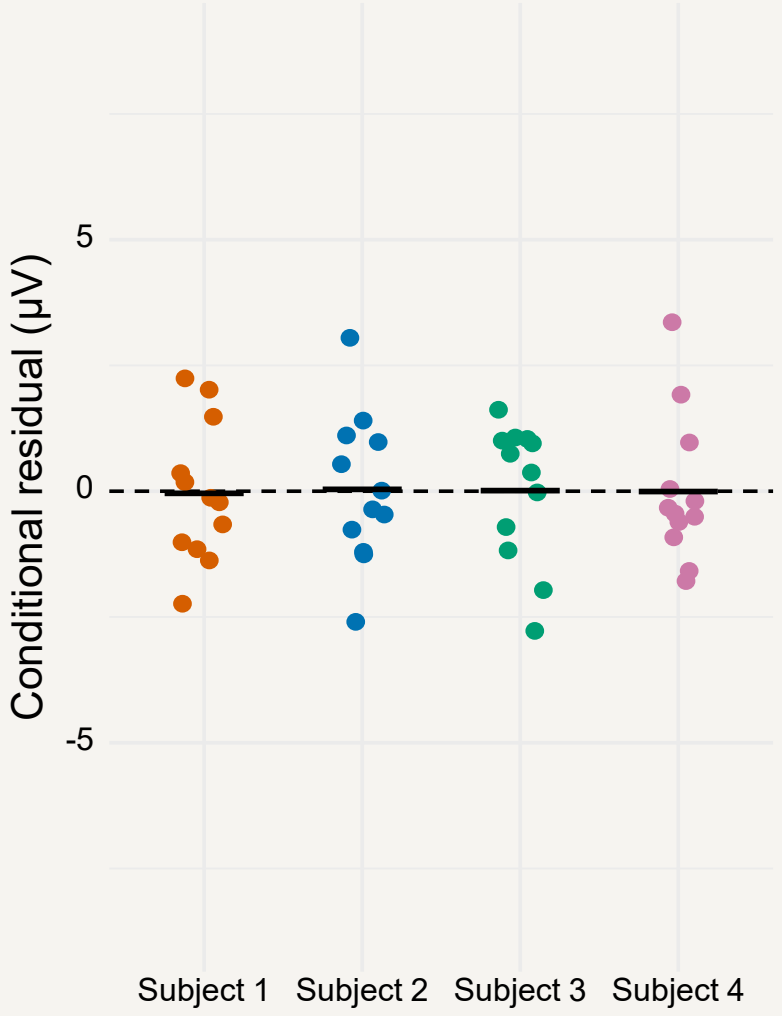
- *Review of linear regression & its assumptions*
- *The problem: ERP trials aren't independent*
- ***The fix: Overview of linear mixed effects models***
- *Fixed vs. random effects*
- *Partial pooling*
- *Maximum likelihood*

LME solution to non-independence of residuals

Each participant has their own intercept



Conditional residuals by participant



Participant means now sit at ~0; what's left is trial-level noise

Quick recap

- ERP trials are nested within participants — ERPs from the same participant are more similar to each other (“correlated with each other”) than to another participant's trials
- Linear regression meets assumptions of independence by averaging — collapsing to one value per condition per participant
- LMEs instead model the nesting directly, at the trial level, without collapsing data

The linear mixed effects model

Level 2 (between-subjects):

$$\beta_{0j} = \gamma_{00} + \gamma_{01}\text{Predictor}_j + u_{0j}$$

- i = trial-level estimates; j = participant-level estimates; γ = fixed-effect means across the sample
- u_{0j} = each participant's own deviation from the grand mean (the random intercept, one value per participant)

Benefits of LMEs

- Includes all collected clean ERP data (doesn’t require casewise deletion)
- Allows you to model intra-individual variability

Participant-level/between-subjects effects

Linear regression or Repeated measures ANOVA

$$\text{Mean amplitude} = \text{intercept} + \text{Predictor} + \text{error}$$

- **Age**
- **Group** (e.g., clinical diagnosis vs. control, bilingual vs. monolingual)
- **Behavioral/cognitive measures** (e.g., working memory, vocabulary)

LME

Level 2 (between-subjects):

$$\beta_{0j} = \gamma_{00} + \gamma_{01}\text{Predictor}_j + u_{0j}$$

Level 1 (within-subjects):

$$y_{ij} = \beta_{0j} + \beta_{1j}\text{Predictor}_{ij} + \varepsilon_{ij}$$

Trial-level/within-subjects effects

Repeated measures ANOVA

$$\text{Mean amplitude} = \text{intercept} + \text{Predictor} + \text{error}$$

LME

Level 2 (between-subjects):

$$\beta_{0j} = \gamma_{00} + \gamma_{01}\text{Predictor}_j + u_{0j}$$

Level 1 (within-subjects):

$$y_{ij} = \beta_{0j} + \beta_{1j}\text{Predictor}_{ij} + \varepsilon_{ij}$$

Categorical factors (e.g., Angry vs. Happy) and **one number per participant per cell**

- **Condition** (e.g., congruent vs. incongruent, standard vs. deviant, word vs. pseudoword, emotional valence)
- **Task** (e.g., passive viewing vs. active response
- **Hemisphere or cluster** (left vs. right)

Trial-level/within-subjects effects

Repeated measures ANOVA

$$\text{Mean amplitude} = \text{intercept} + \text{Predictor} + \text{error}$$

LME

Level 2 (between-subjects):

$$\beta_{0j} = \gamma_{00} + \gamma_{01}\text{Predictor}_j + u_{0j}$$

Level 1 (within-subjects):

$$y_{ij} = \beta_{0j} + \beta_{1j}\text{Predictor}_{ij} + \varepsilon_{ij}$$

Continuous, trial-varying, single-trial-level predictors

- **Cloze probability/predictability**: continuous variable for whether a stimulus is expected
- **Semantic distance**: similarity between prime and target, or between a word and context
- **Visual salience**
- **Word length**
- **Reaction time**
- **Trial number**

Benefits of LMEs

- Includes all collected clean ERP data (doesn't require casewise deletion)
- Allows you to model intra-individual variability
- Allows you to model MORE predictors
  - Continuous trial-level predictors

02

Concepts & theory

- *Review of linear regression & its assumptions*
- *The problem: ERP trials aren't independent*
- *The fix: Overview of linear mixed effects models*
- ***Fixed vs. random effects***
- *Partial pooling*
- *Maximum likelihood*

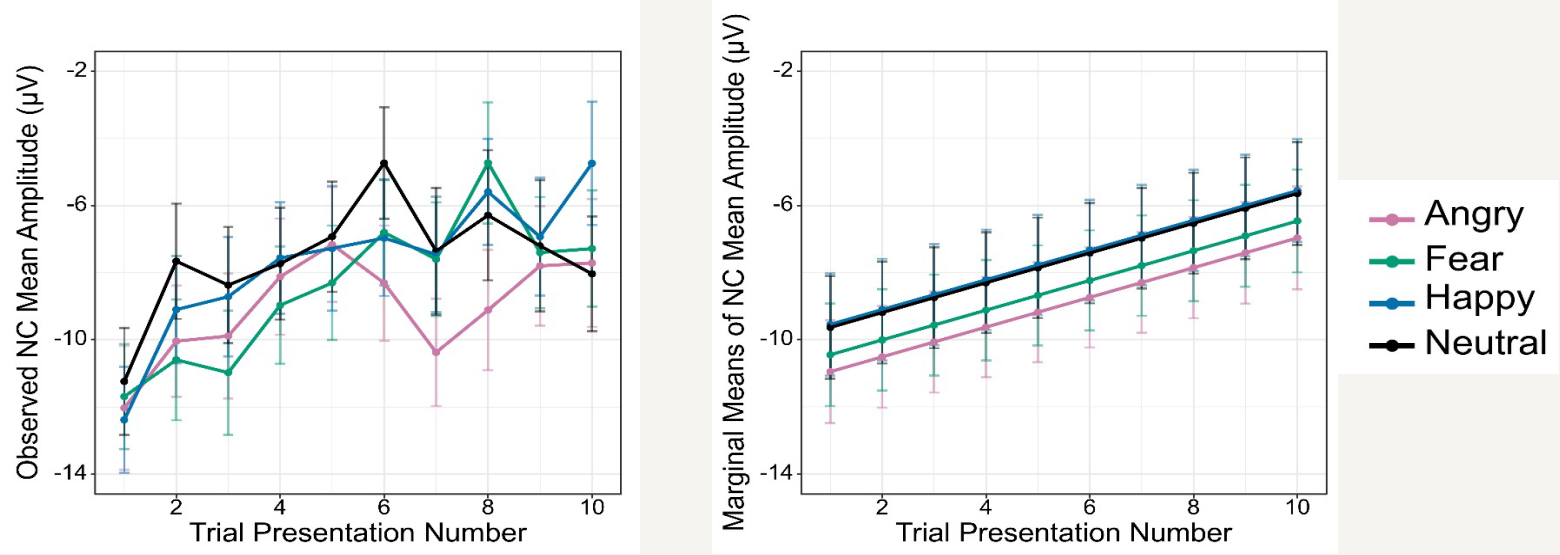
Fixed vs. random effects

Fixed Effects

- Assumed non-random and consistent across the population
- Effects you want to test or interpret (e.g., condition, age, group)
- Modeled the same way in regression and LME

Examples of common fixed effects in LMEs

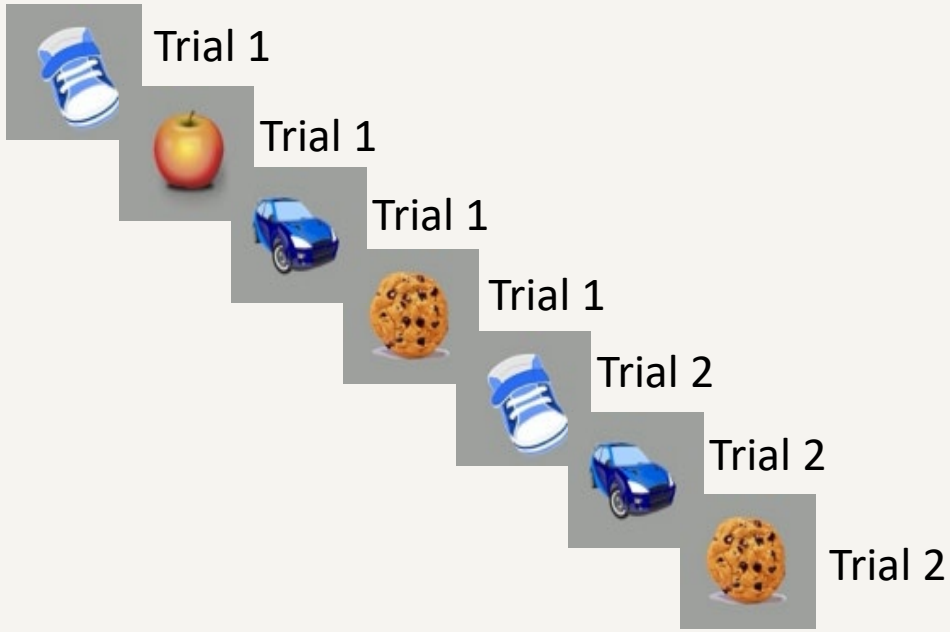
- Participant-level predictors (e.g., age, vocabulary, clinical diagnosis)
- Trial-varying predictors
  - Behavioral response (e.g., correct/incorrect)
  - Trial presentation number (e.g., see Borgström et al., 2016 for habituation example)



Examples of common fixed effects in LMEs

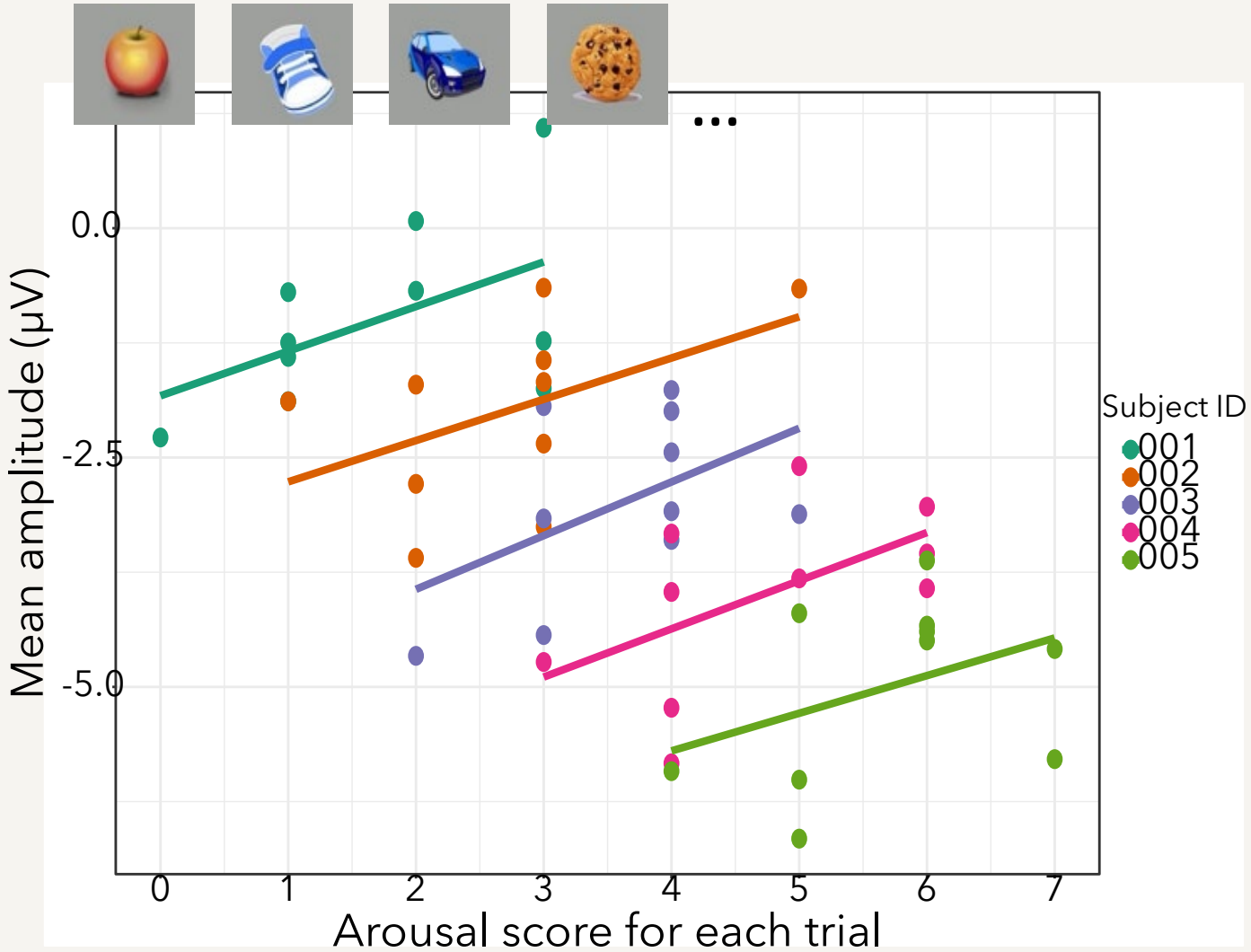
- Trial presentation number (e.g., see Borgström et al., 2016 for habituation example)

Heise et al. 2022



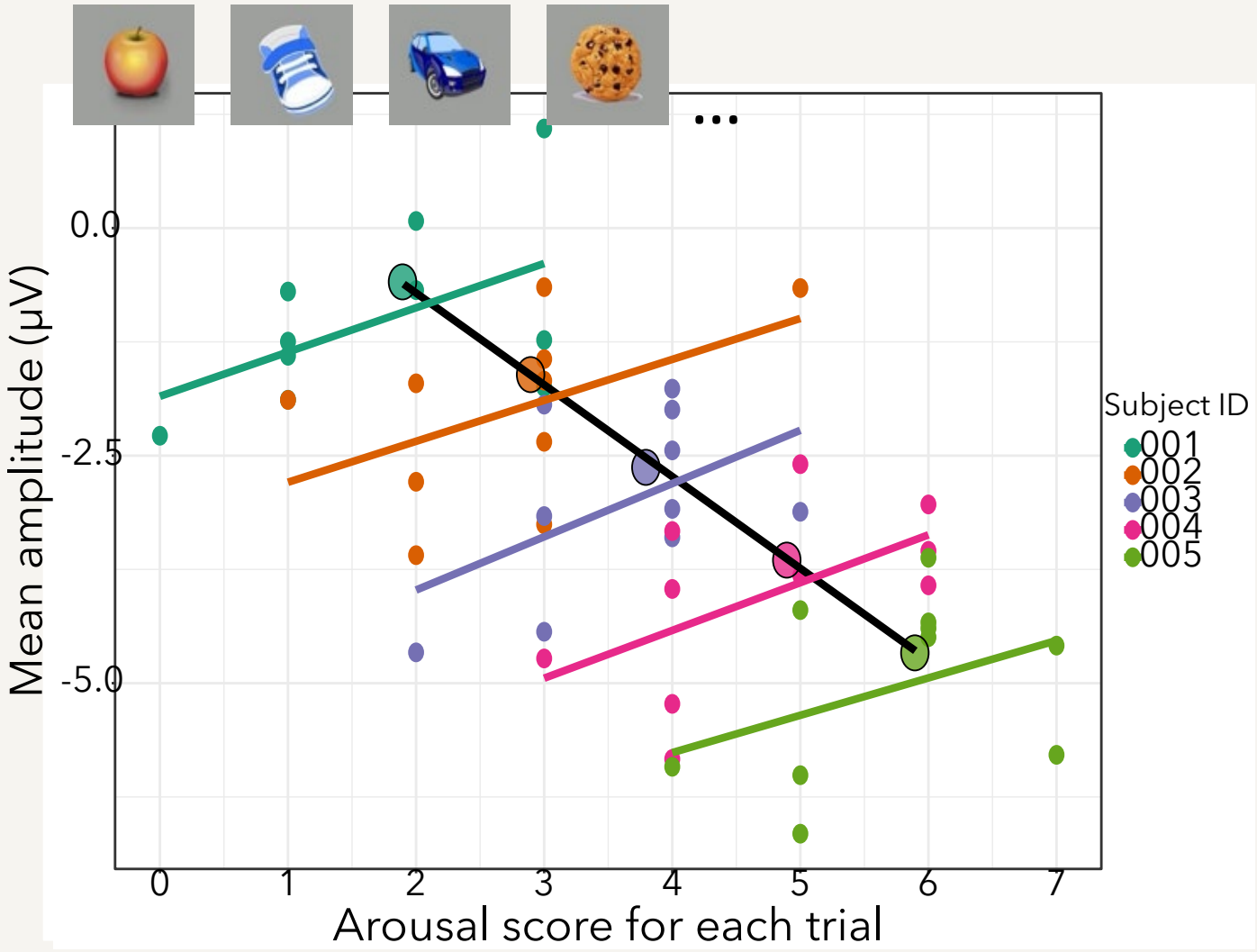
- Stimuli repeat in a block-randomized way
- Stimulus-specific adaptation / growing familiarity with that face, sound, or word

Within- & between-subjects trial-level predictors



Data generating using R package 'correlation' (Makowski et al., 2020)

Within- & between-subjects trial-level predictors



Data generating using R package 'correlation' (Makowski et al., 2020)

Benefits of LMEs

- Includes all collected clean ERP data (doesn’t require casewise deletion)
- Allows you to model intra-individual variability
- Allows you to model MORE predictors
  - Continuous trial-level predictors
  - Within-subject/trial-level effects

Fixed vs. random effects

Fixed Effects

- Assumed non-random and consistent across the population
- Effects you want to test or interpret (e.g., condition, age, group)
- Modeled the same way in regression and LME

Random Effects

- Assumed to be sampled from a population but not consistent
- Usually not of direct interest themselves (nuisance variables)
- Modeled to account for extra variability that would otherwise bias your effect of interest
- e.g., participant, stimulus/actor

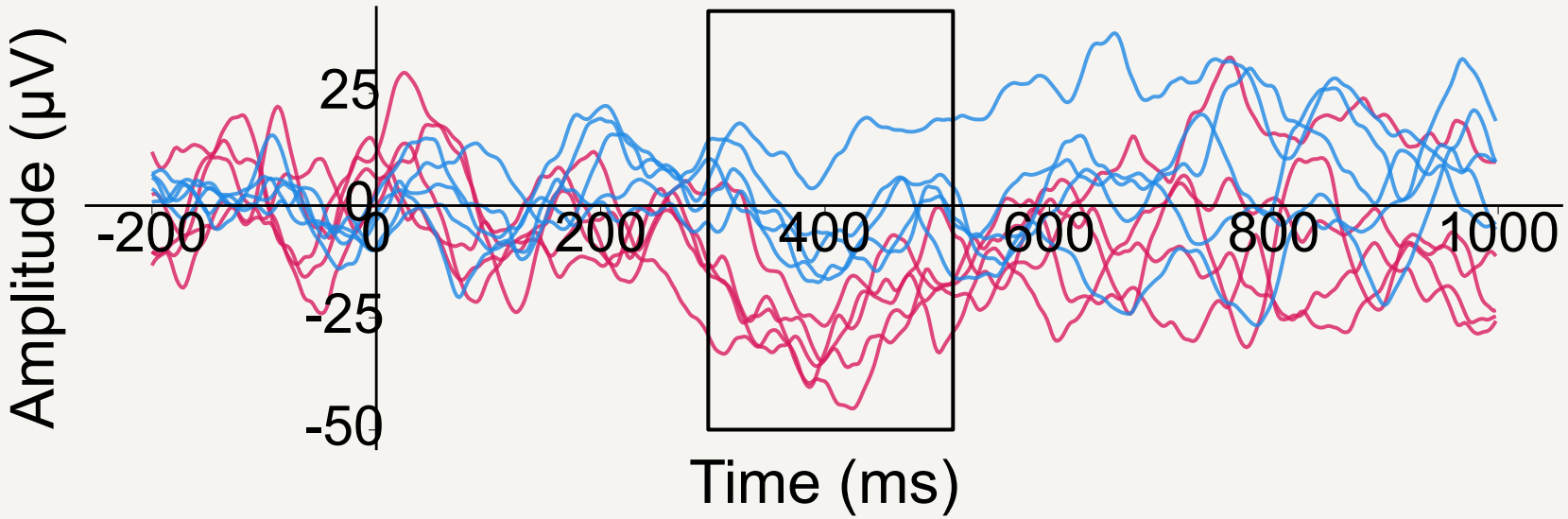
When your model contains both fixed and random effects, you have "mixed effects" in your model

Examples of common random effects

- Participant

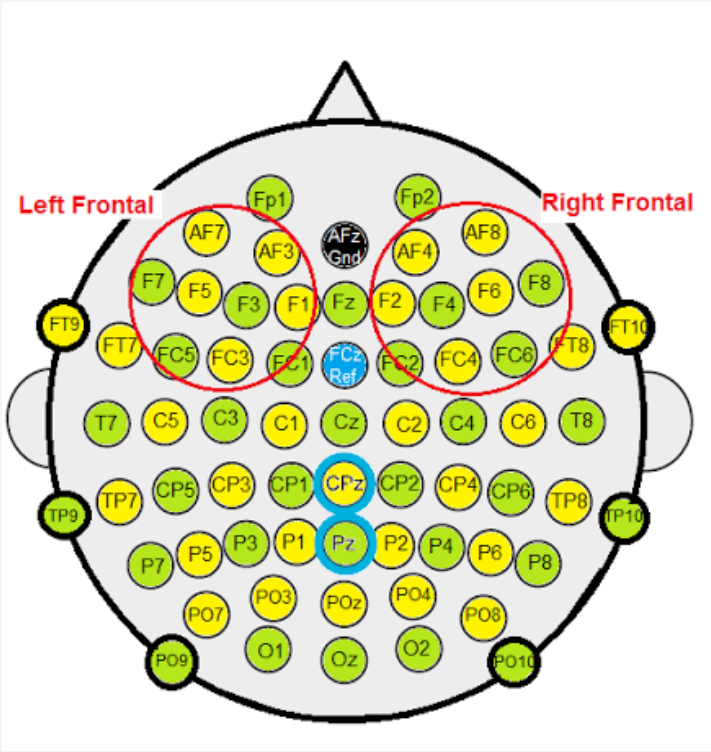
Subject ID (random effect): Trials from the same participant have higher or lower average mean amplitude

- Subject 1
- Subject 2



Examples of common random effects

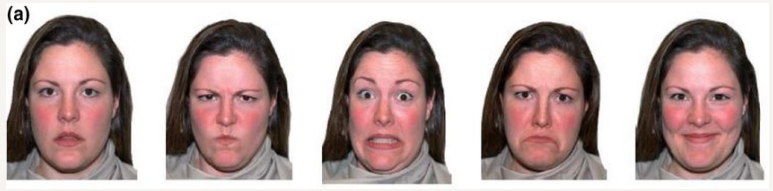
- Electrode/channel
  - Depending on:
    - 1) whether the coefficient for the channel itself is of interest
    - 2) number of channels analyzed
      - Volpert-Esmond et al. (2021) recommend that random effects should have 5 or more categories (e.g., 5+ channels or actors)
      - Suggestions vary depending on field



Examples of common random effects

- Stimulus features shared across trials

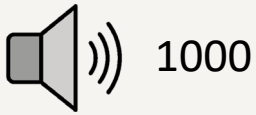
Actor A



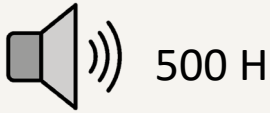
Actor B



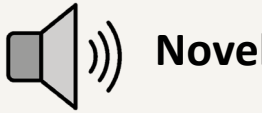
Kammermeier, Marina & Perez, Lena & König, Lilith & Paulus, Markus. (2019). Attachment security and attention to facial emotional expressions in preschoolers: An eye-tracking study. British Journal of Developmental Psychology. 38. 167-185. 10.1111/bjdp.12313.



1000 Hz Standard



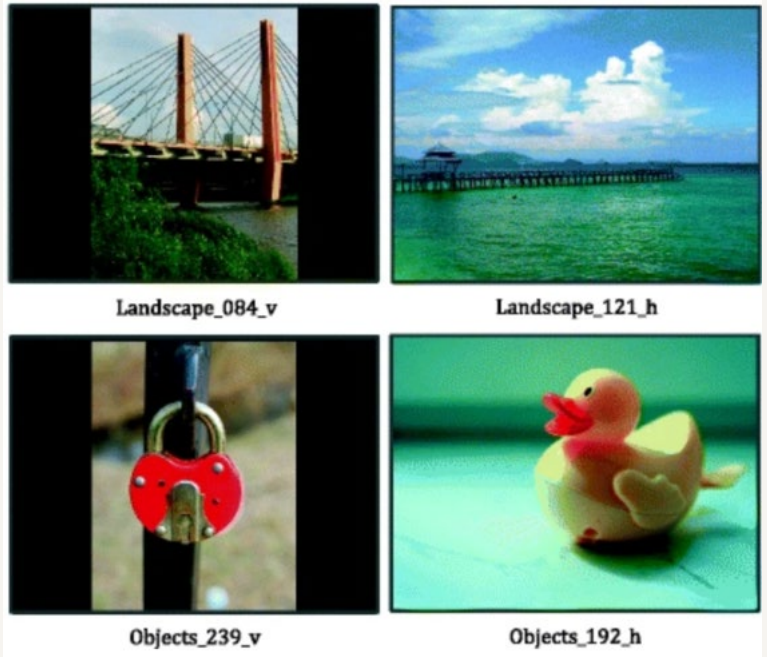
500 Hz Target Deviant



Novel deviant







Nencki Affective Picture set

Which of these variables is typically modeled as a RANDOM effect in an LME modeling ERPs?

- Condition
- Task difficulty
- Age group
- Participant ID
- None of the above

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Which of these variables is typically modeled as a RANDOM effect in an LME modeling ERPs?

Condition 0%

Task difficulty 0%

Age group 0%

Participant ID 0%

None of the above 0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Which of these variables is typically modeled as a RANDOM effect in an LME modeling ERPs?

Condition 0%

Task difficulty 0%

Age group 0%

Participant ID 0%

None of the above 0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

LEARNING CHECK

Which of these is typically modeled as a RANDOM, not fixed, effect in an ERP LME?

A. Condition

B. Participant ID

C. Age group

D. Task difficulty

Which of the following determine whether a variable is modeled as a fixed or random effect?

- Number of factor levels (e.g., # of electrodes analyzed)
- Whether factor levels are a sample from a larger population you want to generali…
- Whether you're interested in interpreting the coefficient of the predictor
- All of the above

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Which of the following determine whether a variable is modeled as a fixed or random effect?

Number of factor levels (e.g., # of electrodes analyzed) 0%

Whether factor levels are a sample from a larger population you want to generalize to 0%

Whether you're interested in interpreting the coefficient of the predictor 0%

All of the above 0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Which of the following determine whether a variable is modeled as a fixed or random effect?

Number of factor levels (e.g., # of electrodes analyzed) 0%

Whether factor levels are a sample from a larger population you want to generalize to 0%

Whether you're interested in interpreting the coefficient of the predictor 0%

All of the above 0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Which of the following factors determine whether a variable is a fixed or random effect?

A. Number of categories (e.g., number of electrodes analyzed)

B. Whether the categories (e.g., the electrodes or actors you used) are a sample from a larger population you want to generalize to

C. Whether you're interested in interpreting the coefficient of the predictor

D. All of the above

Benefits of LMEs

- Includes all collected clean ERP data (doesn’t require casewise deletion)
- Allows you to model intra-individual variability
- Allows you to model MORE predictors
  - Continuous trial-level predictors
  - Within-subject/trial-level effects
  - Random effects

02

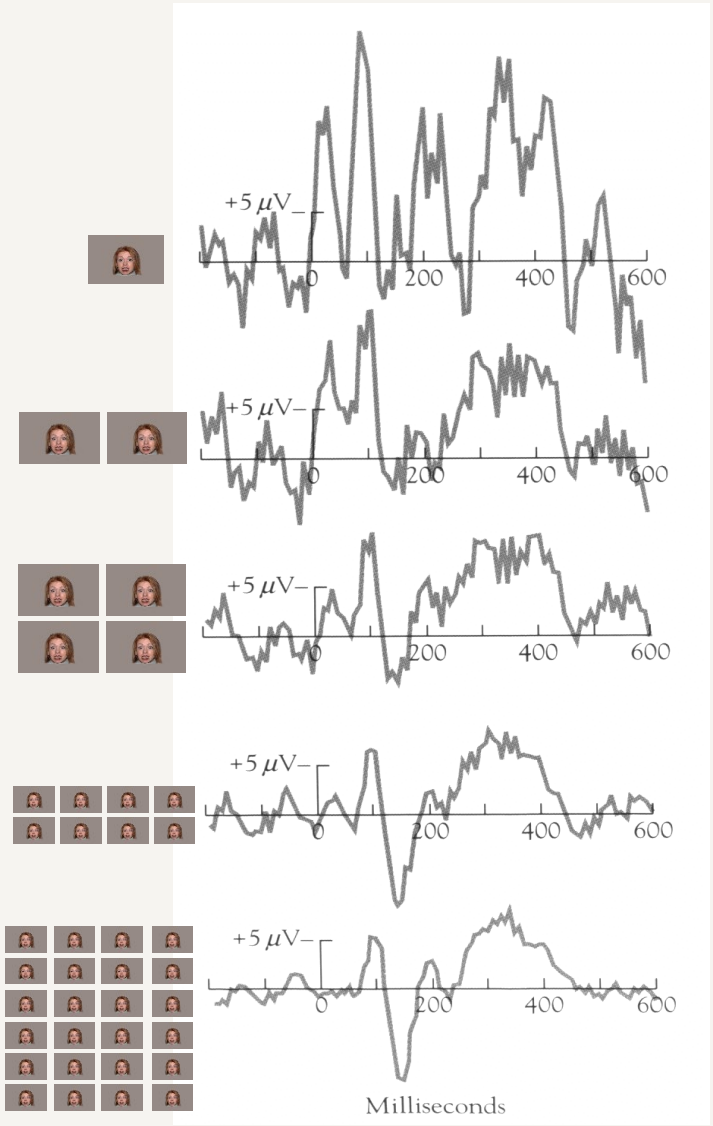
Concepts & theory

- *Review of linear regression & its assumptions*
- *The problem: ERP trials aren't independent*
- *The fix: Overview of linear mixed effects models*
- *Fixed vs. random effects*
- ***Partial pooling***
- *Maximum likelihood*

Partial pooling

- You never observe the true value of a participant's mean amplitude, only a noisy estimate built from however many trials that participant contributed.
  - A participant with 4 clean trials gives you a very noisy estimate.
  - A participant with 60 clean trials gives you a precise, trustworthy estimate.

One Trial



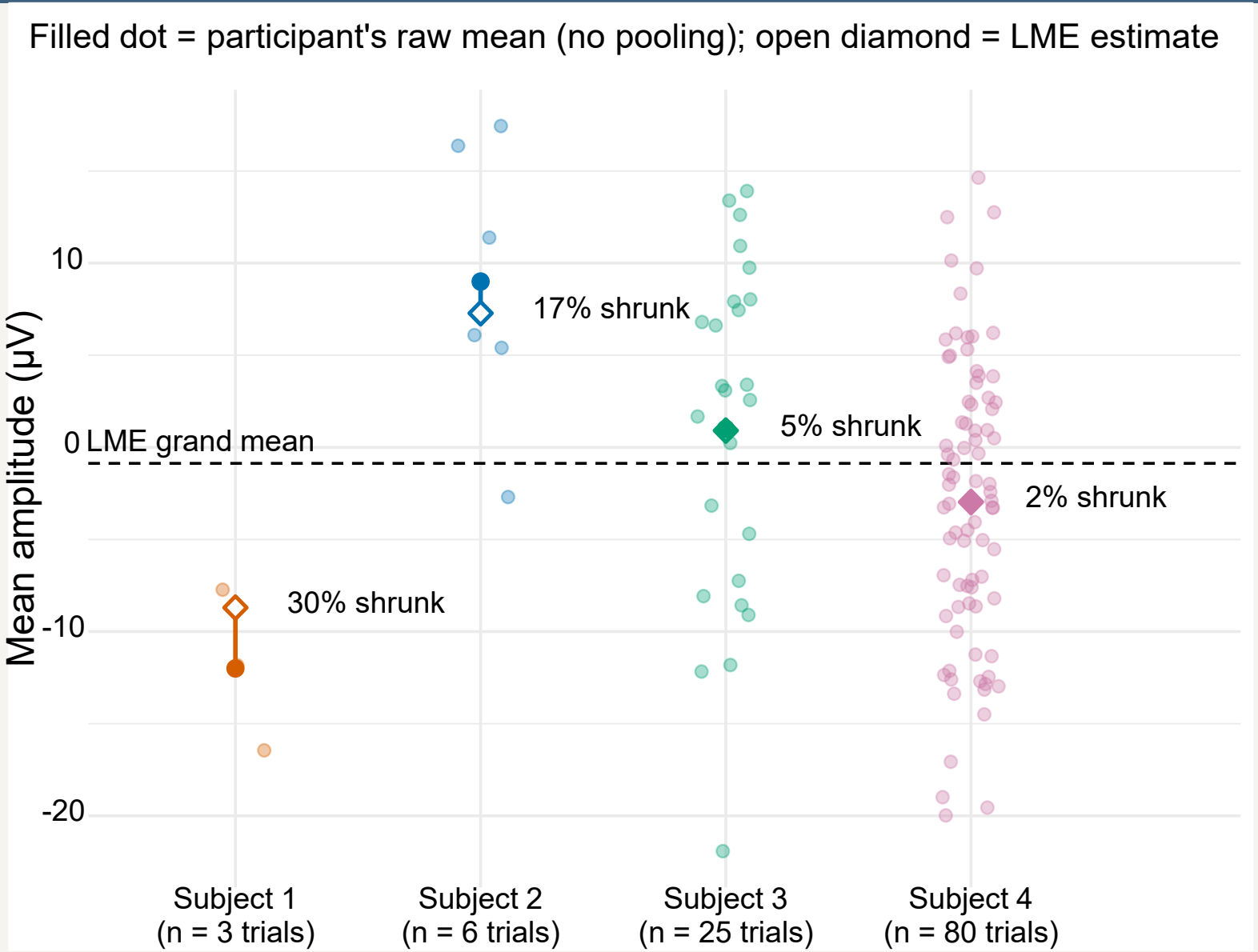
Partial pooling

- In LMEs, participants contribute different amounts of data (i.e., different numbers of trials)
- Participants with less data have noisier estimates of their mean (remember this is why they were being casewise deleted in regression!)
- Participants with fewer usable trials stay in the model — their estimates “shrink” toward the group mean rather than being excluded

Partial pooling

- **Partial pooling**: each participant's estimate is drawn toward the group mean, weighted by how much data that participant contributes ("shrinkage")
  - No pooling: fit a separate model per participant — overfits
  - Complete pooling: treat all trials as one group — ignores differences between subjects
- In LMEs, participants with very few trials are still included in the model, but are automatically down-weighted rather than excluded

Partial pooling



Benefits of LMEs

- **Includes all collected clean ERP data (doesn't require casewise deletion)**
- Allows you to model intra-individual variability
- Allows you to model MORE predictors
  - Continuous trial-level predictors
  - Within-subject/trial-level effects
  - Random effects

02

Concepts & theory

- *Review of linear regression & its assumptions*
- *The problem: ERP trials aren't independent*
- *The fix: Overview of linear mixed effects models*
- *Fixed vs. random effects*
- *Partial pooling*
- ***Maximum likelihood***

How LMEs handle missing data: restricted maximum likelihood

- In LMEs, there are missing data (missing trials), so how are those handled?

Casewise deletion (ANOVA, regression)



- **Participant 3 is excluded** (fewer than 10 trials)
- Unbiased **only if data are MCAR**

Maximum likelihood (LME)

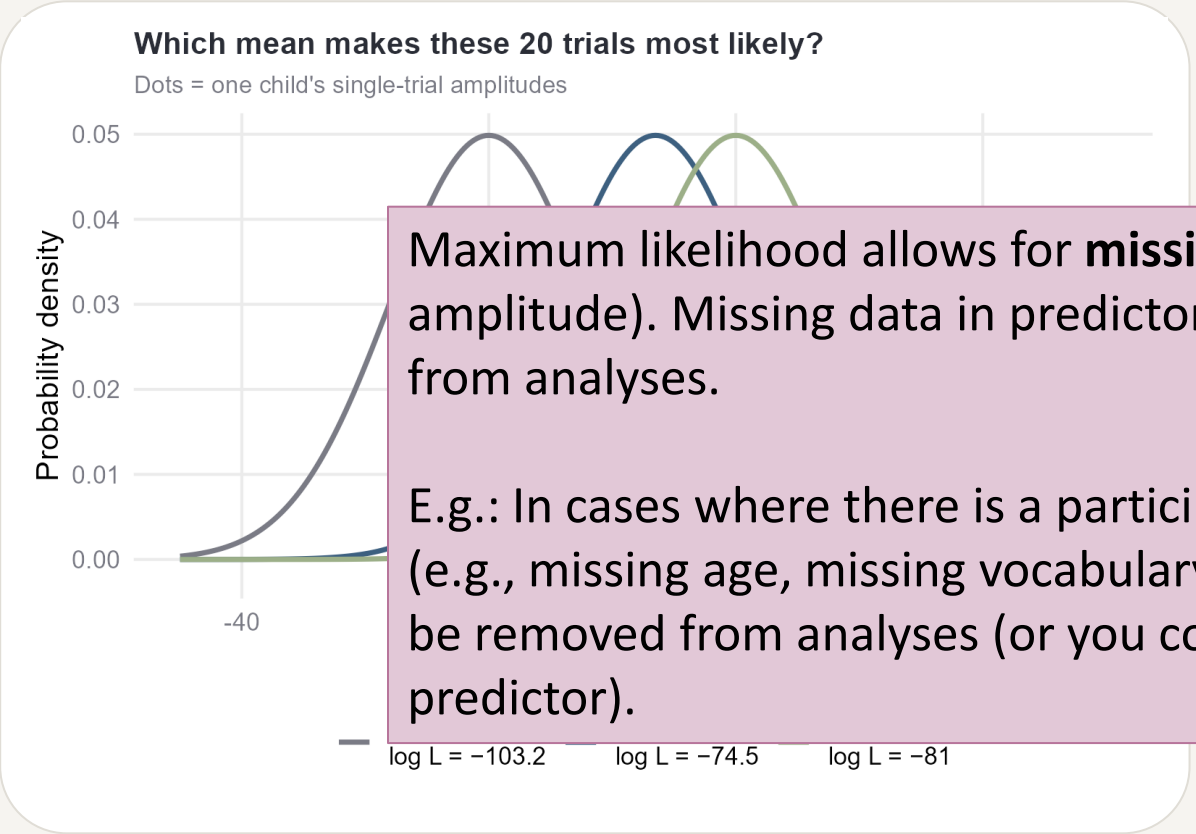


- **Every clean trial counts:** Participant 3 is retained, with less weight (partial pooling)
- **Nothing is imputed:** ML uses only the observed trials

Maximum likelihood returns unbiased estimates when data are MCAR or MAR!

Heise, Mon & Bowman (2022), Baraldi & Enders (2010); Little et al. (2016)

What is maximum likelihood?



Likelihood: What parameters were most likely to have generated our observed data?

e.g., With the data we have, is it most likely that the data-generating process has a mean of 2 µV? 3 µV? 4 µV?

Maximum likelihood allows for **missing data in the outcome** (mean amplitude). Missing data in predictors means that the trial is excluded from analyses.

E.g.: In cases where there is a participant-level predictor with missing data (e.g., missing age, missing vocabulary score), that participant's data would be removed from analyses (or you could use multiple imputation on the predictor).

tude)

under those

e the estimates

In an LME, all parameters are estimated together: fixed effects plus the participant, actor, and trial-level SDs. lme4's optimizer climbs the likelihood step by step.

Converged = it found the peak.

Singular = the peak sits at a boundary (an SD of 0 or a correlation of ±1).

Benefits of LMEs

- **Includes all collected clean ERP data (doesn't require casewise deletion)**
- Allows you to model intra-individual variability
- Allows you to model MORE predictors
  - Continuous trial-level predictors
  - Within-subject/trial-level effects
  - Random effects

03

Demonstration of LMEs vs. ANOVA in simulated data

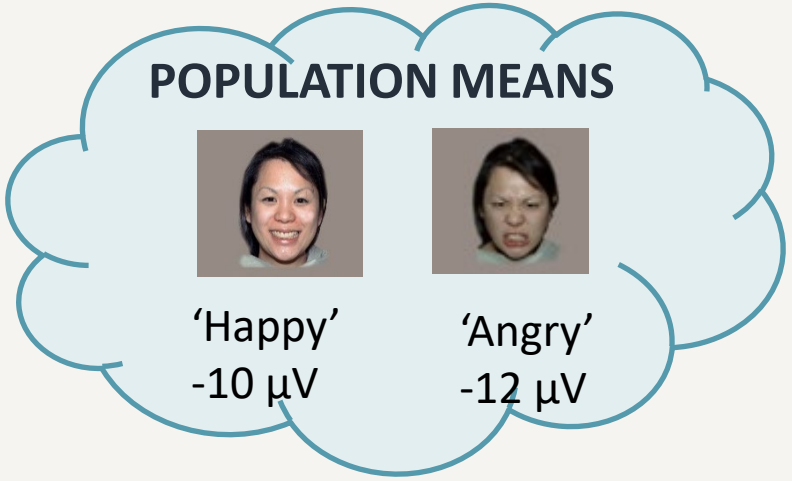
Analyzing simulated mean amplitude with LME & ANOVA



Smaller (less negative) NC amplitude



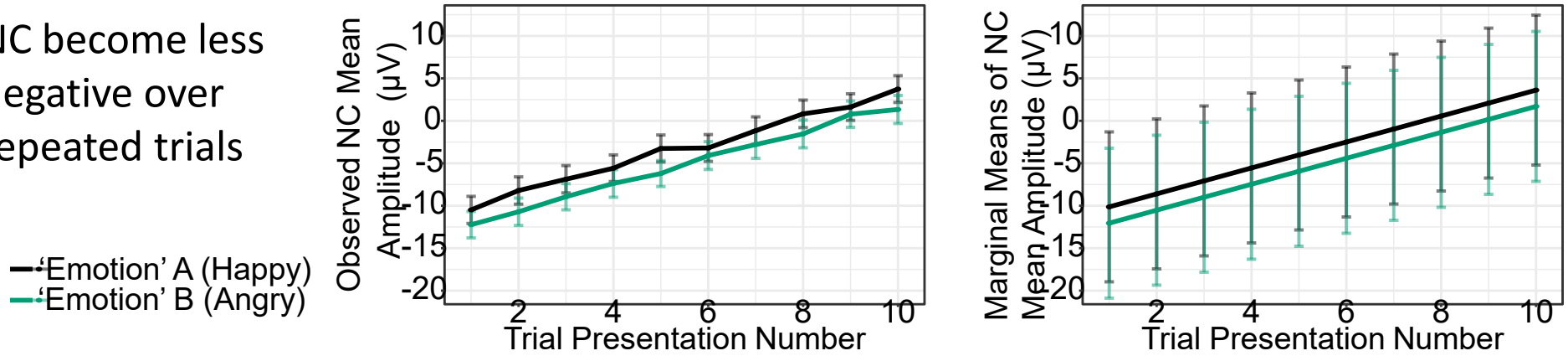
Bigger (more negative) NC amplitude



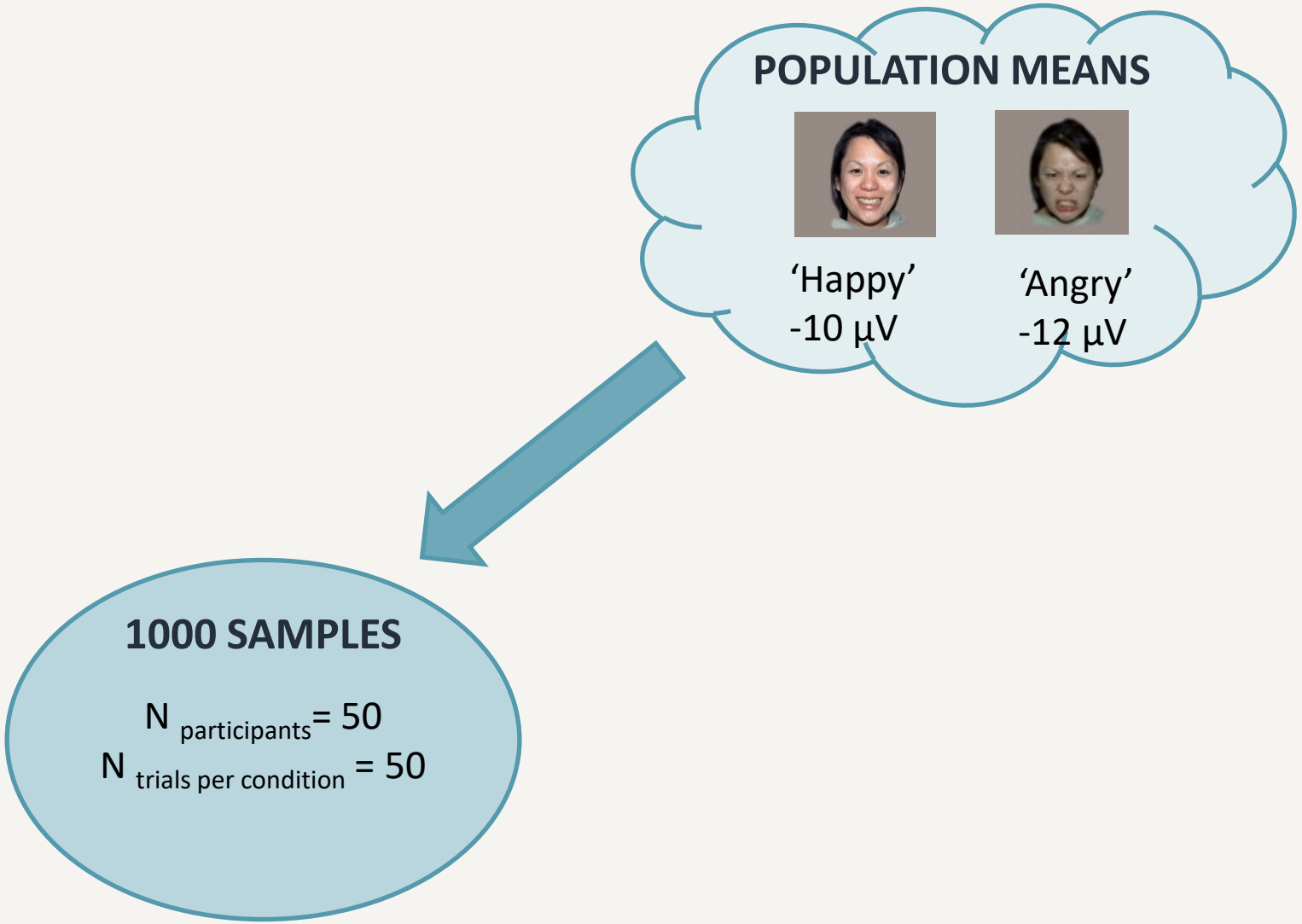
1. Simulate NC ERP component based on parameters in infant ERP literature (Leppänen et al., 2007; Reynolds & Richards, 2005; Smith et al., 2020)

- Two hypothetical conditions
- Amplitude (µV) variability in two 'age groups' (Di Lorenzo et al., 2020)
- Amplitude (µV) 'habituation' with repeated trials (less negative mean amplitude for later trials) (Bergström et al., 2016)

NC become less negative over repeated trials



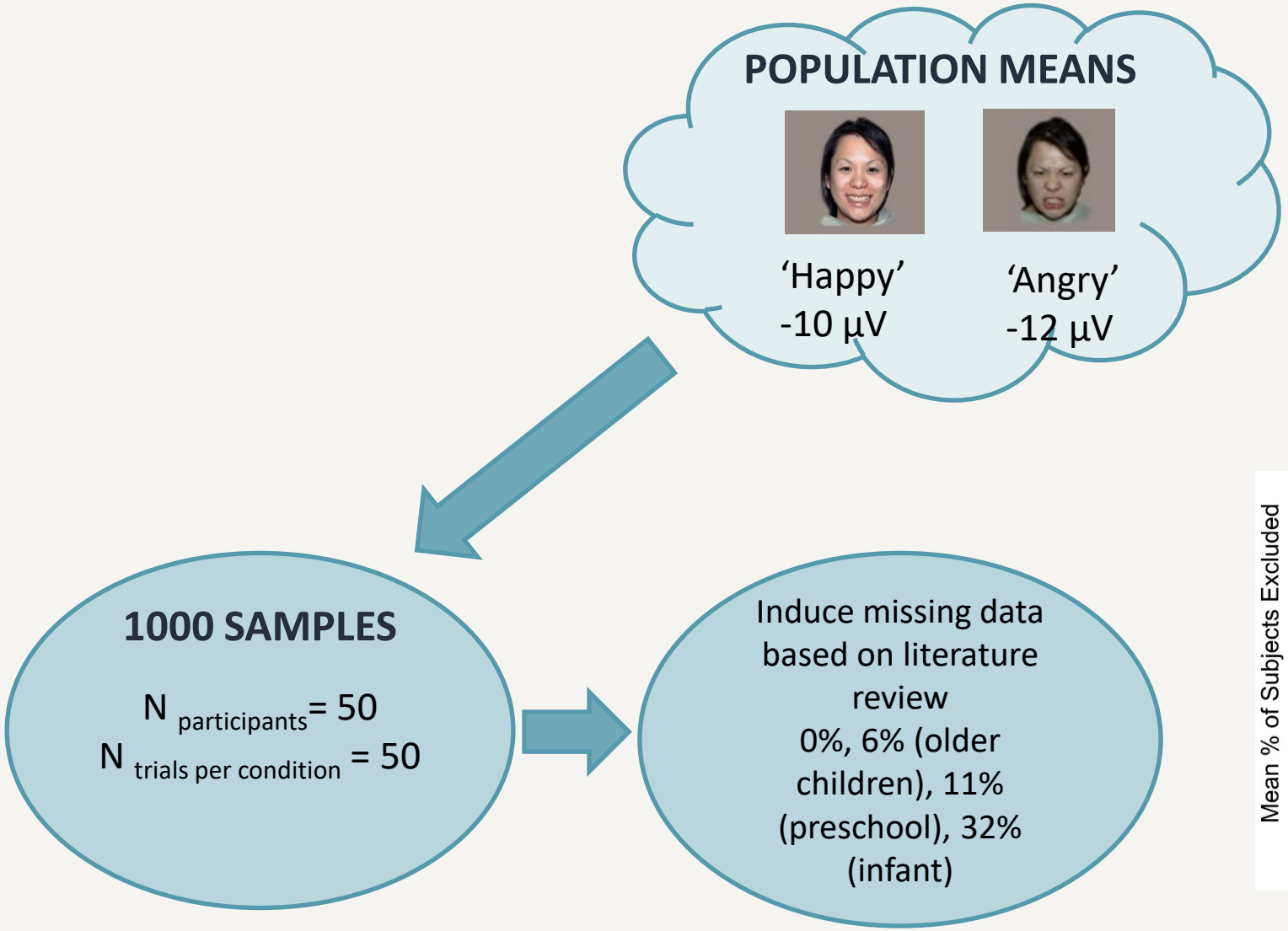
Analyzing simulated mean amplitude with LME & ANOVA



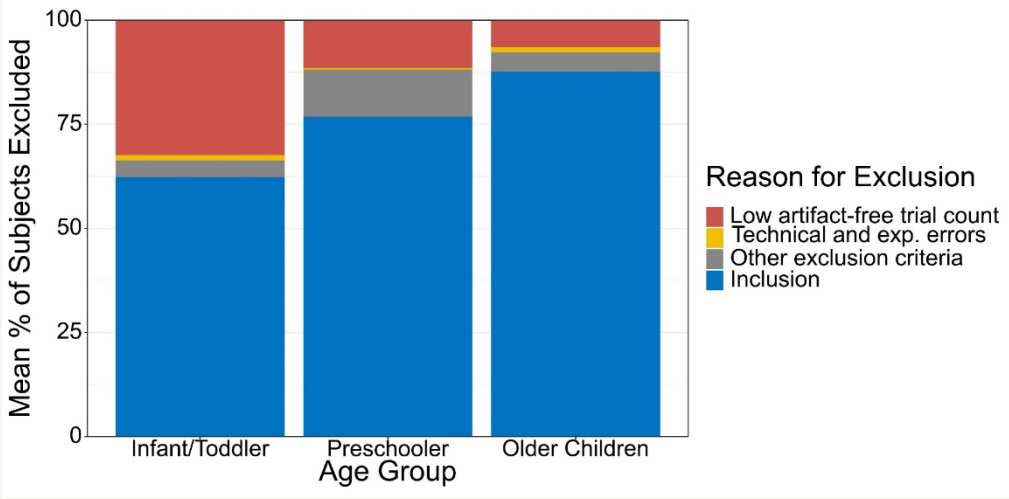
1. Simulate NC

2. Simulate 1000 samples ("studies") from this population of values, each with 50 participants

Analyzing simulated mean amplitude with LME & ANOVA



1. Simulate NC
2. Simulate samples from population parameters
3. **Induce missing data by removing 10+ trials in one or more conditions, such that 0%, 6%, 11%, and 32% of participants would be deleted**



Analyzing simulated mean amplitude with LME & ANOVA



Smaller (less negative) NC amplitude

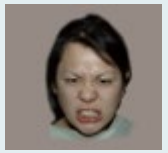


Bigger (more negative) NC amplitude

POPULATION MEANS



'Happy'
-10 µV



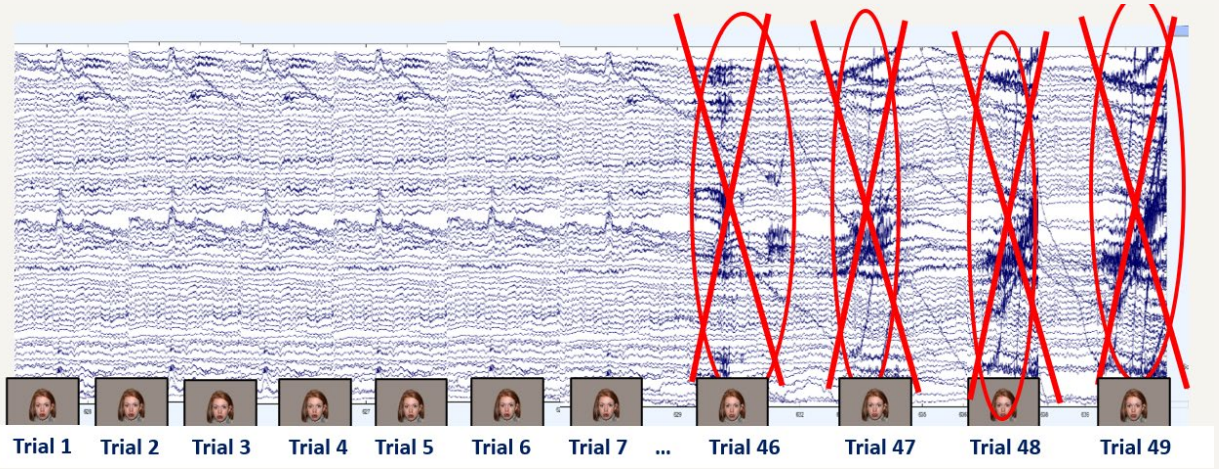
'Angry'
-12 µV

1000 SAMPLES

$N_{participants} = 50$

$N_{trials\ per\ condition} = 50$

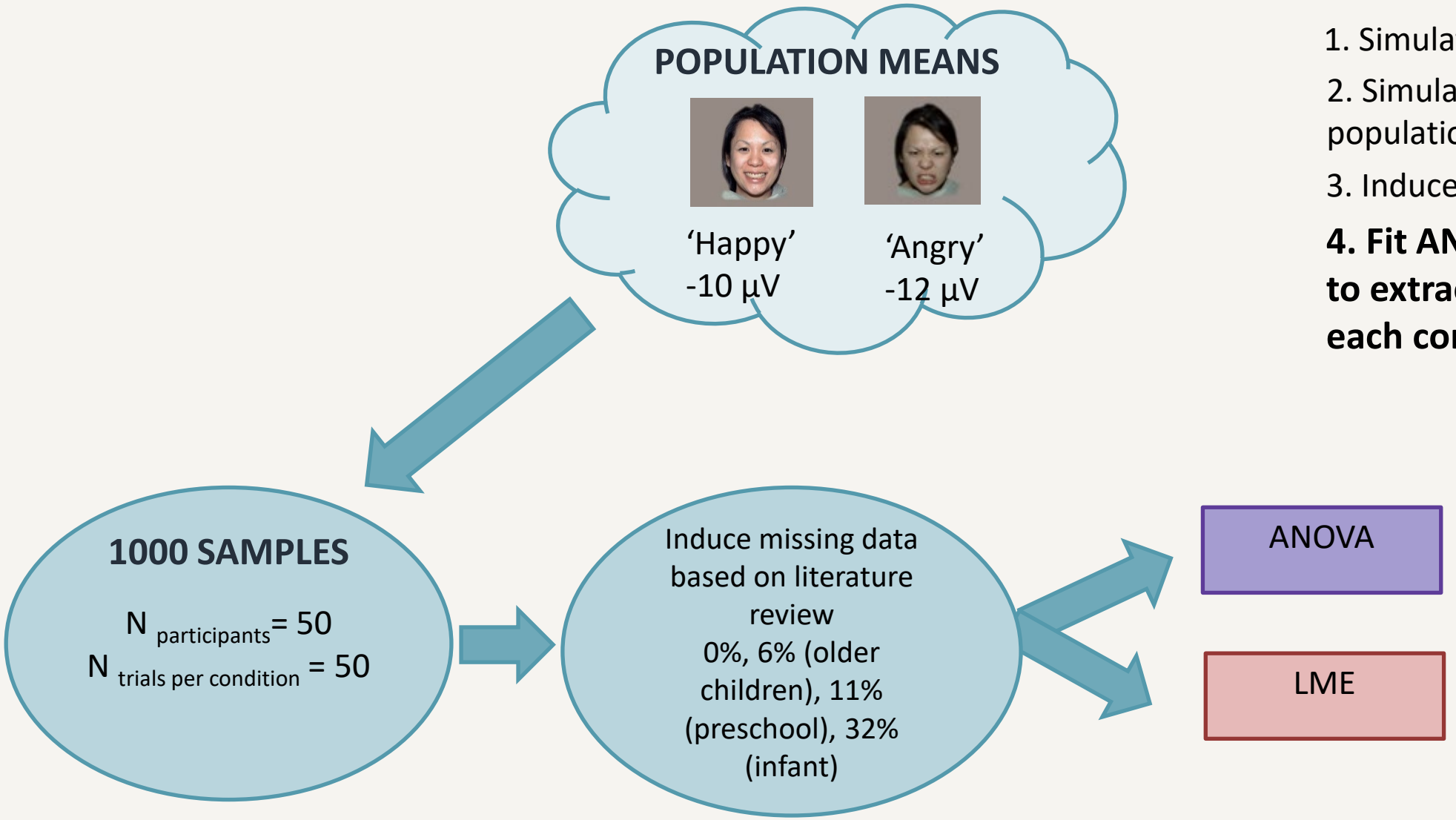
Induce missing dat based on literature review
0%, 6% (older children), 11% (preschool), 32% (infant)



1. Simulate NC
2. Simulate samples from population parameters

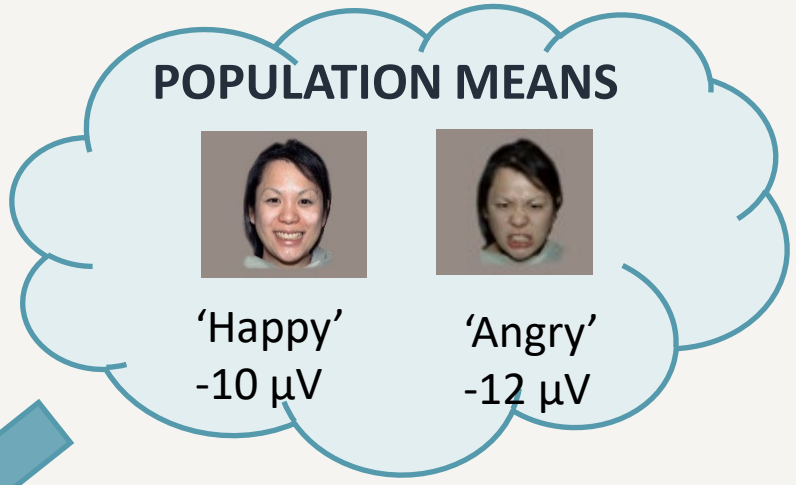
3. Induce missing data by removing 10+ trials in one or more conditions, such that 0%, 6%, 11%, and 32% of participants would be deleted in ANOVA

Analyzing simulated mean amplitude with LME & ANOVA

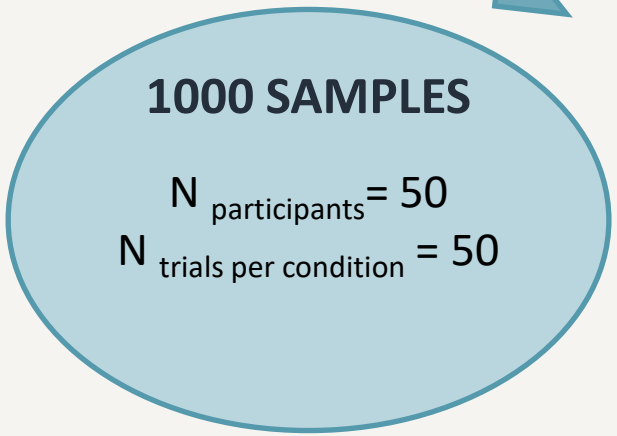


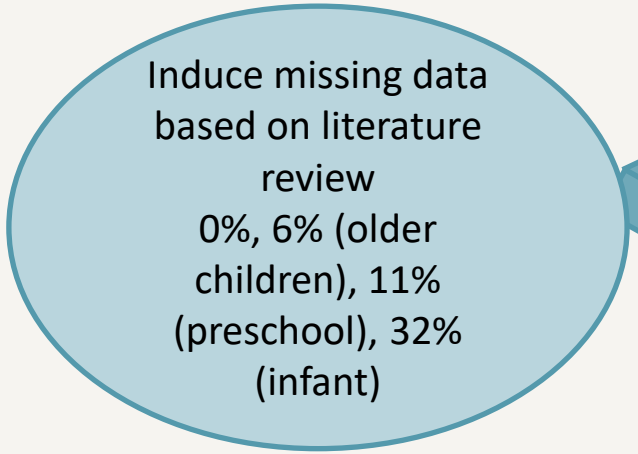
1. Simulate NC
2. Simulate samples from population parameters
3. Induce missing data
4. **Fit ANOVA and LME models to extract mean amplitude of each condition**

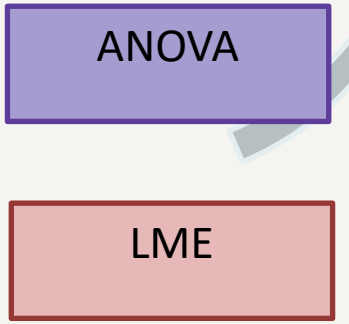
Analyzing simulated mean amplitude with LME & ANOVA



1. Simulate NC
2. Simulate samples from population parameters
3. Induce missing data
4. Fit ANOVA and LME models
5. **Compare model mean amplitude estimates to simulated population values**







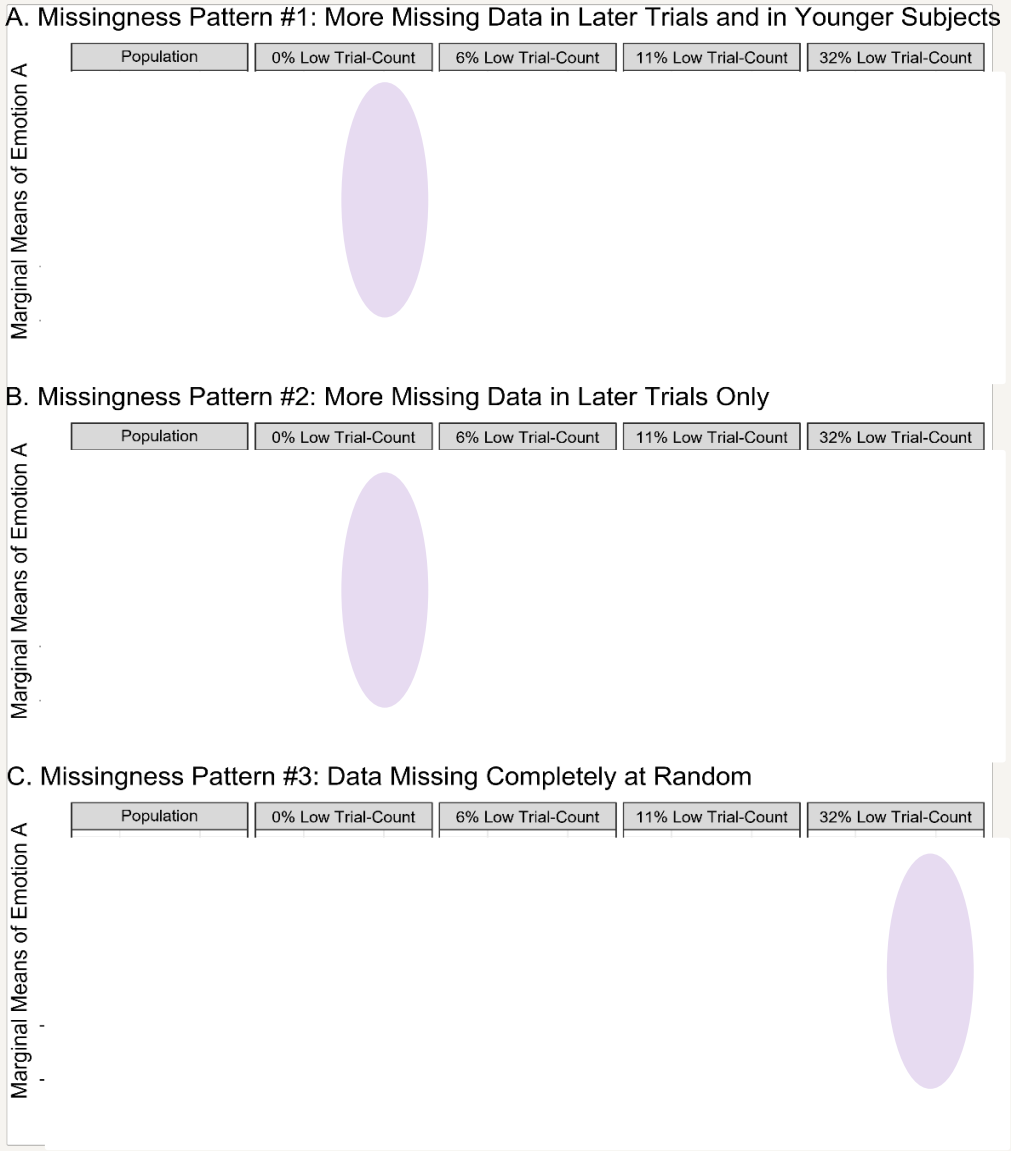
Metrics for model evaluation

- **Power to detect the true effect**
  - In what percent of samples was a statistically significant effect detected?
- **Error (RMSE) from the true population value**
  - How much variability is there in estimates (e.g., how far is the spread around the true population value?)
- **Bias from the true population value**
  - Are estimates of the condition difference accurate or systematically different than what we simulated?

Analyzing simulated mean amplitude with LME & ANOVA

ANOVA yielded inaccurate, biased estimates, that got more biased at higher percentages of participants with low trial-counts

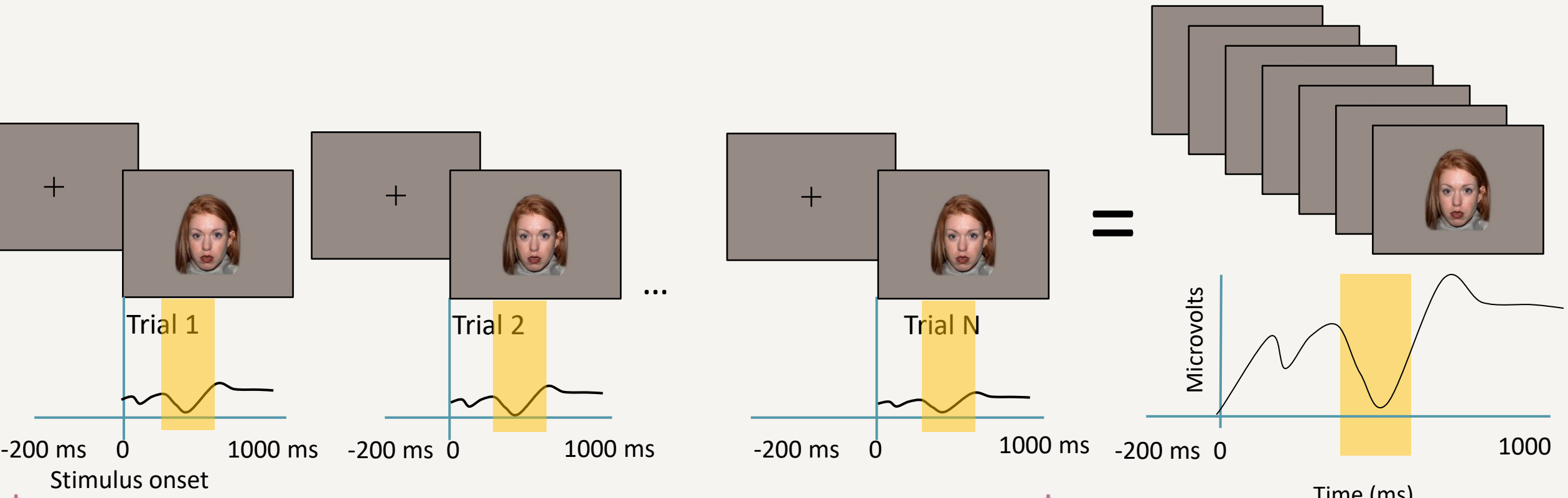
LME yielded accurate, unbiased estimates at all percentages of participants with low trial counts



Benefits of LMEs

- Includes all collected clean ERP data (doesn't require casewise deletion)
- Allows you to model intra-individual variability
- Allows you to model MORE predictors
  - Continuous trial-level predictors
  - Within-subject/trial-level effects
  - Random effects
- Lower error & unbiased estimates compared to regression/ANOVA

Why do LMEs and ANOVA/regression produce the same results?



Simulation results: LME remains unbiased

- ANOVA marginal means became increasingly biased as the percentage of casewise-deleted / low-trial-count participants increased
- LME marginal means stayed accurate to the population value at every missingness level tested — 0%, 6%, 11%, and 32%
- Even under close-to-ideal MCAR conditions, ANOVA still showed greater variability in error than LME

When can casewise deletion bias your results, even with a reasonable trial-count cufoff?

- Never -- casewise deletion only reduces power
- When excluded participants or trials differ systematically from those retained
- Only once more than 50% of the sample has been excluded
- Only in between-subject designs
- None of the above

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

When can casewise deletion bias your results, even with a reasonable trial-count cufoff?

Never -- casewise deletion only reduces power

0%

When excluded participants or trials differ systematically from those retained

0%

Only once more than 50% of the sample has been excluded

0%

Only in between-subject designs

0%

None of the above

0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

When can casewise deletion bias your results, even with a reasonable trial-count cufoff?

Never -- casewise deletion only reduces power

0%

When excluded participants or trials differ systematically from those retained

0%

Only once more than 50% of the sample has been excluded

0%

Only in between-subject designs

0%

None of the above

0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

LEARNING CHECK

When can casewise deletion bias your results, even with a “reasonable” trial-count cutoff?

A. Never — casewise deletion only reduces power

B. When excluded participants or trials differ systematically from those retained

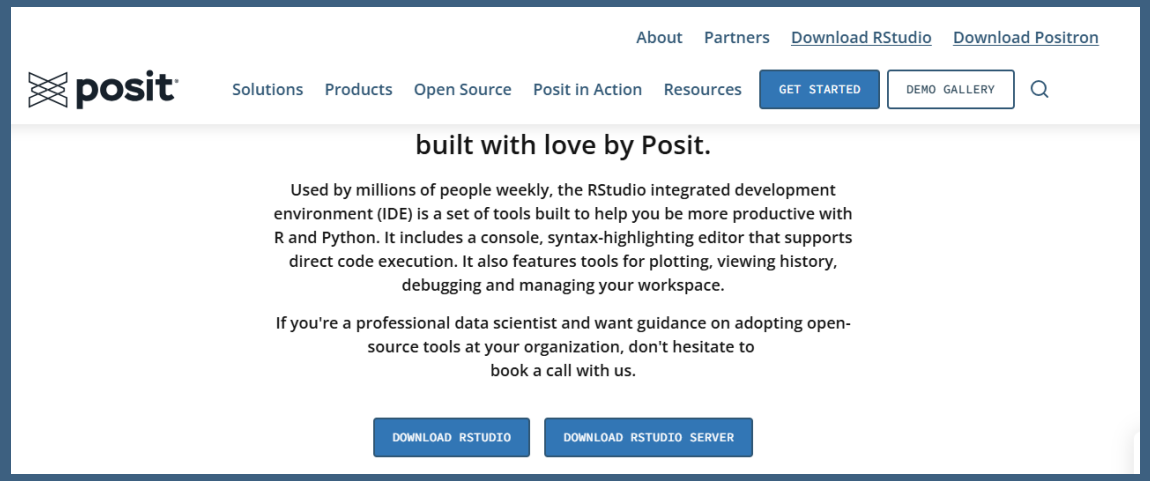
C. Once more than 50% of the sample has been excluded

D. Only in between-subjects designs, not within-subjects designs

Break!

After the break, you can either:

1) code along as we fit mixed effects models and run power analyses
    - You'll need Rstudio
    - install.packages(c("lme4", "lmerTest", "performance", "see", "emmeans", "ggplot2", "tidyverse", "faux", "mixedpower"))
2) scroll through a knitted Rmd file if you'd rather not code along & fidget spin



04 Model specification

Let’s fit some models!

Code along: NC elicited by face stimuli

https://github.com/mjheise

Open `LME_ex1_emotionfaces_codealong.Rmd`

How we'll work

- Go over a few slides about each step in the model-building process, and then run that step's code together
- Polls along the way

Importance of correctly specifying LME

Model misspecification: the model is a poor match for the data/ the model doesn't correctly represent the real-world process generating the data due to incorrect assumptions or a flawed design

Model misspecification can have a big impact on interpreting results:

- Leaving out random effects (underfitting) → inflated Type I error
  - Random-intercepts-only models falsely rejected the null 40–50% of the time at α = .05 for within-items designs (Barr et al., 2013).
  - Across simulations, omitting a random effect raised the average Type I error to .078, vs. .055 for fixed effect predictors (LeBeau et al., 2018).
- Adding random effects the data can't support → convergence problems, lower power (Liu et al., 2024)
  - Maximal models are often overparameterized (Bates et al., 2015).

Fitting LME models in R

So, let’s assume we’ve done everything right in our study design and now we’re ready to analyze our data…

A 9-Step guide to building your LME

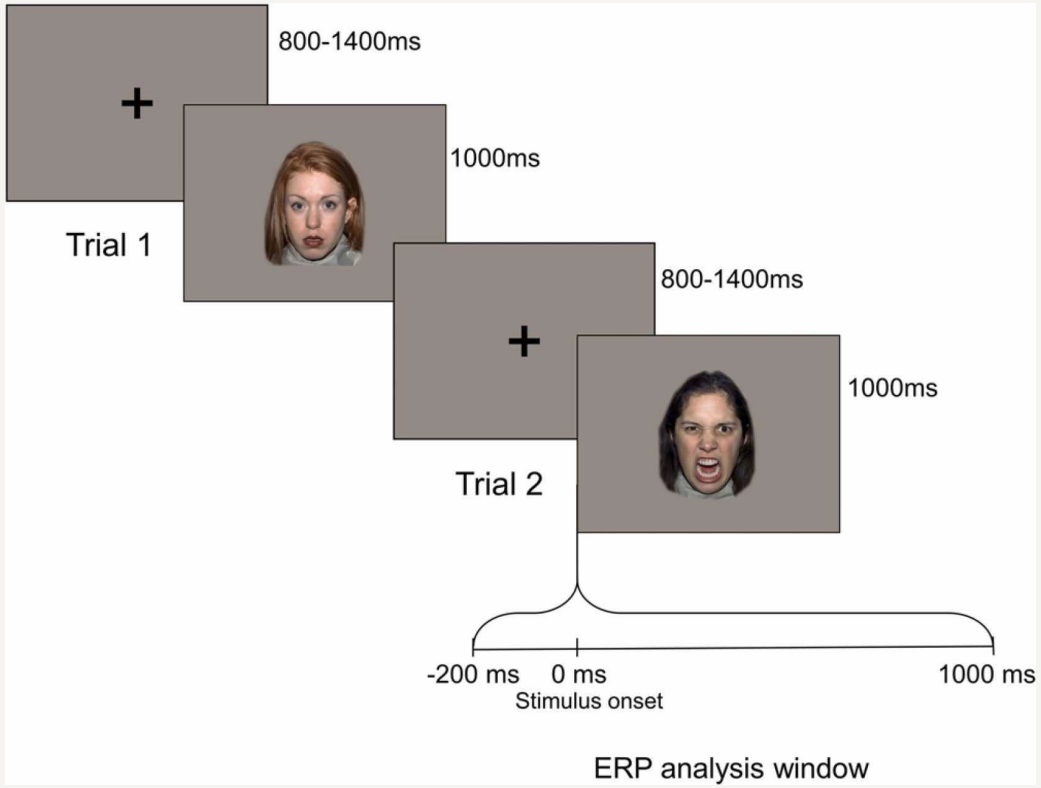
1. Define the research question
2. List all relevant variables
3. Draw the study design schematic
4. Categorize variables by type & level
5. Assign fixed and random effects
6. Consider data transformations & center predictors (optional)
7. Fit maximal model and simplify until convergence is reached
8. Check model assumptions
9. Report & visualize results



Tips discussed starting on pg. 58

Study design

- Participants: 40 children aged 3–5 years
- Stimuli: faces showing two emotions, happy and angry, each posed by the same 5 actors
- Trials: each actor × emotion is shown 10 times, so 5 actors × 2 emotions × 10 = 100 trials per child
  - Fully crossed: every child sees every actor show every emotion the same number of times
- ERP outcome: Nc (negative central) mean amplitude on every trial



Data exploration

Step 1: Define the research question

- What is your outcome variable?
  - **Can your outcome be trial-level mean amplitude?**
    - Your predictors (trial presentation number, child age, etc.) are predicting the outcome, so we can't explain missingness in mean amplitude *unless* it's the outcome
- Does your question require modeling within-subject variability, trial-level effects (e.g., habituation), or handling unequal trial counts across participants?
- Are data likely to be systematically missing (i.e., MAR)?

Which of the following research questions CAN'T be answered using mixed effects models?

Does N400 amplitude decrease as the cloze probability of each word increases? **(A)**

On trials where pupil dilation is larger than the child's own average, is the P3 l… **(B)**

Do more anxious children show a larger ERN on error trials relative to correct … **(C)**

Is the mismatch negativity smaller in toddlers with a family history of autism? **(D)**

Do 2-year-olds who show a larger N400 have larger vocabularies at age 4? **(E)**

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Which of the following research questions CAN'T be answered using mixed effects models?

(A) Does N400 amplitude decrease as the cloze probability of each word increases?

0%

(B) On trials where pupil dilation is larger than the child's own average, is the P3 larger?

0%

(C) Do more anxious children show a larger ERN on error trials relative to correct trials?

0%

(D) Is the mismatch negativity smaller in toddlers with a family history of autism?

0%

(E) Do 2-year-olds who show a larger N400 have larger vocabularies at age 4?

0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Which of the following research questions CAN'T be answered using mixed effects models?

(A) Does N400 amplitude decrease as the cloze probability of each word increases?

0%

(B) On trials where pupil dilation is larger than the child's own average, is the P3 larger?

0%

(C) Do more anxious children show a larger ERN on error trials relative to correct trials?

0%

(D) Is the mismatch negativity smaller in toddlers with a family history of autism?

0%

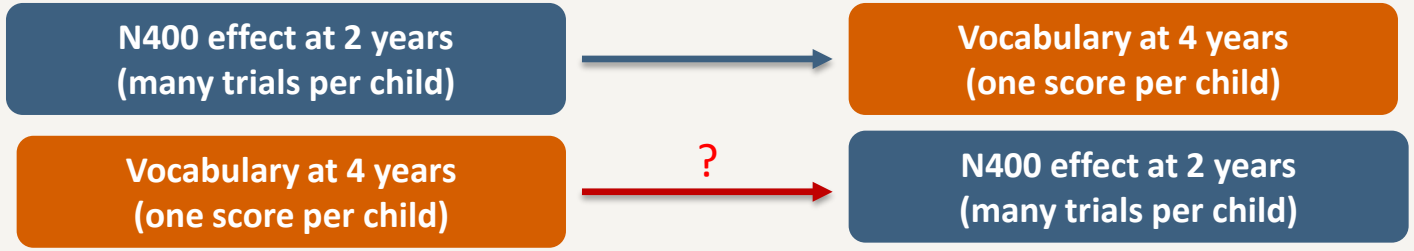
(E) Do 2-year-olds who show a larger N400 have larger vocabularies at age 4?

0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Why can't we use LMEs when ERPs are the predictor

- LMEs can be used for longitudinal research questions where predictors at Time 1 predict mean amplitude at Time 2 and for concurrent research questions (e.g., questions you'd answer with regression, ind. samples t-tests, correlation)
- Using trial-level ERP mean amplitude at Time 1 to predict a behavioral outcome at Time 2 doesn't make sense *theoretically* (The outcome/dependent variable can't be the predictor in the model)



Some possible solutions for this issue

A. LME random slopes → two-step

Fit the trial-level LME with a random slope for congruency, pull out each child's N400 effect, then regress vocabulary on it

✓ Keeps every child with any clean trials
✗ Few-trial children are pulled toward the group mean

✗ Step 2 treats each child's N400 effect as known exactly: standard errors are too small

B. Condition averages → path model or regression

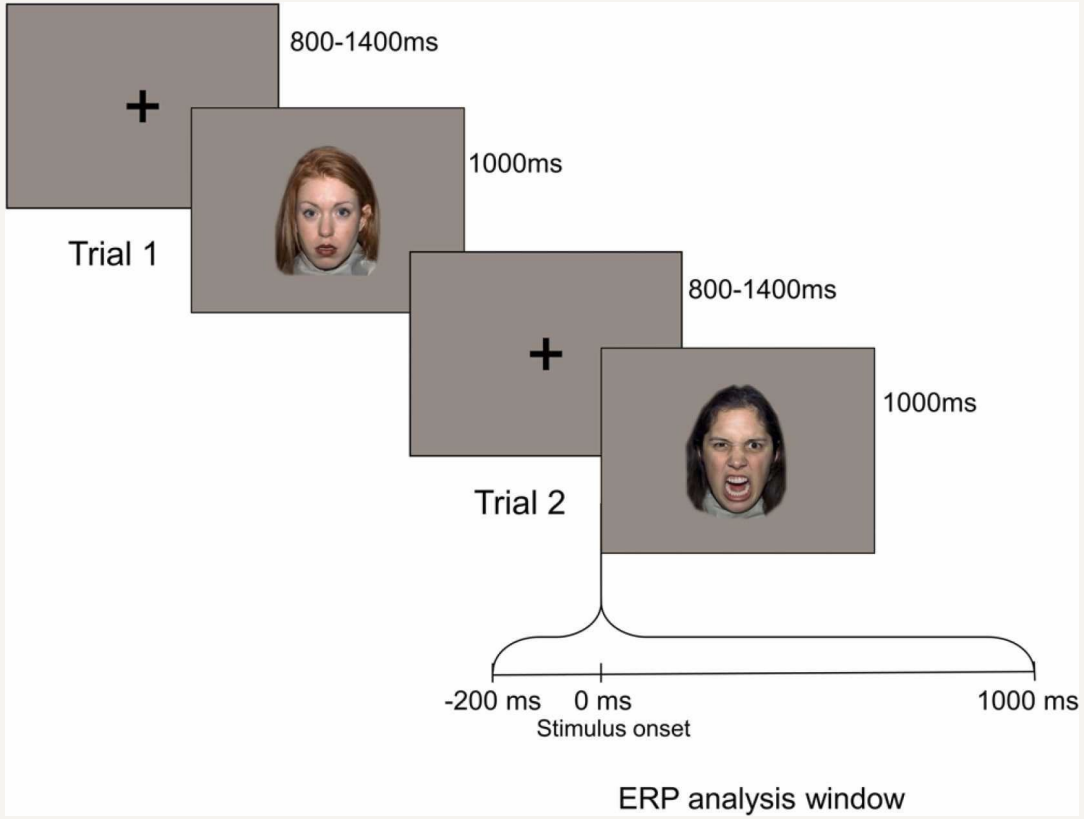
Average each child's trials per condition (incongruent − congruent), maybe casewise delete participants with low data quality (e.g., using trial-count or SME – Luck et al. 2021) then fit a path model (e.g., lavaan)

✓ Easy to add mediators or more outcomes; FIML handles attrition

✗ Averaging brings back trial cutoffs & potential bias from casewise deletion

Step 1: Define the research question

- 1. Is there a difference in NC mean amplitude elicited by Happy vs. Angry faces?
- 2. Does the magnitude of the NC emotion difference change with age (3 to 5 years)?



A 9-Step guide to building your LME

1. Define the research question
2. **List all relevant variables**
3. Draw the study design schematic
4. Categorize variables by type & level
5. Assign fixed and random effects
6. Consider data transformations & center predictors (optional)
7. Fit maximal model and simplify until convergence is reached
8. Check model assumptions
9. Report & visualize results

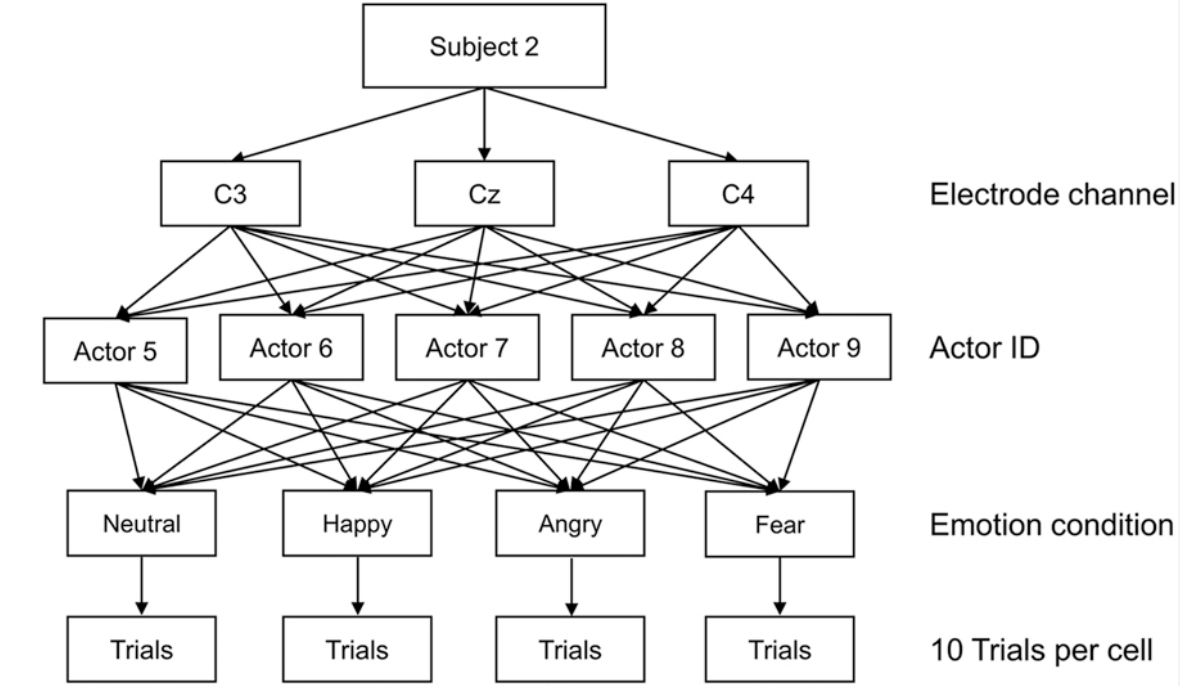
Step 2: List all relevant variables

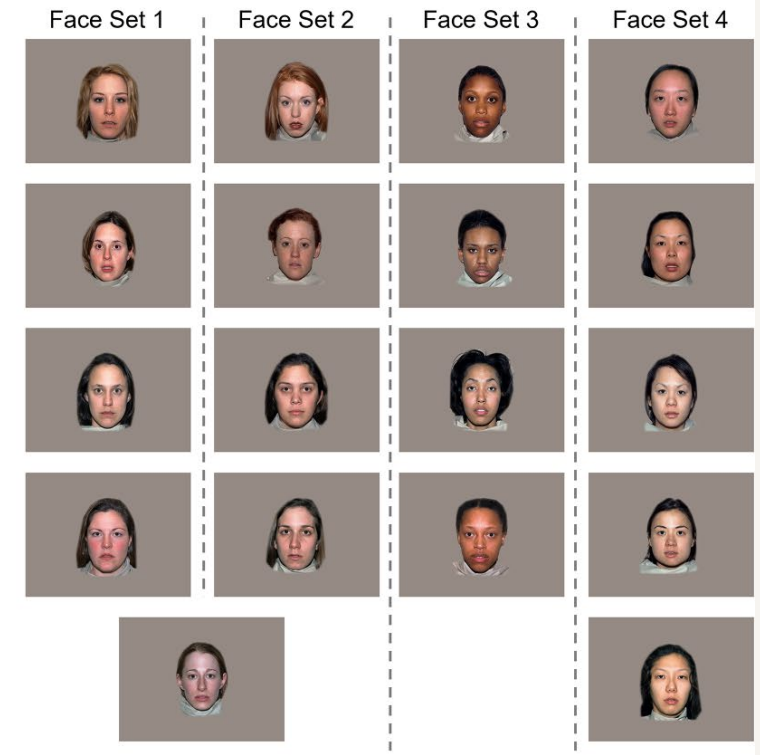
- Relevant variables are anything that predicts mean amplitude, or both mean amplitude and the probability that a trial is missing
- This step would be ideal to do before data collection!

A 9-Step guide to building your LME

1. Define the research question
2. List all relevant variables
3. **Draw the study design schematic**
4. Categorize variables by type & level
5. Assign fixed and random effects
6. Consider data transformations & center predictors (optional)
7. Fit maximal model and simplify until convergence is reached
8. Check model assumptions
9. Report & visualize results

Step 3: Draw the study design schematic





A 9-Step guide to building your LME

1. Define the research question
2. List all relevant variables
3. Draw the study design schematic
4. **Categorize variables by type & level**
5. Assign fixed and random effects
6. Consider data transformations & center predictors (optional)
7. Fit maximal model and simplify until convergence is reached
8. Check model assumptions
9. Report & visualize results

Step 4: Categorize variables by type & level

Type

- Continuous/numeric: variable measured on numeric scale where values have a meaningful order and the distance between them means something (e.g., receptive vocabulary, Big 5 personality traits)
- Categorical/factor : variable whose labels are distinct groups (e.g., condition, race/ethnicity)

Level

- Participant-level variables: any variable that has the same value across trials (one value per participant; e.g., age, group, individual-differences measures)
- Trial-level variables: any variable that has different values across trials (e.g., condition, electrode, reaction time, stimulus characteristic [e.g., color, shape, salience], presentation number)

A 9-Step guide to building your LME

1. Define the research question
2. List all relevant variables
3. Draw the study design schematic
4. Categorize variables by type & level
5. **Assign fixed and random effects**
6. Consider data transformations & center predictors (optional)
7. Fit maximal model and simplify until convergence is reached
8. Check model assumptions
9. Report & visualize results

Step 5: Assign fixed effects

- **Fixed effects**: your predictors of theoretical interest, plus any nuisance variables with few categories (e.g., 3 electrodes of interest)

What are the fixed effects in our research question "Is there a difference in NC mean amplitude between Happy & Angry faces?"

- Emotion
- Presentation Number
- Age
- Interactions between any of these (Emotion x Presentation number, Emotion x age, Age x presentation number?)

Step 5: Assign fixed and random effects

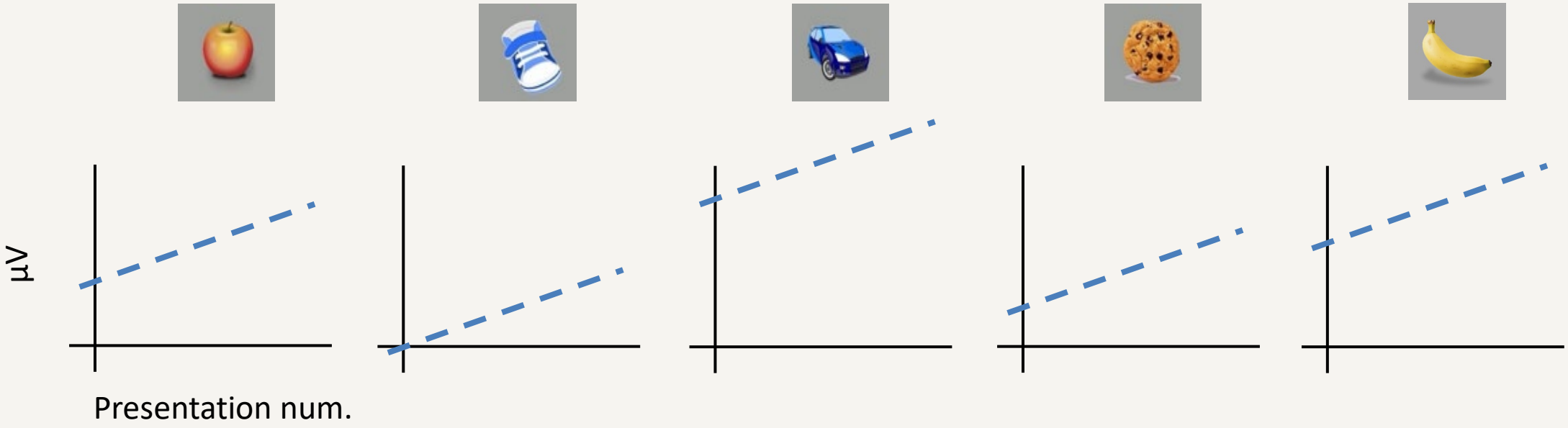
- **Random effects**: grouping variables with many categories (>5, e.g., 40 children, 5 actors; Volpert-Esmond et al., 2021) that you want to generalize beyond

What are the random effects in our research question "Is there a difference in NC mean amplitude between Happy & Angry faces?"

- If you ran the same experiment with 50 NEW participants, you want results to be the same
- If you ran the same experiment with a new stimulus set (e.g., NIMstim versus TED face sets), you want results to be the same

Step 5: Random intercepts vs. random slopes

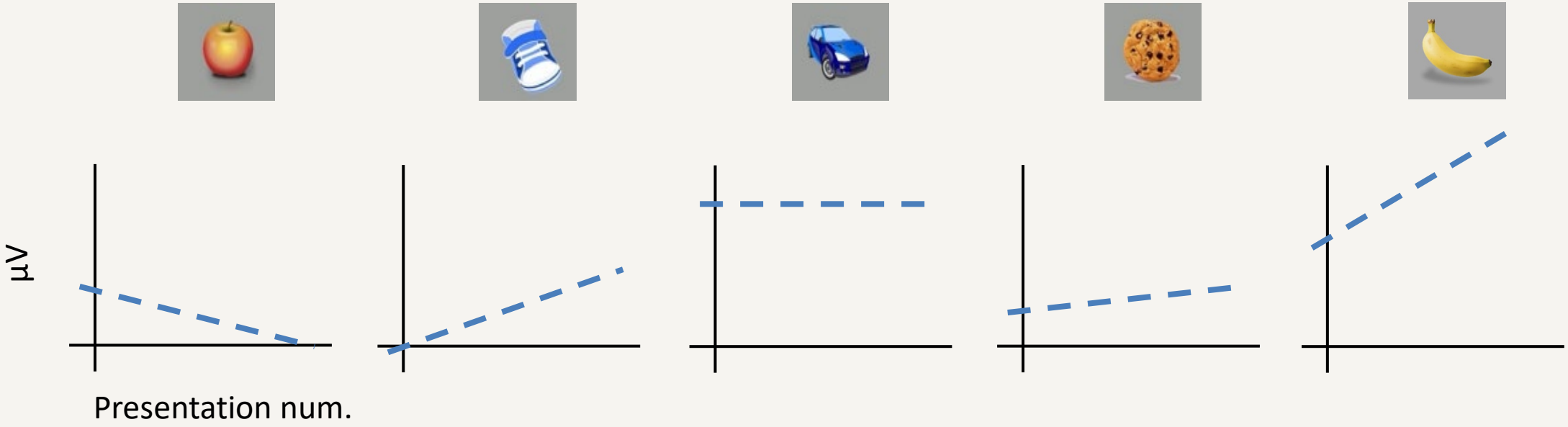
- **Random intercept**: for all categories of your random variable, the slope of the relation between the variable and the outcome is the same in direction and magnitude, but where the line of best fit crosses the Y axis varies



Random intercept for stimuli

Step 5: Random intercepts vs. random slopes

- **Random intercept & random slope**: both the slope and where the line of best fit crosses the Y axis varies (different stimuli/participants/etc. can show different relationships)
  - *You can fit random slopes without a random intercept or fit random intercepts and slopes that are uncorrelated, but this is not common unless you have a good theoretical reason*

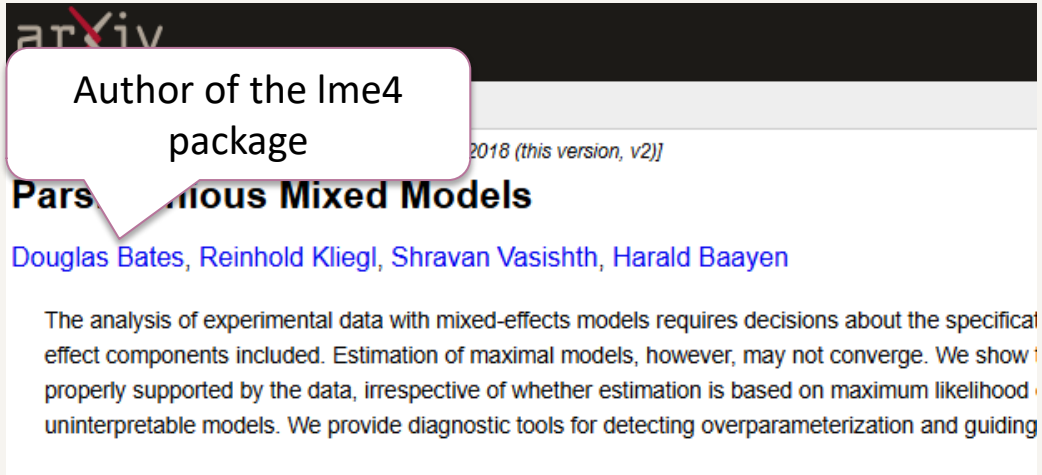


Random slopes & intercepts for stimuli

Step 5: How many random slopes should I have?

- There's lots of papers about fitting mixed effects models & model misspecification if the wrong structure is fit
- "Keep it maximal" versus "parsimony"?
  - "Maximal model that theoretically makes sense."





Step 5: Possible random effects in our design

Grouping variable	How the data could theoretically vary by predictor _____	Syntax
Participant (SubjectID)		
Actor (ActorID)		
Participant × Actor (optional)		

Pair & share: which random slopes would you prioritize?

Think · 1 min on your own → **Pair** · 3 min with a neighbor → **Share** · with the group

1. **Looking at the table: which random slopes would you prioritize for our emotion study, and why?**

2. **If the maximal model doesn't converge, which slope would you drop first?**
   Which slopes are least plausible, or least related to the research question?

3. **In your own study: what are your random effects?**

Step 5: Possible random effects in our design

Grouping variable	How the data could theoretically vary by predictor _____	Syntax
Participant (SubjectID)	Do some participants have higher or lower mean amplitudes on average?	`(1 | SubjectID)`
	Does the effect of emotion (Happy vs. Angry) differ across participants?	`(1 + Emotion | SubjectID)`
	Does habituation (change across presentations) differ across participants?	`(1 + PresentNum | SubjectID)`
	Does the change in the emotion effect across presentations differ across participants?	`(1 + Emotion * PresentNum | SubjectID)`
	Does the age effect differ across participants? Not possible: each child has only one age, so age is a fixed effect	✗ `(1 + Age | SubjectID)`
Actor (ActorID)	Do some actors' faces produce larger or smaller mean amplitudes overall?	`(1 | ActorID)`
	Does the effect of emotion differ across actors (e.g., a more intense angry pose)?	`(1 + Emotion | ActorID)`
	Does habituation differ across actors (e.g., amplitude drops faster for some faces)?	`(1 + PresentNum | ActorID)`
	Does the change in the emotion effect across presentations differ across actors?	`(1 + Emotion * PresentNum | ActorID)`
	Does the age effect differ across actors? (Each actor is seen by children of many ages)	`(1 + Age | ActorID)`
Participant × Actor (optional)	Do particular participants respond differently to particular actors?	`(1 | SubjectID:ActorID)`

Random slopes we might prioritize: `(1 + Emotion | SubjectID)` and `(1 + Emotion | ActorID)`: does the emotion effect differ across participants? Across actors?

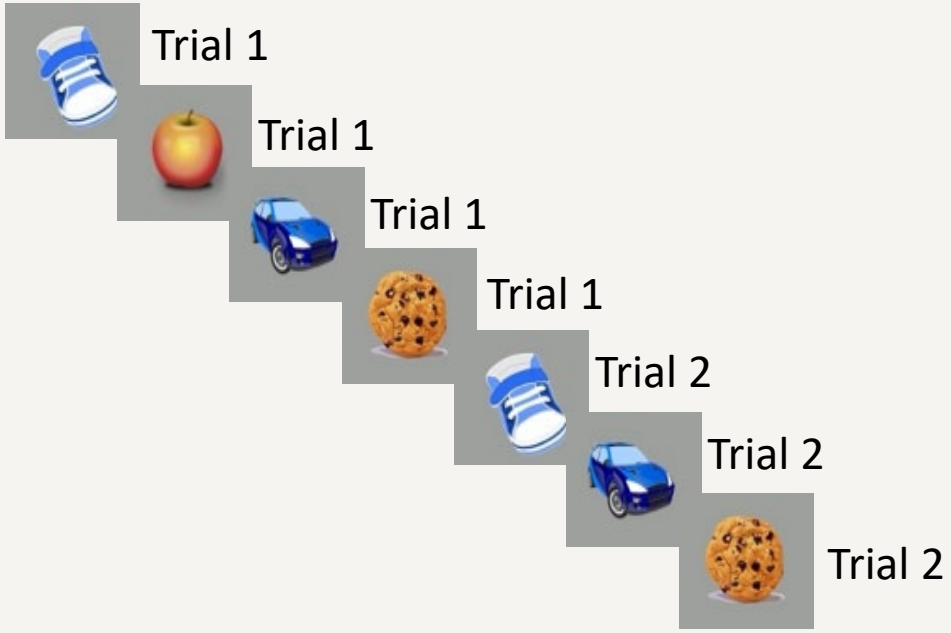
A 9-Step guide to building your LME

1. Define the research question
2. List all relevant variables
3. Draw the study design schematic
4. Categorize variables by type & level
5. Assign fixed and random effects
6. **Consider data transformations & center predictors (optional)**
7. Fit maximal model and simplify until convergence is reached
8. Check model assumptions
9. Report & visualize results

Step 6: Consider data transformations/centering

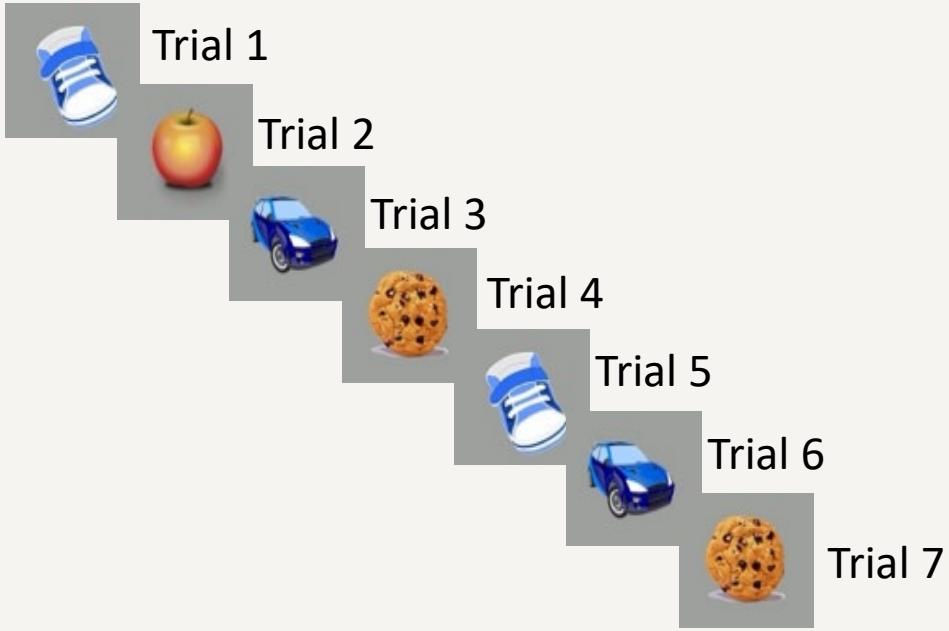
- Trial presentation number (e.g., see Borgström et al., 2016 for habituation example)

Option 1 (used in Heise et al. 2022)
Stimulus presentation number



- Stimuli repeat in a block-randomized way
- Stimulus-specific adaptation / growing familiarity with that face, sound, or word

Option 2
Global presentation number



- All stimuli are unique
- Captures time on task/ task-general habituation

Pair & share: coding trial presentation number

Think · 1 min on your own → **Pair** · 3 min with a neighbor → **Share** · with the group

- **What are some other possible ways to code trial presentation number?**
- **Are there other study designs that don't seem amenable to using presentation number as the # of presentations of that unique stimulus?**
- **Are there other factors that you would consider in deciding between stimulus presentation number (option 1) versus global presentation number (option 2)?**

Brainstorm: what else shapes how you code trial number?

Count presented trials, not retained trials

Number trials from the event markers before artifact rejection: rejected trials were still seen

Breaks, blocks & timing

Breaks and attention-getters can cause recovery. Consider block, trial within block, or minutes since start if timing varies

Shape of the change

Habituation may occur over the first few trials but then level off. Compare linear, log, and quadratic terms (Step 6) and check residual plots (Step 8)

Including random slopes

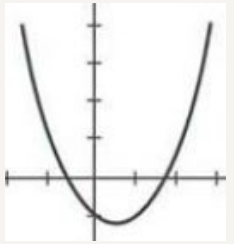
Random slope for trial number (rates differ across people); Condition × trial number if, e.g., deviants habituate less

Where is zero?

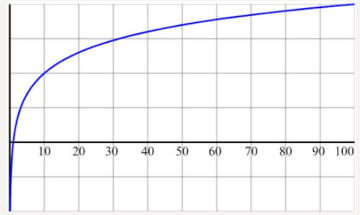
First presentation = 0 vs. average presentation = 0 changes the intercept and the effects that interact with presentation number: e.g., the emotion effect becomes 'at the first trial' vs. 'at a typical trial' (Centering, Step 6)

Step 6: Possible data transformations for trial presentation number

- In Heise et al. (2022), amplitude decline across the session was modeled as a **linear** decay of trial presentation number and this appeared to be approximately linear
  - LMEs can include transformations of predictors:
    - **Linear**: presentationNumber
    - **Quadratic**: presentationNumber^2
    - **Log**: log(presentationNumber)
  - A downside of including non-linear effects that they're less interpretable/intuitive to readers
- First steps for projects after data cleaning should be plotting data to help inform/model distribution



quadratic



log

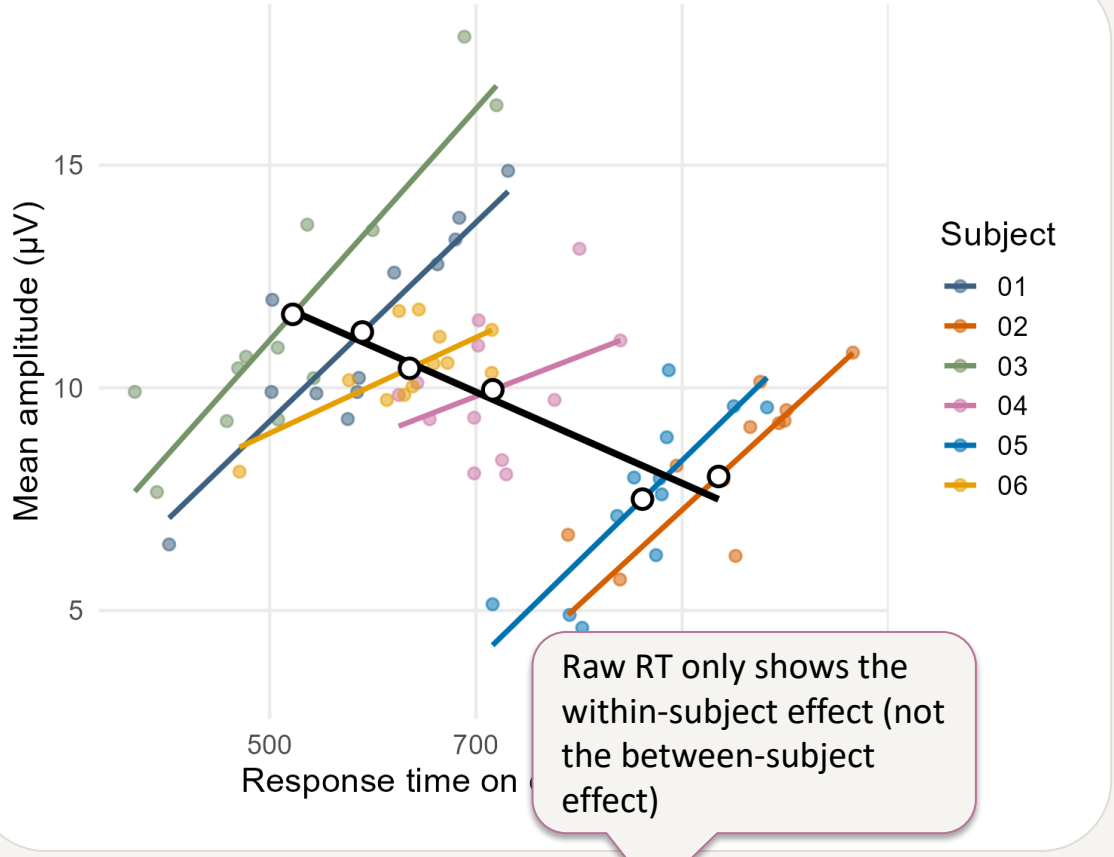
Step 6: Centering

- Centering is just subtracting a constant from every value of a variable.
  - This redefines the 0 point for that predictor to be whatever value you subtracted
  - Centering shifts the scale over, but retains the units
- Centering is generally recommended, and is especially useful for interpreting interactions
- All centering happens on the final dataset AFTER participant exclusion!
  - E.g., If some participants' final EEG data isn't included in analyses (e.g., in looking at EEG cap placement after the visit you realize that it's very shifted and not over your target scalp ROI so you decide to exclude the participant; a new experimenter messed up when giving instructions to the participant so you don't think the participant did the task correctly), they are not included in calculating the mean for centering

Step 6: Centering

- **Grand-mean centering (GMC)**: subtract the overall sample mean from a participant-level (level-2) predictor
  - Coefficients describe deviation from the “average” participant
- **Centering within cluster (CWC)**: subtract each participant's own mean from a trial-level (level-1) predictor
  - Coefficients describe relatively higher or lower trials within that participant
- There isn't a universal rule for how to center predictors, you can use GMC or CWC or both or neither based on your research question and what you want the coefficient to represent.

Step 6: Centering predictors example – Response Time



Raw RT only shows the within-subject effect (not the between-subject effect)

Sub	RT (ms)	Each participant's mean RT	Within (CWC) RT – participant mean	Between (GMC) Participant mean – GM
01	502	590	−88	−120
01	585	590	−5	−120
01	503	590	−87	−120
02	940	935	5	225
02	840	935	−95	225
02	994	935	59	225

Grand mean = mean of the participant means, so each participant counts equally

Raw RT only

+0.020 µV/ms

One slope: only the within-subject effect shows up

Within (CWC)

+0.020 µV/ms

Slower-than-usual trial → larger amplitude

Between (GMC)

−0.015 µV/ms

Slower participants → smaller amplitude

Simulated illustration; see Volpert-Esmond et al. (2021), Int. J. Psychophysiology

Step 6: Scaling predictors

Also called z-scoring or standardizing.

Centering

$x - mean$

- Changes the meaning of the intercept (and lower-order terms in interactions)
- Keeps the units: coefficients are still in µV per unit of the predictor

Example: For each 1-month increase in age, mean amplitude decreases by 0.5 µV

Scaling (z-scoring)

$(x - mean) / SD$

- Changes the intercept and the units: coefficients are now per 1 SD of the predictor
- Puts predictors measured in different units on a common scale

Example: If the SD of age is 10 months: for each 1-SD increase in age, mean amplitude decreases by 0.5 × 10 = 5 µV

- Another use of scaling is to scale all (numeric/continuous) predictors AND the outcome (mean amplitude) to estimate "effect size" in LMEs.
- Effect size is a standardized estimate of the relation between variables (Ferron et al., 2008; Snijders and Bosker, 2012) – relative contribution of each predictor on mean amplitude

Pair & share: centering

Think · 1 min on your own → **Pair** · 3 min with a neighbor → **Share** · with the group

1. **Pick a predictor from our simulated study (Age, Presentation Number). Is it trial-level or participant-level?**

2. **Where is “zero” for that predictor right now? Is that value meaningful?**

3. **How would you center it, and what would the intercept mean afterward?**

After grand-mean centering a subject-level predictor like age, what does the intercept represent?

- The predicted outcome for a hypothetical participant at the sample's average age
- The predicted outcome for a participant aged 0
- The outcome averaged across all trials, regardless of age
- Centering doesn't change how the intercept is interpreted

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

After grand-mean centering a subject-level predictor like age, what does the intercept represent?

The predicted outcome for a hypothetical participant at the sample's average age

0%

The predicted outcome for a participant aged 0

0%

The outcome averaged across all trials, regardless of age

0%

Centering doesn't change how the intercept is interpreted

0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

After grand-mean centering a subject-level predictor like age, what does the intercept represent?

The predicted outcome for a hypothetical participant at the sample's average age

0%

The predicted outcome for a participant aged 0

0%

The outcome averaged across all trials, regardless of age

0%

Centering doesn't change how the intercept is interpreted

0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

After grand-mean centering a participant-level predictor like age, what does the intercept represent?

A. The predicted outcome for a hypothetical participant at the sample's average age

B. The predicted outcome for a participant aged zero

C. The outcome averaged across all trials, regardless of age

D. Centering doesn't change how the intercept is interpreted

Step 6: Effects vs. dummy coding categorical predictors

Example: Emotion with two categories, Happy = −10 µV and Angry = −12 µV

Dummy/treatment coding (R's default): one category (e.g., Angry) is a reference group and βs correspond to the difference between the reference category and all other categories (e.g., Happy, Fearful)

Effects coding: +/-1 compares each group to the overall (grand) mean; +/-0.5 β is the difference between groups

Coding	Happy	Angry	Intercept =	β for Emotion =
Dummy / treatment coding (R's default)	0	1	**−10** Happy (reference) mean	**−2** Angry − Happy
Effects coding, ±0.5	−0.5	+0.5	**−11** grand mean	**−2** Angry − Happy
Effects coding, ±1	−1	+1	**−11** grand mean	**−1** Angry − grand mean (half the difference)

Good when there's a true reference (e.g., "neutral" emotion, control group that doesn't receive instructions

Good when groups don't have a clear reference

Marginal means: another alternative is to use dummy coding, and then probe condition differences using marginal means of the fitted model (but all coefficients correspond to the reference group)

Coding Emotion: we'll come back to this

For now: R's default dummy coding. Angry = 0, Happy = 1, so Angry is the reference group.

Step 7
Fit and simplify the model

→

Step 8
Check assumptions

→

Step 9
Interpret & visualize the final model

→

Circle back
Refit with effects coding (±0.5, ±1)

A 9-Step guide to building your LME

1. Define the research question
2. List all relevant variables
3. Draw the study design schematic
4. Categorize variables by type & level (trial vs. subject)
5. Assign fixed and random effects
6. Consider data transformations & center predictors (optional)
7. **Fit maximal model and simplify until convergence is reached**
8. Check model assumptions
9. Report & visualize results

Step 7: Fit the maximal model

1. **Identify the maximal model (that makes sense for our study design)**
   - For each variable (anything we've identified as a fixed or random effect), think about what allowing for random slopes/intercepts means (e.g., do we expect there to be variability in participant's response to XYZ…)

2. **Adjust optimizer & iterations to increase chance of convergence**
   - model <- lmer(Reaction ~ Days + (Days | Subject), data = sleepstudy
     control = lmerControl(optimizer = "bobyqa",
     optCtrl = list(maxfun = 20000)))

This is the best optimizer (underlying algorithm that is looking for the best fitting parameters for your model)

Increase iterations (calculate and test the model's likelihood up to 20,000 times to find the optimal solution), this is good for troubleshooting this warning: **Warning: Convergence code 1 or Max number of iterations reached**

Step 7: Fit the maximal model

1. **Identify the maximal model (that makes sense for our study design)**
   - For each variable (anything we've identified as a fixed or random effect), think about what allowing for random slopes/intercepts means (e.g., do we expect there to be variability in participant's response to XYZ…)

2. **Adjust optimizer & iterations to increase chance of convergence**
   - model <- lmer(Reaction ~ Days + (Days | Subject), data = sleepstudy,
     control = lmerControl(optimizer = "bobyqa",
     optCtrl = list(maxfun = 20000)))

3. **Model simplification steps**
   - If your model doesn't converge, you should systematically remove one random effect at a time
     - First you start with taking out random slopes
       - If you have multiple random slopes, take one random slope out at a time and rerun your model considering only one random slope left out at a time.
       - E.g., if you had random slope 1 and random slope 2, try first taking out random slope 1 and leaving in random slope 2; then try taking out random slope 2 and leaving in random slope 1; then take out both if the model still doesn't converge
   - Bare minimum LME is a random intercept model with subject ID. Because if you aren't modeling within-subject variability, you no longer have a nested data structure for your outcome/DV (full pooling, which is bad)

Step 7: Warnings that the model is too complex

Both non-convergence and singular fit mean the model is more complex than the data can support.

Non-convergence

```
Model failed to converge with max|grad| = 0.0042
```

What it means: the optimizer searched for the values (fixed effects, random-effect SDs, correlations) that best fit your data, but couldn't confirm it found them.

Tips to see if the model could converge: center or rescale predictors, or try another optimizer (bobyqa, allFit()).

Singular fit

```
boundary (singular) fit: see help('isSingular')
```

What it means: part of the random-effects structure sits at an impossible boundary: an SD of exactly 0 or a correlation of ±1.

Easy to miss: you still get βs, SEs, and p-values.

Why it's a problem:

- Estimates aren't believable; a variance of 0 explains nothing
- SEs, Satterthwaite df, and LRTs work poorly
- Results can shift with small changes to data or optimizer

Common causes

- Small sample size
- Few within-subject and between-subject observations
- High complexity (you've identified a lot of sources of variability)
- Little consistent variability between units of a random effect, relative to the noise within them (e.g., children's trials are so noisy that their true average Nc can't be told apart; the red-haired actress doesn't elicit a consistently different Nc across children)

Step 7: Simplify our model in R

How we simplified

1 **Remove interaction slopes first.** Random slopes for interactions (e.g., Emotion × PresentNum) are the least plausible and need the most data.

2 **Remove main-effect slopes:** presentation number (habituation) before emotion.

3 **Leave one out.** Remove the actor version and refit; if still singular, put it back and remove the participant version. If neither works, remove both and move on.

4 **Remove actor before participant.** Only 5 actors vs. 40 children: too little information to estimate how actors differ, and our questions are about children.

5 **Remove emotion random slopes last.** The condition effect of emotion is central to our research question, so we care more about capturing subject-level variability in the emotion effect than for the effect of presentation number

```
# Final model: dummy coding (Angry = reference), 0 = first presentation, centered age
fit_final <- lmer(MeanAmp ~ Emotion * Age_gmc + PresentNum_True0 +
                    (1 | SubjectID) + (1 | ActorID), data = sim_data)
```

A 9-Step guide to building your LME

1. Define the research question
2. List all relevant variables
3. Draw the study design schematic
4. Categorize variables by type & level (trial vs. subject)
5. Assign fixed and random effects
6. Consider data transformations & center predictors (optional)
7. Fit maximal model and simplify until convergence is reached
8. **Check model assumptions**
9. Report & visualize results

Step 8: Check assumptions

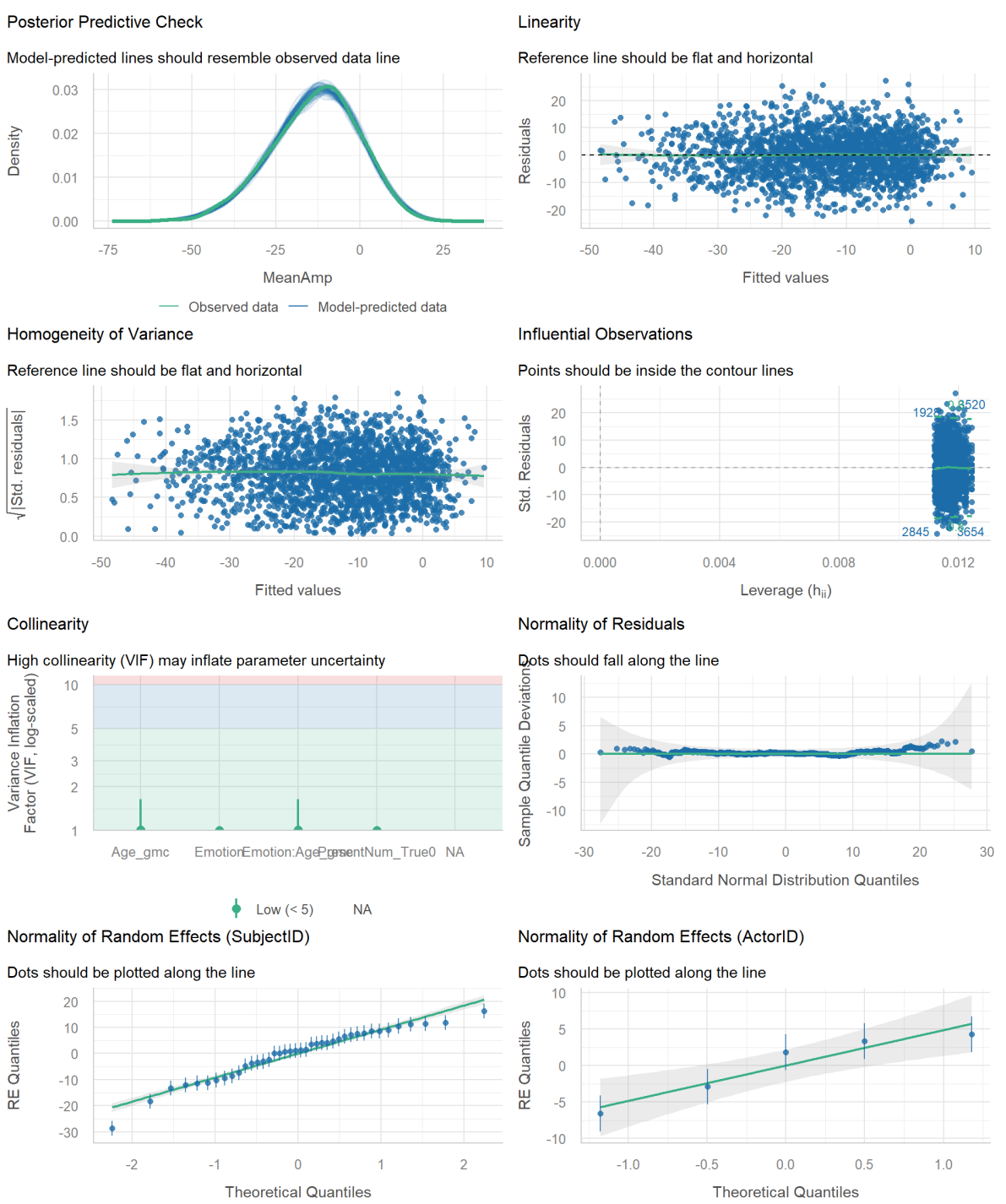
Assumption	Look at	If it's violated
Linearity	Linear relationship between predictors and outcome (mean amplitude), Visually inspect model residuals plotted against fitted values	Add nonlinear terms (e.g., log or quadratic trial number; Step 6)
Homoscedasticity	Outcome is equally distributed across groups (e.g., mean amplitude has equal variance across conditions), Spread of residuals vs. fitted, and by condition or group, Levene's test for homogeneity of variance	Check outliers; model the variance (e.g., location scale model)
Normality	Normal distribution of residuals, visual inspection for n>5000 (Q–Q plot), Shapiro-Wilk test (n between 3–5,000)	Fixed effects are fairly robust (Schielzeth et al., 2020); check outlying trials; transform skewed outcomes
Normal random effects	Q–Q plot of each random effect (one per grouping factor)	Look for outlying participants or stimuli
Independence	Residuals vs. trial order within participant	Add trial number / block; revisit the random effects
Influential points	Cook's distance; participants with extreme random effects	Refit without them and report both
Collinearity	Variance inflation factors (VIF < 5 is fine)	Center predictors, especially for interactions (Step 7); drop redundant predictors

Schielzeth et al. (2020), Methods Ecol. Evol.; Heise et al. (2022)

Step 8: check_model() runs every check in one call

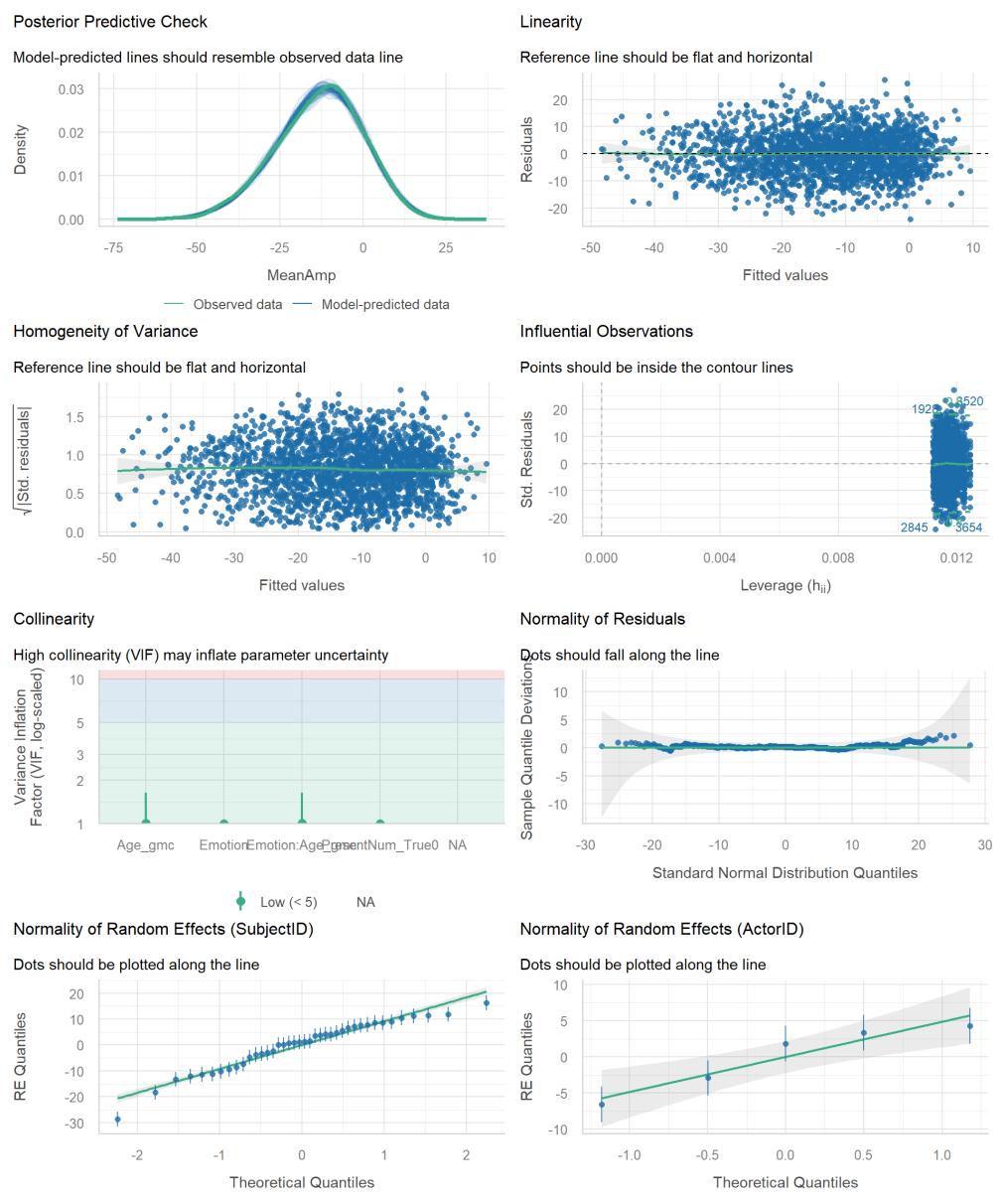
```
# install.packages(c("performance", "see"))
library(performance)
library(see)    # draws the plots

check_model(fit_final)
```



Lüdecke et al. (2021), performance: An R package, JOSS

Step 8: Reading check_model() output



Posterior predictive check

Model-simulated data should look like the observed data

Linearity

Green line flat at 0; no curve

Homogeneity of variance

Even distribution around a flat line

Influential observations

Points inside the Cook's distance contours

Collinearity

VIFs <5

Normality of residuals

Dots along the line

Normality of random effects

One panel per grouping factor; dots along the line. Hard to judge with only 5 actors

Step 8: The individual check_*() functions

Function	What it checks	fit_final
check_convergence()	Did the optimizer converge?	TRUE
check_singularity()	Does the final model have a singular fit?	FALSE
check_collinearity()	VIF for each fixed effect	1.00–1.01
check_outliers()	Influential observations (Cook's distance)	None detected
check_normality()	Test of residual normality	p = .36
check_heteroscedasticity()	Test of constant residual variance	p < .001
icc()	Share of variance due to participants and actors	0.64

VIF (variance inflation factor): how much a coefficient's variance is inflated because its predictor overlaps with the others: $1 / (1 - R^2)$ from predicting it from the other predictors. 1 = no overlap; above 5 is a concern.

Cook's distance: how much the model's predictions change when one observation is left out. Large values (e.g., above 1) flag observations with outsized influence.

Data were simulated with constant trial noise (SD = 8 µV), and the variance panel is flat, but the test flags heteroscedasticity. With 4,000 trials, formal tests can be overly conservative.

ICC: how much of the variability in amplitude comes from grouping factors (participants and actors, 64%)

Lüdecke et al. (2021), performance: An R package, JOSS

A 9-Step guide to building your LME

1. Define the research question
2. List all relevant variables
3. Draw the study design schematic
4. Categorize variables by type & level (trial vs. subject)
5. Assign fixed and random effects
6. Consider data transformations & center predictors (optional)
7. Fit maximal model and simplify until convergence is reached
8. Check model assumptions
9. **Report & visualize results**

Reading the output: model info & random effects

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: MeanAmp ~ Emotion * Age_gmc + PresentNum_True0 + (1 | SubjectID) +      (1 | ActorID)
##    Data: sim_data
##
## REML criterion at convergence: 28196.1
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.4460 -0.6864  0.0108  0.6586  3.5025
##
## Random effects:
##  Groups    Name        Variance Std.Dev.
##  SubjectID (Intercept) 91.55    9.568
##  ActorID   (Intercept) 21.26    4.611
##  Residual              63.91    7.994
## Number of obs: 4000, groups:  SubjectID, 40; ActorID, 5
```

1. **The model:** Emotion × age + presentation number, with random intercepts for children and actors
2. **REML criterion:** only useful for comparing models with the same fixed effects
3. **Scaled residuals:** centered near 0, roughly symmetric, nothing past ±4: no extreme outliers
4. **Random effects:** children differ in overall Nc (SD = 9.6 µV), actors differ (SD = 4.6 µV), and single trials vary around that (SD = 8.0 µV)
5. **Number of obs:** 4,000 trials from 40 children and 5 actors: check this matches your cleaned data

Reading the output: fixed effects

```
## Fixed effects:
##                         Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)           -16.46275    2.57142    9.31259  -6.402 0.000107 ***
## EmotionHappy            1.99846    0.25280 3953.00028   7.905 3.44e-15 ***
## Age_gmc                -2.87091    2.63796   38.52833  -1.088 0.283220
## PresentNum_True0        0.49528    0.04401 3953.00028  11.255  < 2e-16 ***
## EmotionHappy:Age_gmc    1.45851    0.43775 3953.00028   3.332 0.000871 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) EmtnHp Ag_gmc PrN_T0
## EmotionHppy -0.049
## Age_gmc      0.000  0.000
## PrsntNm_Tr0 -0.077  0.000  0.000
## EmtnHppy:A_  0.000  0.000 -0.083  0.000
```

6

1 2 3 4 5

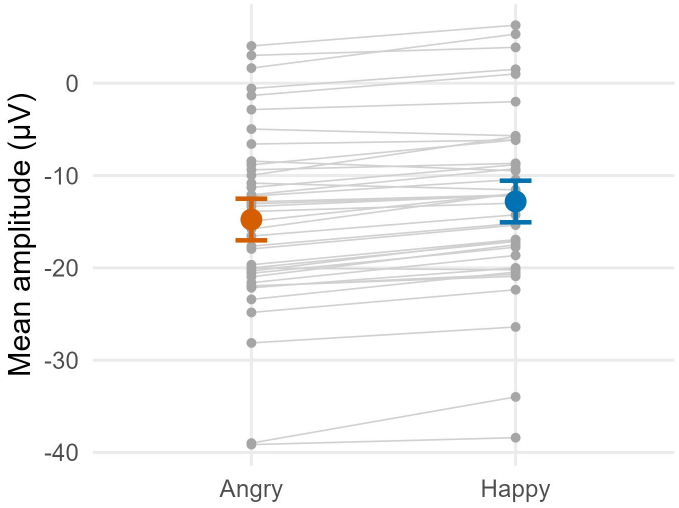
7

1. **Intercept = −16.5 µV:** Nc for Angry faces at the first presentation, for a child of average age
2. **EmotionHappy = 2.0 µV:** Happy faces are 2.0 µV less negative than Angry, for a child of average age ($p < .001$)
3. **Age_gmc = −2.9 µV, p = .28:** for Angry faces, each additional year of age predicts a mean amplitude 2.9 µV more negative (a larger NC) (not significant)
4. **PresentNum = 0.50 µV:** Nc gets less negative with each presentation, for both emotions (habituation)
5. **EmotionHappy:Age_gmc = 1.46 µV, p < .001:** the Happy − Angry difference grows by 1.46 µV per year
6. **df:** ≈ 9 for the intercept (5 actors), ≈ 39 for age (40 children), ≈ 3,953 for trial-level effects
7. **Correlations:** correlations near 0 mean that two estimates/effects are independent of each other; centered variables will have an intercept of 0

Step 9: Visualize and report your results

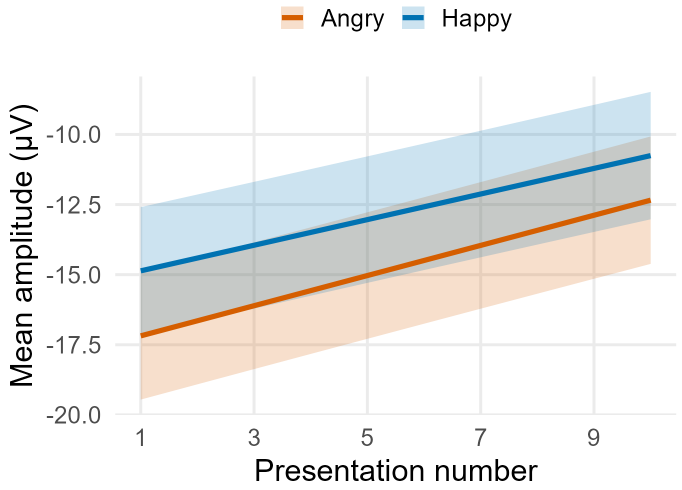
Plot the model's estimates + the raw data: EMMs ± 1 SE, with each child's averages (or single trials) behind them

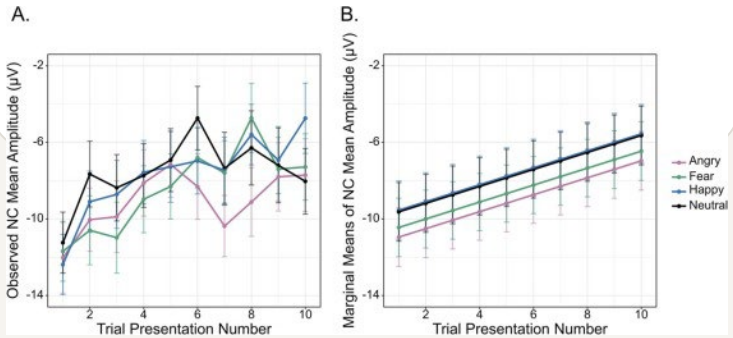
By emotion



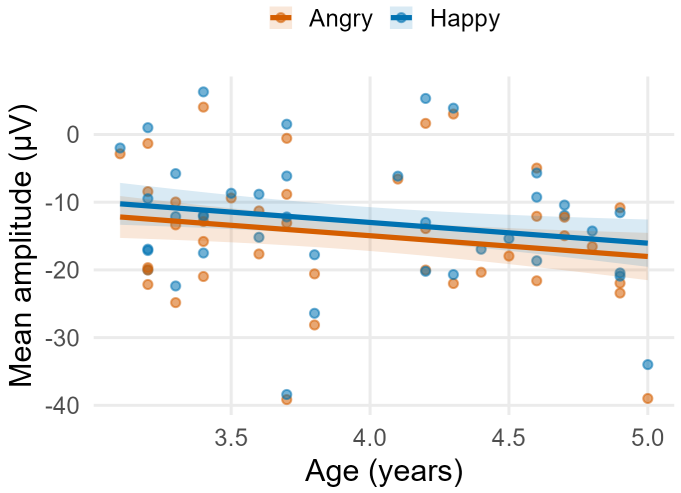
Bars = ±1 SE; gray lines = each child's means

Across presentations





Across age

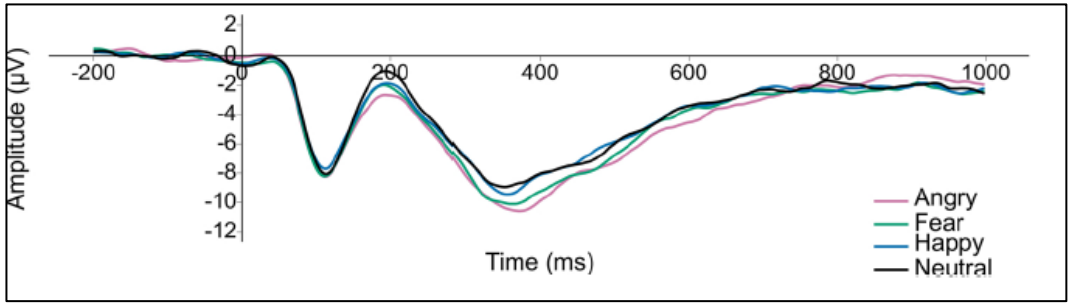


Points = each child's mean

Step 9: Visualize and report your results

A More digestible: average the trials

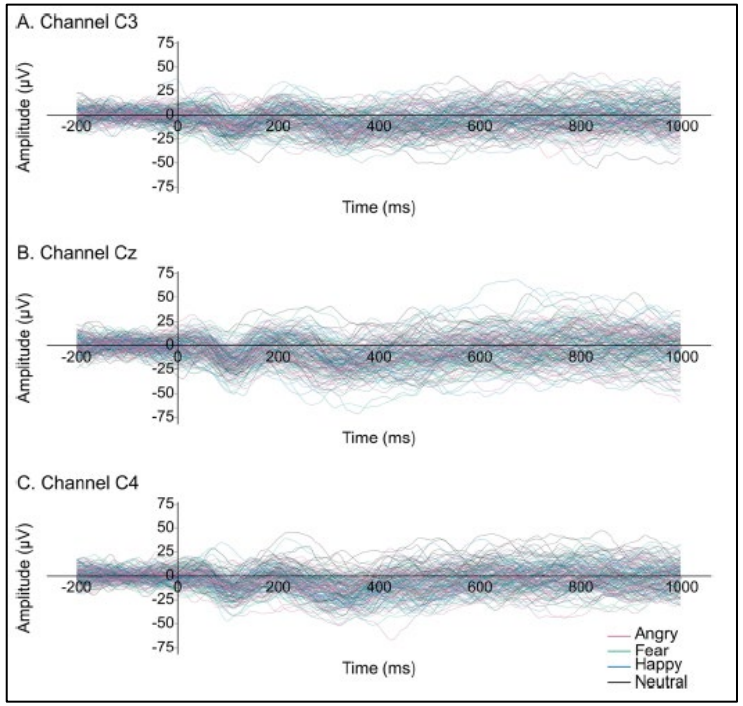
Mean-average across trials and plot the ERPs by condition



+ Familiar and easy for any audience to read
– Hides the trial-to-trial variability your model is fit to

B More transparent: show single trials

Pick a representative participant and plot their trials by condition



+ True to your analysis: shows within-subject variability
– Busier; one participant may not represent everyone

Pair & share: dummy vs. effects coding

Think · 1 min on your own → **Pair** · 3 min with a neighbor → **Share** · with the group

Which coefficients changed, which stayed the same, and why?

Coefficient	Dummy (0 / 1)		Effects (−0.5 / +0.5)		Effects (−1 / +1)	
	b (µV)	t	b (µV)	t	b (µV)	t
Intercept	−16.463	−6.40	−15.464	−6.02	−15.464	−6.02
Emotion (Happy vs. Angry)	1.998	7.91	1.998	7.91	0.999	7.91
Presentation number	0.495	11.25	0.495	11.25	0.495	11.25
Age	−2.871	−1.09	−2.142	−0.81	−2.142	−0.81
Emotion × Age	1.459	3.33	1.459	3.33	0.729	3.33

Dummy vs. effects coding

Final emotion model, fit with each coding of Emotion (0 = first presentation; age centered)

Coefficient	Dummy (0 / 1)		Effects (−0.5 / +0.5)		Effects (−1 / +1)	
	b (μV)	t	b (μV)	t	b (μV)	t
Intercept	−16.463	−6.40	**−15.464**	−6.02	**−15.464**	−6.02
Emotion (Happy vs. Angry)	1.998	7.91	1.998	7.91	**0.999**	7.91
Presentation number	0.495	11.25	0.495	11.25	0.495	11.25
Age	−2.871	−1.09	**−2.142**	−0.81	**−2.142**	−0.81
Emotion × Age	1.459	3.33	1.459	3.33	**0.729**	3.33

Changes with effects coding: the effect for Angry faces becomes the effect averaged over both emotions

Halved with ±1: Happy and Angry are 2 units apart, so b and SE are halved, and t stays the same

Effects coding	Intercept	Emotion	Emotion × Age
−0.5 / +0.5	Grand mean	Full difference (Happy − Angry)	Full difference in age slopes
−1 / +1	Grand mean	Half the difference (each emotion vs. the mean)	Half the difference in age slopes

Dummy vs. effects coding: what changes, and why?

Dummy → effects coding

Lower-order effects change meaning

- **Dummy:** the age effect is for Angry faces, the reference group
- **Effects:** it's averaged over both emotions (= dummy effect + ½ × the interaction)
- **So their t and p change too:** it's a different question

In our data: Age: −2.87 → −2.14 µV per year, t = −1.09 → −0.81

±0.5 → ±1

Only the emotion terms rescale

- **±0.5:** Happy and Angry are 1 unit apart, so b is the full Happy − Angry difference
- **±1:** they're 2 units apart, so b (and SE) for emotion and its interactions are halved
- **t and p are identical**

In our data: Emotion: 2.00 → 1.00 µV, t = 7.91 for both

Never changes

Same model, different parameters

- Predictions and model fit
- Random effects
- t and p of the highest-order interactions

In our data: Emotion × Age: t = 3.33 in all three

Effects coding (±0.5) when you want effects averaged over conditions (the usual "main effect")

Dummy coding when the reference group matters in its own right, e.g., a control group or baseline condition

Prefer ±0.5 over ±1: the coefficient is the full difference between conditions. With ±1, double it

Effects coding with 3 or more groups

k groups → k − 1 coded columns. The intercept is the grand mean (the average of the group means) in every version below

3 groups: sum coding

`contr.sum(3)`

	E1	E2
Angry	1	0
Happy	0	1
Neutral	−1	−1

E1: Angry − grand mean

E2: Happy − grand mean

Neutral: not in the output; its deviation is −(E1 + E2)

4 groups: sum coding

`contr.sum(4)`

	E1	E2	E3
Angry	1	0	0
Fearful	0	1	0
Happy	0	0	1
Neutral	−1	−1	−1

E1–E3: each group − grand mean

Neutral: −(E1 + E2 + E3)

3 groups: planned contrasts

the ±0.5 idea for 3 groups

	C1	C2
Angry	+⅓	−½
Happy	+⅓	+½
Neutral	−⅔	0

C1: emotional faces (average of Angry & Happy) − Neutral

C2: Happy − Angry

Sum coding: each coefficient is a group vs. the grand mean, not a difference between two groups. **Planned contrasts:** each coefficient is the comparison you chose

Marginal means don't depend on coding

Another option: keep dummy coding and get any effect from marginal means (EMMs), which don't depend on coding.

Emotion: emmeans()

```
emm_coding <- emmeans(fit_final, ~ Emotion,
                      lmer.df = "asymptotic")
emm_coding
pairs(emm_coding, reverse = TRUE)   # Happy - Angry
```

1 2

```
 Emotion emmean   SE  df asymp.LCL asymp.UCL
 Angry    -14.2 2.56 Inf     -19.3     -9.21
 Happy    -12.2 2.56 Inf     -17.3     -7.21

 contrast      estimate    SE  df z.ratio p.value
 Happy - Angry        2 0.253 Inf   7.905  <.0001
```

3

Age: emtrends()

```
emt_coding <- emtrends(fit_final, ~ Emotion, var = "Age_gmc",
                       lmer.df = "asymptotic")
emt_coding
emtrends(fit_final, ~ 1, var = "Age_gmc",
         lmer.df = "asymptotic")
```

```
 Emotion Age_gmc.trend   SE  df asymp.LCL asymp.UCL
 Angry           -2.87 2.64 Inf     -8.04      2.30
 Happy           -1.41 2.64 Inf     -6.58      3.76
```

4

```
 1       Age_gmc.trend   SE  df asymp.LCL asymp.UCL
 overall         -2.14 2.63 Inf     -7.29      3.01
```

5

1. **emmean:** the model-estimated Nc for each emotion, at the average age and presentation number (4.5), not the raw means
2. **SE, df, CI:** df = Inf because lmer.df = "asymptotic" uses z tests (fast with 4,000 trials); asymp.LCL/UCL = the 95% CI
3. **contrast:** Happy − Angry = 2.00 µV, z = 7.9, p < .0001, the same as the dummy-coded EmotionHappy coefficient
4. **Age_gmc.trend:** the age slope for each emotion: Angry −2.87 = the dummy-coded Age_gmc; Happy −1.41 = Age_gmc + the interaction
5. **overall:** the age slope averaged over both emotions, −2.14 = the effects-coded Age_gmc; the CI includes 0

Standardized β as an effect size

What is an effect size?

A standardized estimate of the relation between variables: how big an effect is, in units that don't depend on how each variable was measured (μV, years, presentations)

Ferron et al. (2008); Snijders & Bosker (2012)

Why standardized β is good enough

Most journals now require reporting an effect size

No consensus effect size for LMEs: d and R^2 assume one source of variability; an LME has several (children, actors, trials)

Easy: z-score the outcome and continuous predictors and refit the same model; t and p don't change

Comparable: every β is in SDs of amplitude, so you can compare predictors within your model

scale() every variable and refit: β = change in amplitude (in SDs) per 1-SD change in the predictor

Predictor	b (μV)	β	95% CI for β	p
Emotion (Happy vs. Angry)	1.998	**0.076**	[0.057, 0.095]	< .001
Presentation number	0.495	**0.108**	[0.089, 0.127]	< .001
Age	−2.142	**−0.095**	[−0.332, 0.142]	.420
Emotion × Age	1.459	**0.032**	[0.013, 0.052]	< .001

More problems with dummy/effects coding

Emotion_z is −1 / +1, so Happy − Angry = 2 units apart × β = 0.15 SD. Or keep ±0.5 and scale only continuous variables
Emotion_z x age_z also needs to be doubled (.06)

Standardized betas: within vs. across models

Within one model, β compares predictors: presentation number (0.11) ≈ age (−0.10) > emotion (0.08) > emotion × age (0.03)

Across analyses, it's more complicated: the same 2.00 µV Happy − Angry difference analyzed two ways, with three effect sizes:

Analysis	Test	Divides the effect by	Effect size
LME	b = 2.00, SE = 0.25, t(3953) = 7.91, p < .001	Total SD of single-trial amplitude (13.1 µV)	**β = 0.15**
Paired t-test on each child's averages	M = 2.00, 95% CI [1.49, 2.50], t(39) = 8.00, p < .001	**d_z:** SD of each child's Happy − Angry difference (1.6 µV)	**d_z = 1.26**
		d_av: average SD of the two condition means (~9.5 µV) *like Cohen's d for independent groups: ignores the pairing*	**d_av = 0.21**

The LME's β divides by all single-trial variability (children, actors, trial noise). Averaging minimizes children's and actors' differences and shrinks the noise, so the same effect looks much larger.

Use β to compare predictors within your model

Report b in µV

Don't compare β to d from averaged data, or use it in a power analysis that expects d

Step 9: What to report in your results

Table 1.

Predictor	*b*	*SE*	95% CI	*t*	df	*p*	β
Intercept	−15.46	2.57	[−21.25, −9.68]	−6.02	9.27	< .001	
Emotion (Happy – Angry)	2.00	0.25	[1.50, 2.49]	7.91	3953	< .001	0.08
Age (years, centered)	−2.14	2.63	[−7.46, 3.18]	−0.81	38	.420	−0.10
Presentation number	0.50	0.04	[0.41, 0.58]	11.25	3953	< .001	0.11
Emotion × Age	1.46	0.44	[0.60, 2.32]	3.33	3953	< .001	0.03

Random effects (bottom of tab_model())			
σ^2 (residual)	63.91	τ_{00} SubjectID	91.55
τ_{00} ActorID	21.26	ICC	0.64
N SubjectID	40	N ActorID	5
Observations	4,000	Marginal / Conditional R^2	0.026 / 0.648

In text: Averaged over emotions, age was not significantly associated with Nc amplitude, *b* = −2.14, *SE* = 2.63, *p* = .420, β = −0.10 (see Table 1).

Table with model output
fixed-effect β and SE, random-effect variances, residual variance

Model fit indices and/or model simplification steps

Assumption checks

Exclusion criteria & number of included trials & participants

Significance tests
How df were calculated & multiple comparisons if applicable

Step 9: Three ways to report your model

Verbal description

"Single-trial Nc amplitude was predicted by emotion, presentation number, and their interaction, with random intercepts for participants and actors."

Pros

+ Readable by anyone, including reviewers outside your field and people who use different statistical software
+ Gives clear information about what each term means

Cons

– Easy to be vague: "random effect"?
– Can be hard to reproduce model exactly from words alone

Wilkinson notation (R syntax)

```
MeanAmp ~ Emotion * PresentNum
  + (1|Subject)+(1|Actor)
```

Pros

+ Unambiguous: readers can rerun the model

Cons

– Can be a shortcut: copying syntax ≠ understanding the model
– Shorthand hides details: * expands to interactions; || drops correlations

Raudenbush & Bryk equations

L1: $Amp_{ti} = \pi_{0i} + \pi_1 Emo_{ti} + \pi_2 Pres_{ti} + \ldots + e_{ti}$

L2: $\pi_{0i} = \beta_{00} + r_{0i}$

Pros

+ Makes levels explicit: which predictors are trial- vs. participant-level, what varies randomly
+ Shows cross-level interactions clearly

Cons

– Unfamiliar to many ERP readers
– Crossed random effects notation is hard to read/decipher (when stimuli are crossed instead of nested)

Wilkinson & Rogers (1973), Applied Statistics; Raudenbush & Bryk (2002); Meteyard & Davies (2020); Heise, Mon & Bowman (2022)

Step 9: Report what the next study needs

Report anything you'd want if you were planning a power analysis based on this study's findings:

What to report	
Fixed effects, including nuisance terms	Full fixed-effects table: unstandardized betas (μV) with SEs, e.g., the β for trial presentation number
Random-effect SDs and correlations	Random-effects table: every variance component (participants and stimuli)
Single-trial noise	Residual SD in the random-effects table
Trials presented and kept	Mean, SD, and range per participant and condition, by group
Rejection rate and predictors of missingness	Missingness by age, condition, and trial position
Design counts	Participants, stimuli, repetitions per stimulus, trials per condition
Data quality	Data quality per group (e.g., mean of included trials per participant, SME)
Convergence path	The maximal model and what was dropped

Lunch! 12:00–1:15?

- After lunch: difference waves, then the auditory oddball example with single-trial difference waves. You'll work through the oddball more independently, and we'll regroup after each part

Afternoon agenda

05 **Difference waves**
Short lecture: pairing trials to make single-trial difference waves

06 **LME walkthrough #2**
Auditory oddball with single-trial difference waves (MMN)

07 **Planning our ERP designs for LME analyses**
Trial-level covariates and unique event markers

08 **Power analyses**

09 **Location scale models**

10 **Latency**
What do we know so far?

✓ **Key takeaways & wrap-up**
Questions and discussion

05 Special cases: Difference waves

Difference waves

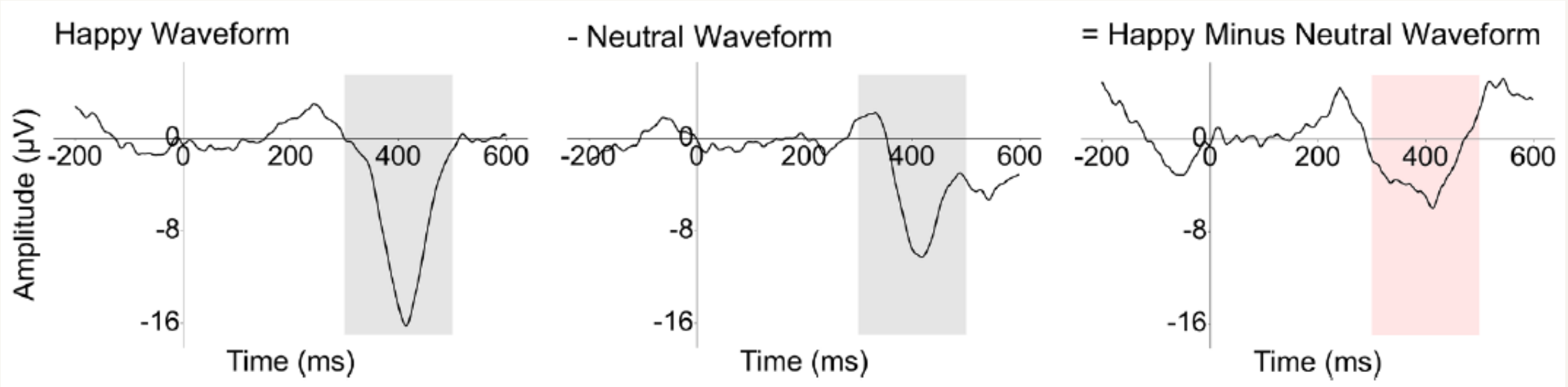


-



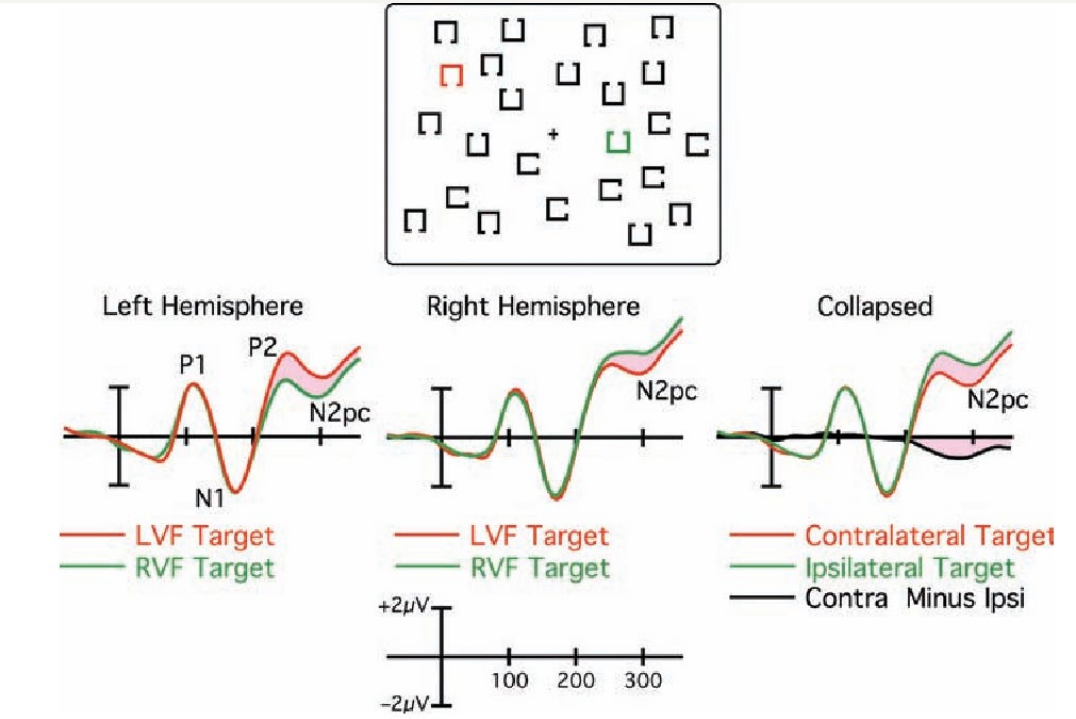
=

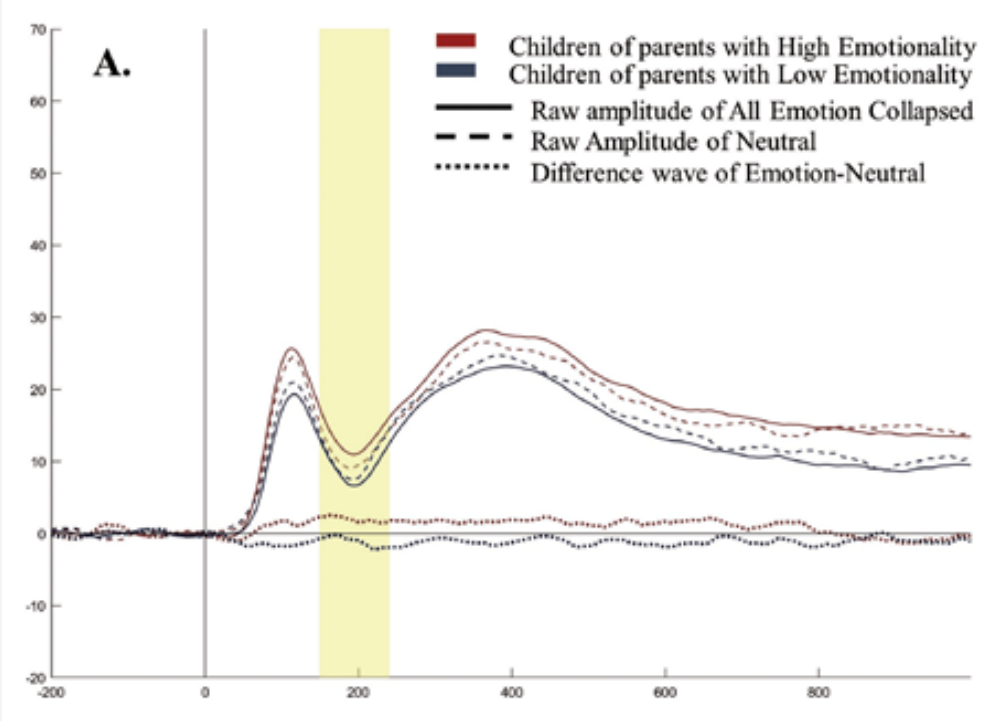
cognitive processing of the Happy emotion separate from overall face processing



Difference waves

- Difference waves are formed by subtracting one ERP from another





Luck, S. J. (2014). *An introduction to the event-related potential technique* (2nd ed.). MIT Press.

Xia, R., Heise, M. J., & Bowman, L. C. (2024). Parental emotionality is related to preschool children's neural responses to emotional faces. *Social Cognitive and Affective Neuroscience*, *19*(1), 1–11. https://doi.org/10.1093/scan/nsad078

Difference waves

- LMEs for ERPs rely on single trials being the smallest unit of analysis
- How do we get a “difference wave trial” comprised of two trials?

Difference waves: Issues in pairing

Possible features to consider in pairing: pair trials that match on anything that changes amplitude

1 Stimulus features

e.g., the actor with ‘red hair’ has a different amplitude from the actor with ‘blonde hair’

2 ERP habituation/decay

Some ERP components habituate over repeated trials

Borgström et al., 2016; Reynolds & Richards, 2019

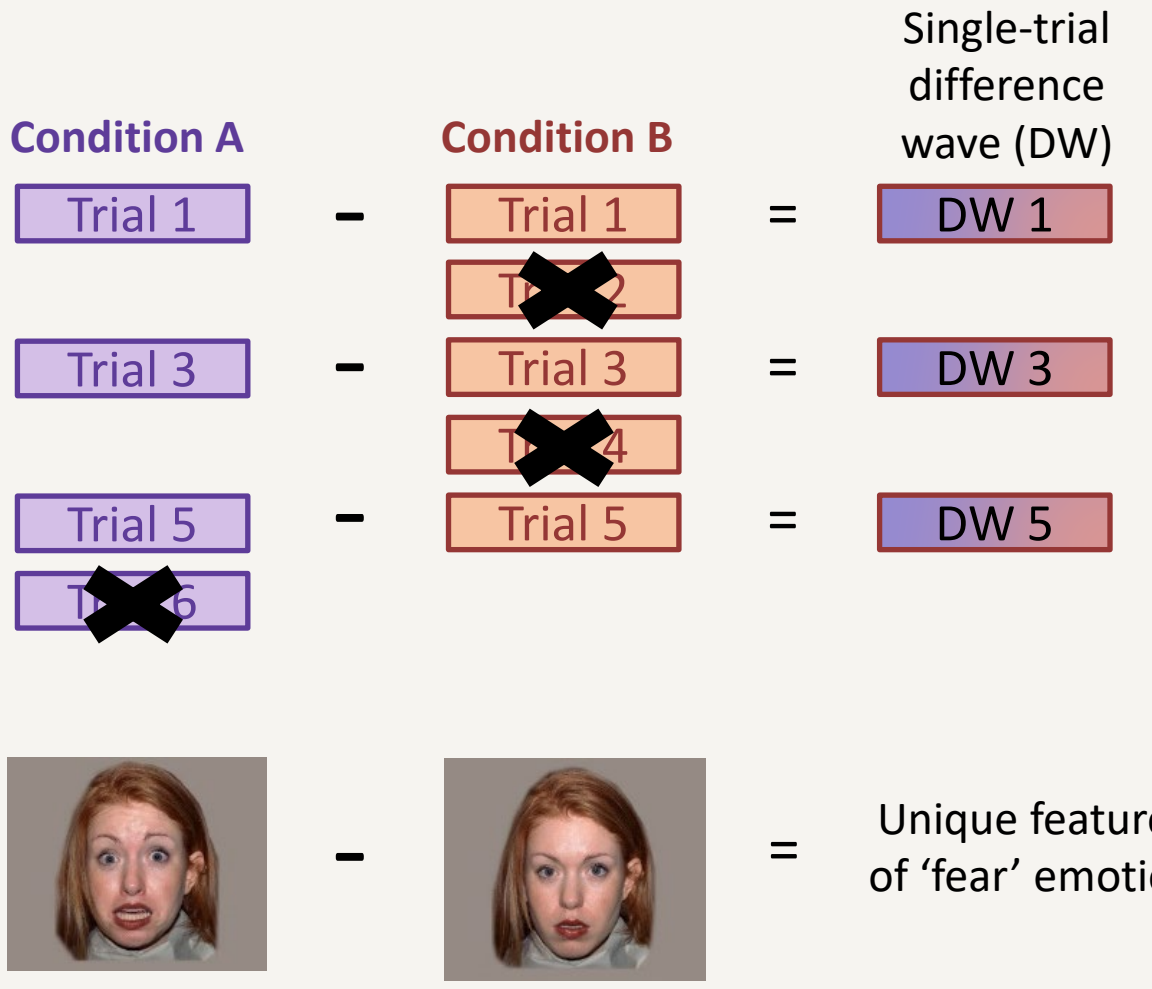
Few papers report their mean amplitude habituation, but Borgström et al. (2016) found **~1.5 µV per trial** in the NC; in preschoolers we saw **0.44 µV per trial**

3 Electrode

If you impute bad channels before exporting single-trial mean amplitude you will have data from every single electrode and can pair easily for DW

1) Exact match

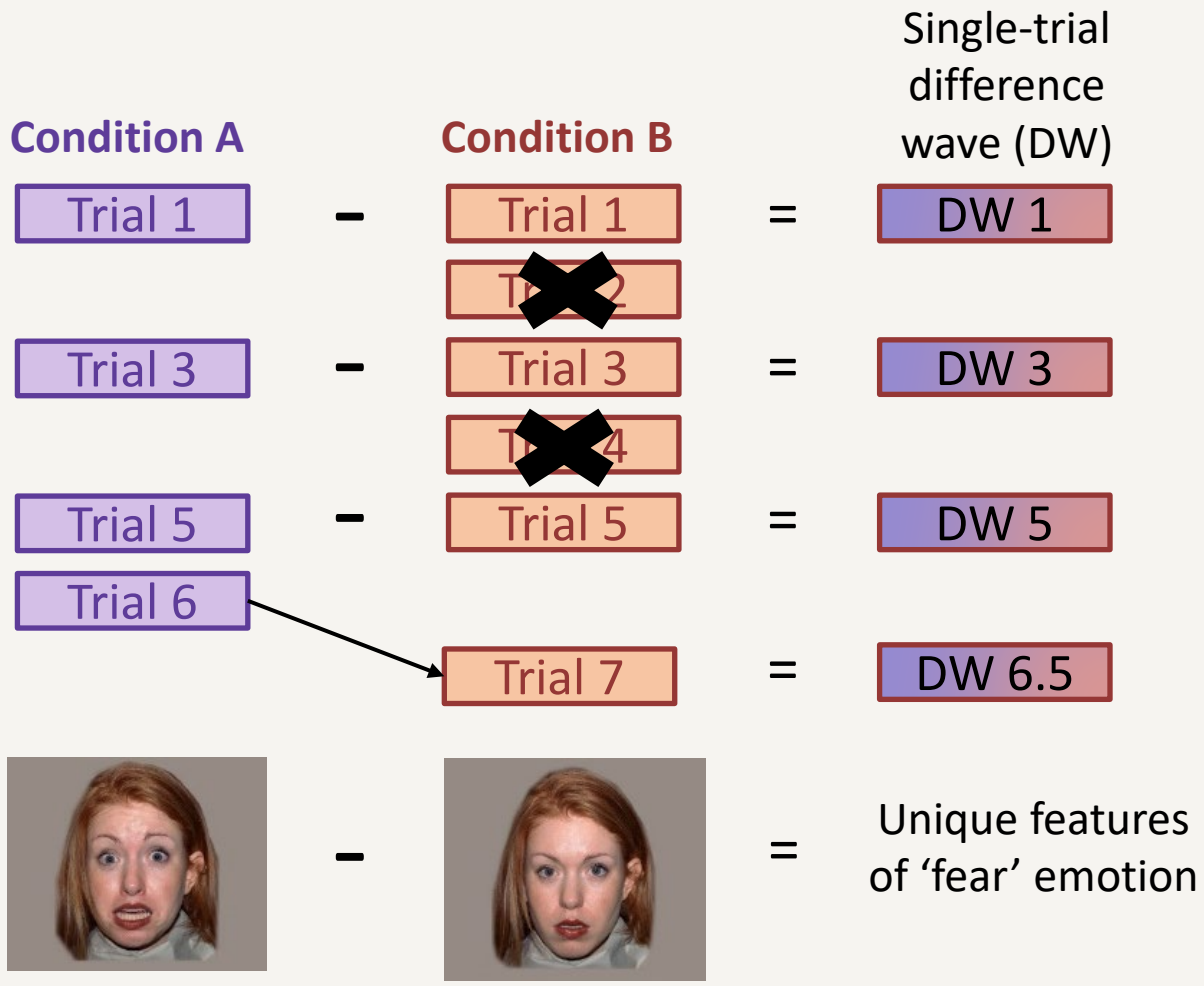
Pair every available trial, drop all trials who do not have a pair



2) Nearest neighbor

Pair trial based on the Mahalanobis distance (distance in multivariate space)

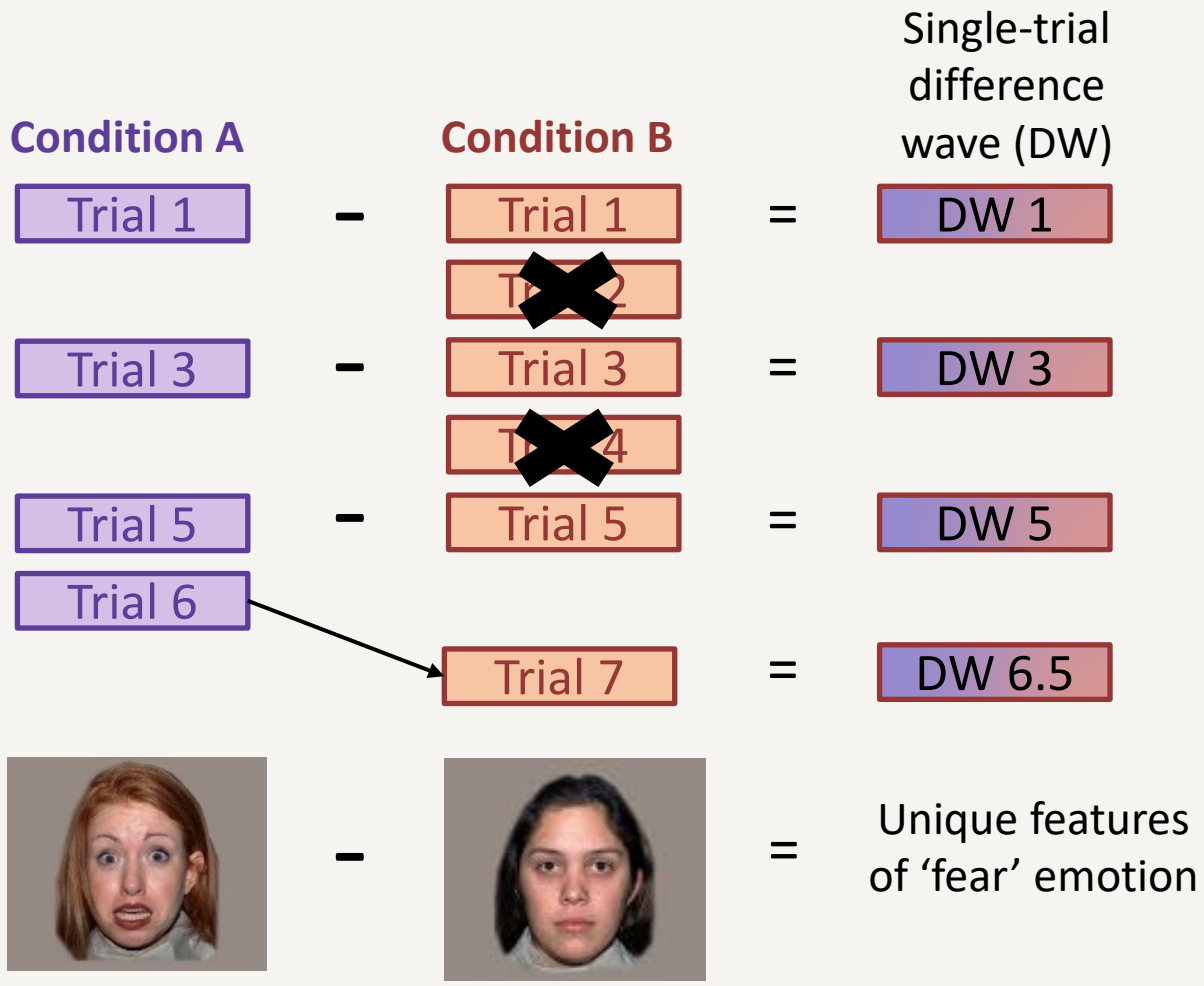
- a) No prioritization
- b) Presentation number prioritization (early trials with other early trials)
- c) Stimulus feature (actor) prioritization



2) Nearest neighbor

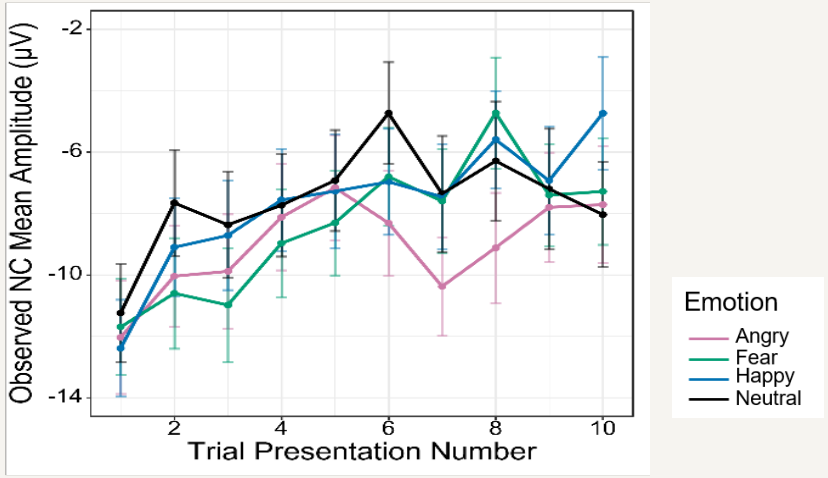
Pair trial based on the Mahalanobis distance (distance in multivariate space)

- a) No prioritization
- b) Presentation number prioritization (early trials with other early trials)
- c) Stimulus feature (actor) prioritization

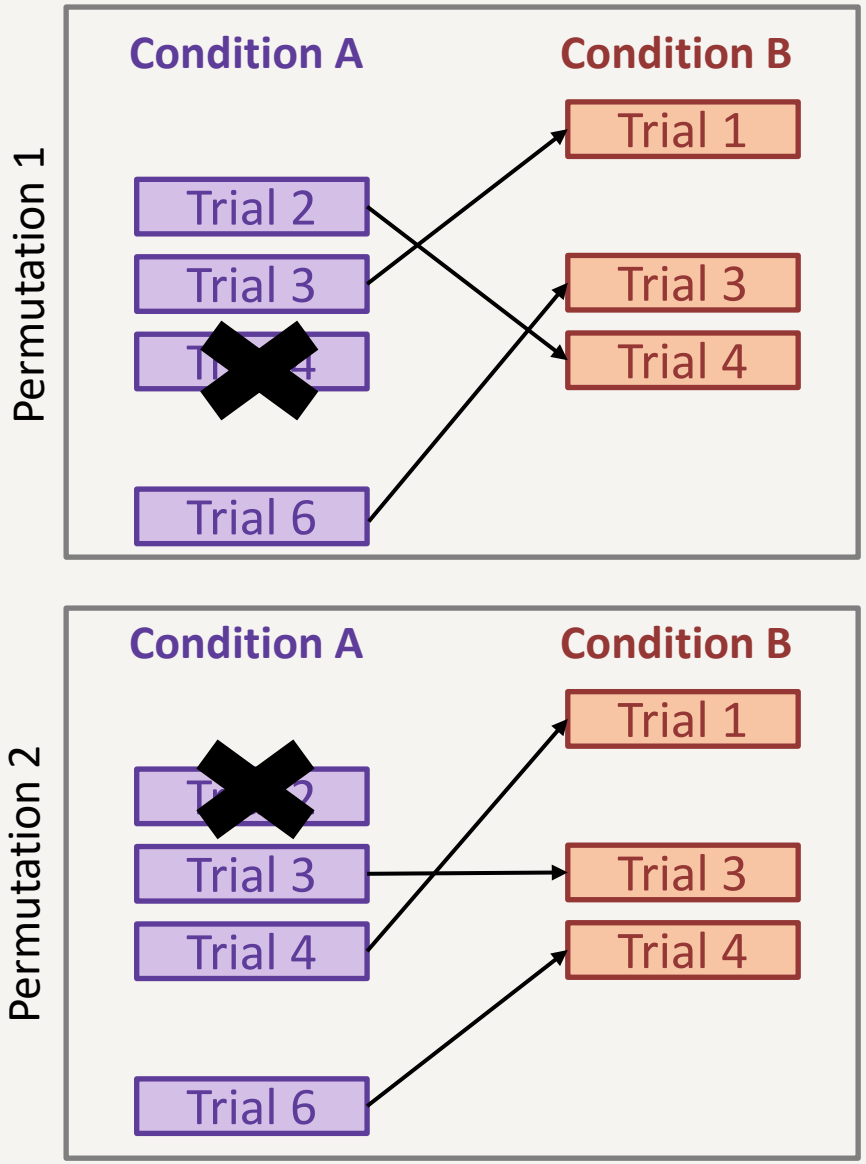


3) Random permutations

- Completely random permutation (regardless of trial number)
  - µV habituation is not perfectly linear, so perhaps it's okay to randomly pair trials



- Does not systematically drop trials when data are unbalanced (all trials are represented in the mean amplitude estimate)



Difference wave LME formula

Research question tested in simulations: Is an NC emotion difference wave modulated by the child's perceptual sensitivity (measured by the early childhood behavior questionnaire)?

In writing our maximal LME model, we don't need to include anything that is "subtracted out" in the creation of the DW

```
lmer(DW_MeanAmp ~ ECBQ + PresentNum + (1 | SubjectID))
```

Pairing approach	Presentation number in the model	Actor
Exact Match	Matched (same in both trials)	Matched → subtracted out
Nearest Neighbor: presentation number priority	Matched	Not matched, and can't be modeled
Nearest Neighbor: stimulus feature priority	Average of the pair (e.g., 1 & 2 → 1.5)	Matched → subtracted out
Nearest Neighbor: no prioritization	Average of the pair	Not matched, and can't be modeled
Random Permutation	Average of the pair (re-paired 10,000 times)	Not matched, and can't be modeled

Why keep presentation number when it's matched? If conditions habituate at different rates, there will be a non-zero coefficient for presentation number (also a coefficient of 0 tells you that the subtraction was successful in "subtracting out" that variance associated with presentation number.

4) Interaction term

Don't pair trials: a **condition × predictor interaction** replace calculating a single-trial difference wave

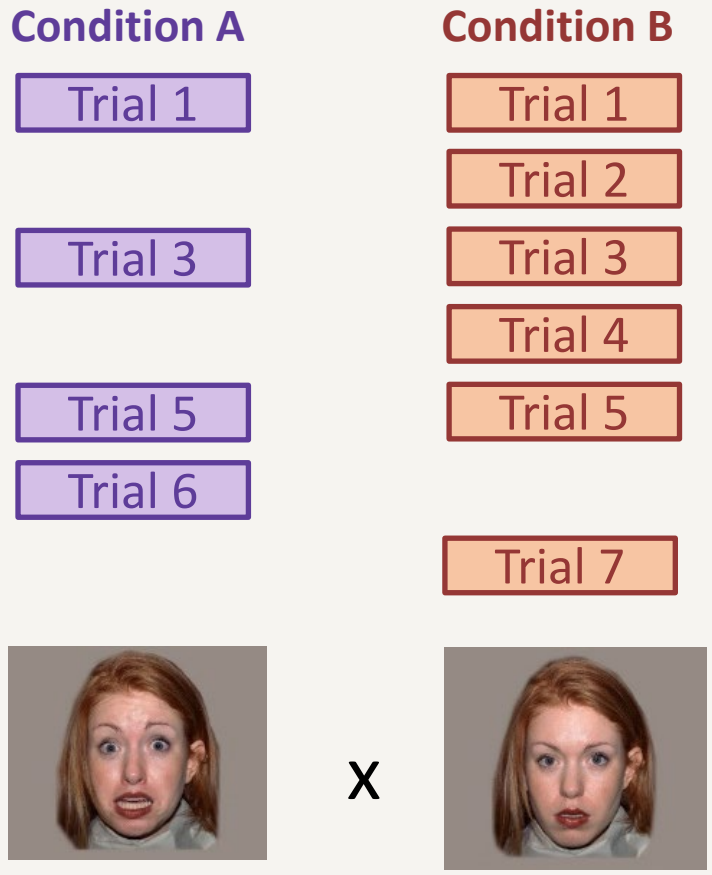
```
lmer(MeanAmp ~ Emotion * ECBQ + Emotion * PresentNum + Electrode +
       (1 | SubjectID) + (1 | ActorID))
```

Interaction interpretation

- A significant interaction means that the relation between a predictor and mean amplitude *differs by condition*

Benefits of interaction term

- No trials lost after artifact rejection
- Raw amplitudes have higher signal-to-noise than difference waves (Luck, 2005)



Difference wave methods: Usable data retained

A. Usable trials

Interaction kept every artifact-free trial; **Exact Match** lost the most (a missing trial in either condition excludes the DW)

B. Usable participants

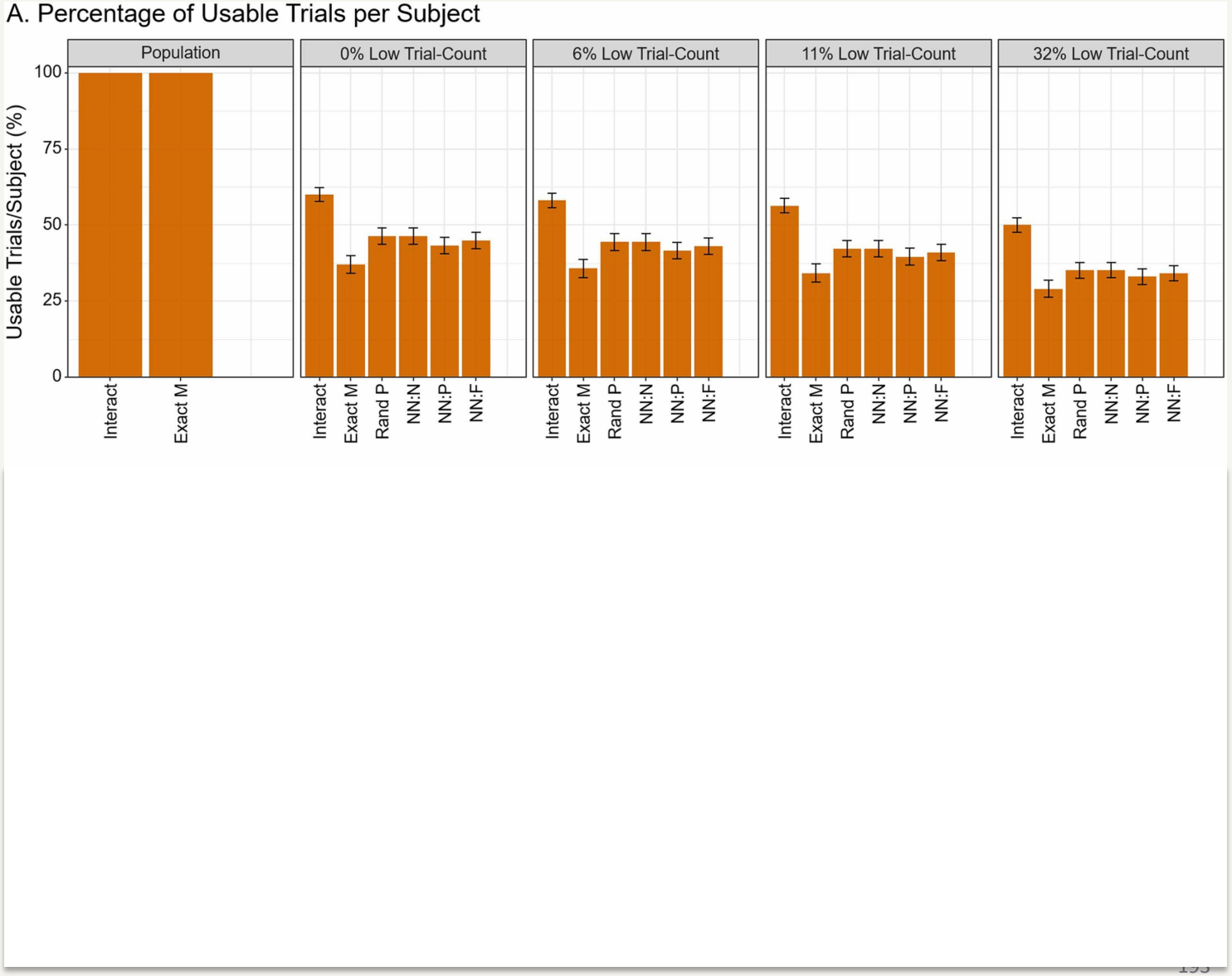
Interaction kept all 48; but even Exact Match kept more than ANOVA's casewise deletion. At 32% low-trial-count:

Interaction retained 100%

Exact Match retained 92%

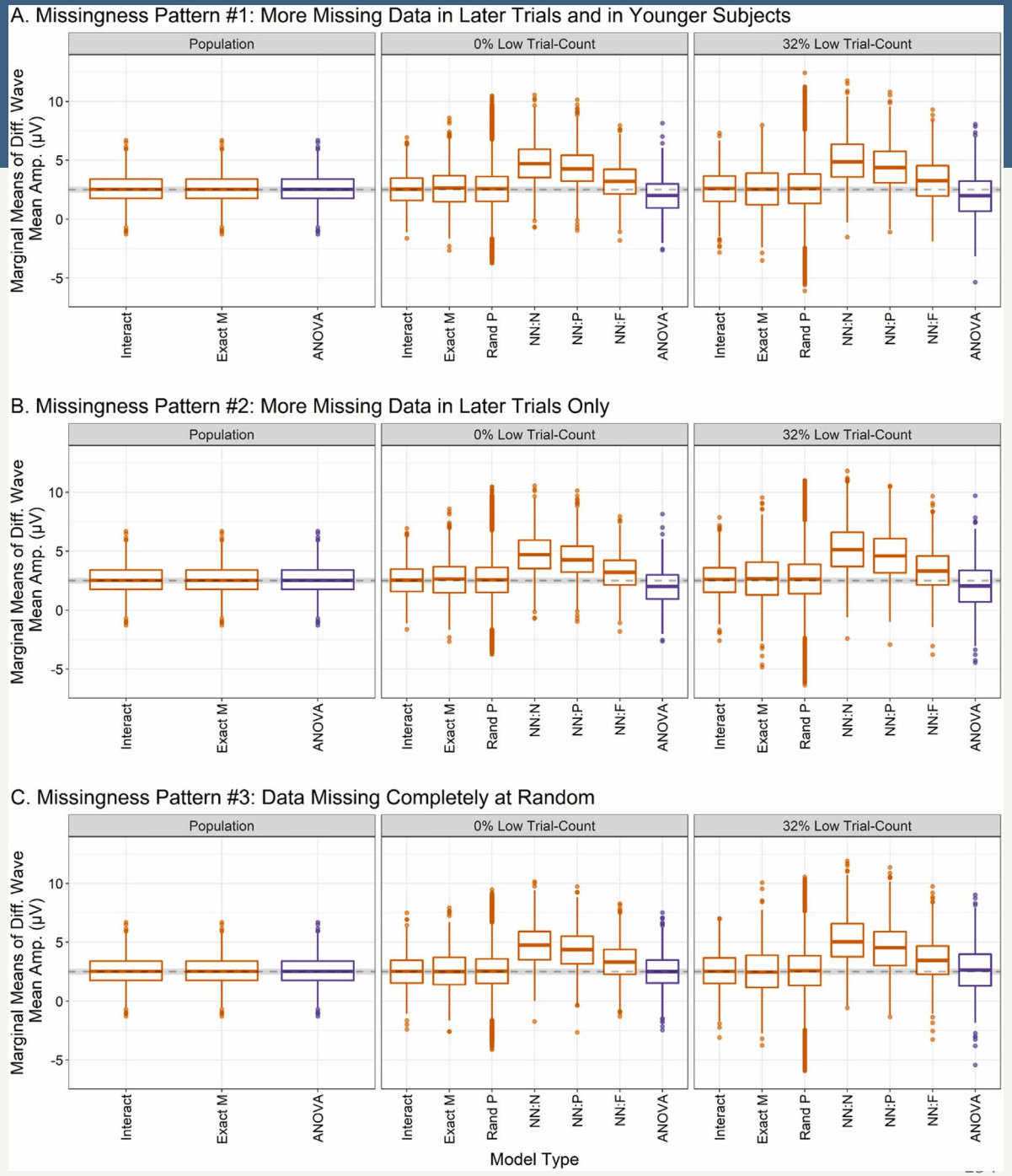
ANOVA retained 67%

Real preschooler data (N = 64): Interaction kept 100%, Exact Match kept 97%; regression kept 89% with a 10-trial cutoff and 73% with a 15-trial cutoff



Difference wave methods: Bias

- Interaction, Exact Match, and Random Permutation approaches accurately estimated the difference wave mean amplitude under realistic decay conditions, and variable-predicted missingness
- LME Nearest Neighbor approaches and ANOVA had biased estimates for at least one of the decay and missing data conditions examined



Simulation results: Power, error & bias

- **Power to detect the true effect**
  - Interaction and Random Permutation performed best (power up to .89 as missing data increased)
  - ANOVA's power dropped the most as low-trial-count participants increased
- **Error (RMSE) from the true population value**
  - Interaction had the lowest error across every simulated condition
- **Bias from the true population value**
  - Interaction, Exact Match, and Random Permutation stayed within ±10% of the true value across conditions
  - ANOVA's bias grew to −23% under realistic decay + missingness
  - The 3 Nearest Neighbor approaches were the most biased overall (up to ±139%) — matching trials by similarity rather than exact features introduced systematic error

Recommendations for your own difference wave analysis

- **Default to LME Interaction or Exact Match — both unbiased in simulation**
  - Interaction: higher power, retains all trials/participants; but tests a predictor × condition interaction, not a literal difference wave — changes your interpretation
  - Exact Match: simpler interpretation (a true trial-level difference wave); pairs on all design features, so it retains fewer trials
- **Run a power analysis before data collection (especially if using Exact Match)**

06

Fitting models: Auditory oddball

Single-trial difference waves

An auditory oddball paradigm

What a participant hears: a frequent standard syllable, occasionally replaced by a rare deviant



S Standard

/ba/ syllable

80% of trials

600 trials per participant

D Deviant

/da/, 6 recordings/speakers/tokens

20% of trials

150 trials per participant (25 per recording/speaker)

▶ Passive listening

Silent movie with subtitles

No task

Auditory change detection is automatic

Simulation loosely based on Mon et al. (2024), Dev. Cogn. Neurosci., and Coy et al. (2024), Psychophysiology

An auditory oddball paradigm

Mismatch negativity (MMN): a negative deflection in the deviant − standard difference wave, ~100–250 ms after stimulus onset



Research question: What is the effect of age on the MMN?

A. Exact Match

Pair each deviant with the standard immediately preceding (e.g., Mon et al., 2024)

B. Interaction (Condition × Age)

MMN mean amplitude predicted by an interaction between age x condition (standard v. deviant)

Your turn: Steps 1–4

1. **Define the research question**
2. **List all relevant variables**
3. Draw the study design schematic
4. **Categorize variables by type & level (trial vs. subject)**
5. Assign fixed and random effects
6. Consider data transformations & center predictors (optional)
7. Fit maximal model and simplify until convergence is reached
8. Check model assumptions
9. Report & visualize results

Steps 1–4

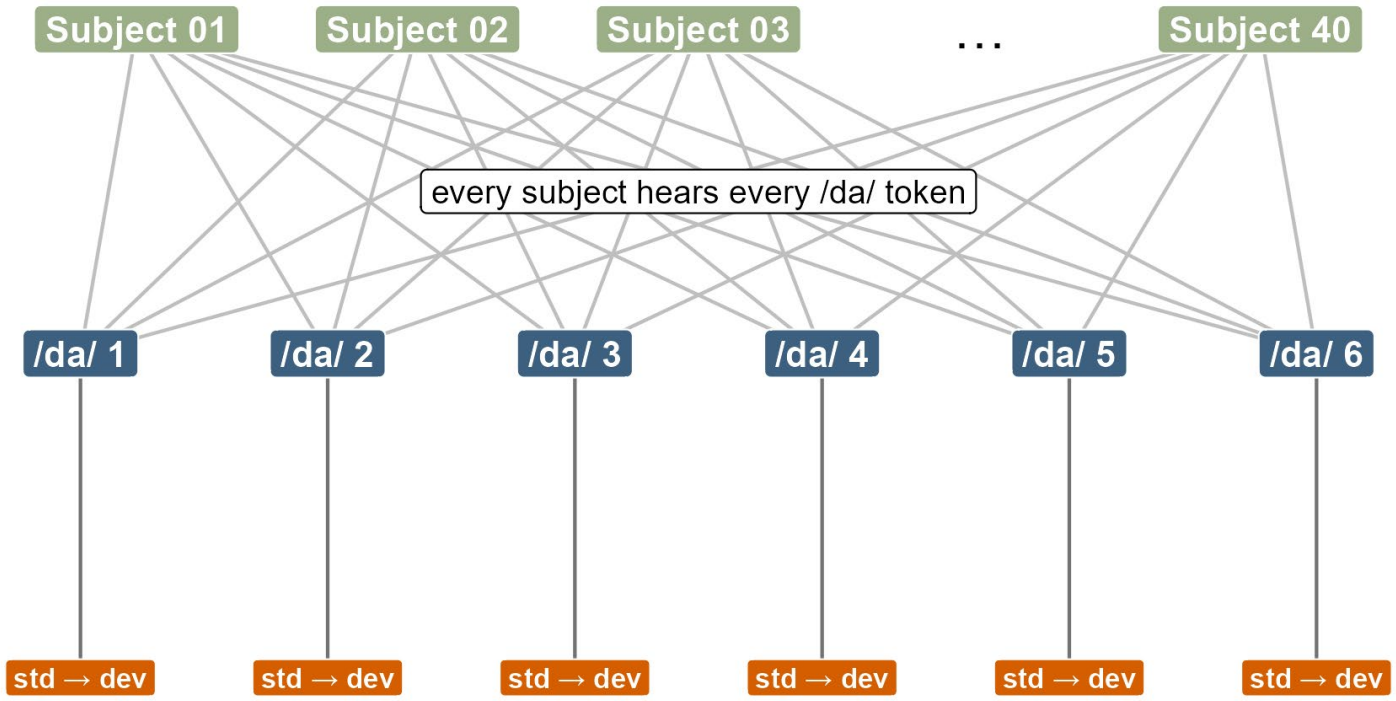
Step 1: Research question: Does the MMN change with age?

Steps 2 & 4: variables — type, level

Variable	What it is	Type, level
MMN	Deviant − the standard right before it (outcome)	continuous, trial-level
AgeYears	Participant's age	continuous, participant-level
DevNum	Deviant presentation number (1–150)	continuous, trial-level
Token	Which /da/ recording/speaker (6)	categorical, trial-level (grouping factor)
SubjectID	Participant	categorical, participant-level (grouping factor)

Simulation loosely based on Mon et al. (2024), Dev. Cogn. Neurosci., and Coy et al. (2024), Psychophysiology

Step 3: Draw the study design schematic



40 × 6

participants × /da/ tokens

every participant hears every token

25

deviants per token/speaker per participant

each paired with the standard before it

6,000

single-trial MMNs

40 × 150 deviant–standard pairs

Simulation loosely based on Mon et al. (2024), Dev. Cogn. Neurosci., and Coy et al. (2024), Psychophysiology

Your turn: Steps 1–4

1. Define the research question
2. List all relevant variables
3. Draw the study design schematic
4. Categorize variables by type & level (trial vs. subject)
5. **Assign fixed and random effects**
6. **Consider data transformations & center predictors (optional)**
7. **Fit maximal model and simplify until convergence is reached**
8. Check model assumptions
9. Report & visualize results

Your turn: Steps 4–7

4 **Categorize variables**

by type (categorical, numeric) & level (trial vs. subject)

5 **Assign fixed & random effects**

6 **Center & scale predictors**

(optional) so the intercept is meaningful, can also help with convergence

7 **Fit the maximal model, then simplify until it converges**

a **Scale & center predictors, and change the optimizer**

```
control = lmerControl(optimizer = "bobyqa", optCtrl =
list(maxfun = 20000))
```

b **Remove random slopes for interactions**

c **Remove main-effect random slopes**

Keep the slope most central to your question until last

d **Remove random intercepts**

Step 5: Assign fixed & random effects

5 Fixed & random effects

- Fixed: age (our question) and presentation number (a covariate).
- Random: participants (MMN size, habituation rate) and tokens. No age slope by participant (one age each).

Simulation loosely based on Mon et al. (2024), Dev. Cogn. Neurosci., and Coy et al. (2024), Psychophysiology

Step 5: Possible random effects in the oddball design

Grouping variable	How the data could theoretically vary by predictor _____	Syntax
Participant (SubjectID)	Do some participants have larger or smaller MMNs on average?	`(1 | SubjectID)`
	Does habituation of the MMN (change across deviants) differ across participants?	`(1 + DevNum | SubjectID)`
	Does the age effect differ across participants? Not possible: each child has only one age, so age is a fixed effect	✗ `(1 + Age | SubjectID)`
Token (/da/ recording)		
Participant × Token (optional)		

Simulation loosely based on Mon et al. (2024), Dev. Cogn. Neurosci., and Coy et al. (2024), Psychophysiology

Step 6: Consider transformations

6 Rescale & center

- When you fit the model or when you're looking at the variance for DevNum, you'll see that the coefficient for DevNum is really small, we could consider rescaling it in units of 10 deviants and centering (so that the intercept represents the middle of the experiment). On the 1–150 scale, the habituation slope and its SD are tiny (0.02 µV per deviant).
- We will center age to be the sample's mean age.

Simulation loosely based on Mon et al. (2024), Dev. Cogn. Neurosci., and Coy et al. (2024), Psychophysiology

Pair & share: which maximal model did you write?

Think · 1 min on your own → **Pair** · 3 min with a neighbor → **Share** · with the group

Two reasonable maximal models, depending on your hypothesis:

A Habituation is the same at every age

```
MMN ~ DevNum_gmc + Age_gmc +
      (1 + DevNum_gmc|SubjectID) +
      (1 + DevNum_gmc + Age_gmc|Token)
```

Tokens can differ in their MMN, habituation, and age effect.

B Habituation rate changes with age

```
MMN ~ DevNum_gmc * Age_gmc +
      (1 + DevNum_gmc|SubjectID) +
      (1 + DevNum_gmc * Age_gmc|Token)
```

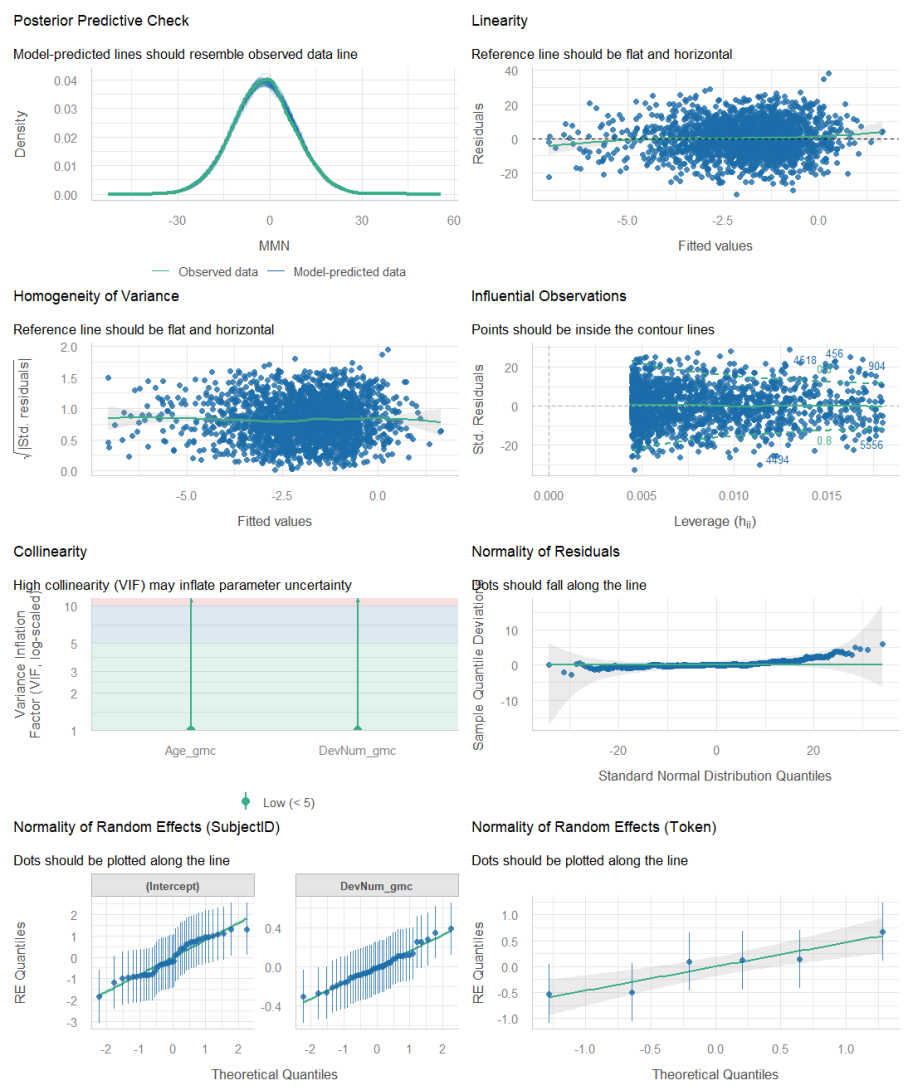
DevNum × Age: do older children's MMNs habituate faster or slower? Tokens could also differ in that interaction.

Simulation loosely based on Mon et al. (2024), Dev. Cogn. Neurosci., and Coy et al. (2024), Psychophysiology

Your turn: Step 8

1. Define the research question
2. List all relevant variables
3. Draw the study design schematic
4. Categorize variables by type & level (trial vs. subject)
5. Assign fixed and random effects
6. Consider data transformations & center predictors (optional)
7. Fit maximal model and simplify until convergence is reached
8. **Check model assumptions**
9. Report & visualize results

Step 8: checking the oddball model's assumptions



✓ **Posterior predictive check**
Simulated MMNs match the observed distribution

✓ **Linearity**
Reference line flat at 0

✓ **Homogeneity of variance**
Flat line, even spread across fitted values

✓ **Influential observations**
All points inside the Cook's distance contours

✓ **Collinearity**
VIFs = 1.00: age varies between participants, deviant number within

~ **Normality of residuals**
Dots on the line; a slightly heavy upper tail

~ **Normality of random effects**
Participants (intercepts and slopes) on the line; only 6 tokens, so hard to judge

check_heteroscedasticity() (check for homogeneity of variance) says $p < .001$, but the plot's line is pretty flat (can be overly sensitive to large numbers of trials)

Simulation loosely based on Mon et al. (2024), Dev. Cogn. Neurosci., and Coy et al. (2024), Psychophysiology

Step 9: interpreting the model

Table 1.

Predictor	*b*	*SE*	95% CI	*t*	df	*p*	β
Intercept	−1.89	0.30	[−2.56, −1.22]	−6.34	9.57	< .001	
Deviant number (per 10)	0.12	0.04	[0.03, 0.21]	2.76	38.96	.009	0.05
Age (years, centered)	−0.08	0.38	[−0.86, 0.69]	−0.22	38	.829	−0.00

Random effects (bottom of tab_model())			
σ^2 (residual)	98.88	τ_{00} SubjectID	1.07
τ_{11} SubjectID.DevNum_gmc	0.05	ρ_{01} SubjectID	−0.45
τ_{00} Token	0.28	ICC	0.02
N SubjectID	40	N Token	6
Observations	6,000	Marginal / Conditional R^2	0.003 / 0.025

Intercept: the MMN mid-session at average age, −1.89 µV (simulated −1.9); df ≈ 10 with only 6 tokens

Habituation: 0.12 µV less negative every 10 deviants (simulated 0.15), ~1.9 µV weaker by the end

Our question: −0.08 µV per year, p = .83: no evidence of an age effect (simulated −0.5; N = 40)

Subject-level variability(σ^2 = 98.88, SD ≈ 9.9 µV). Participants' MMNs differ: SD = √1.07 ≈ 1.03 µV

Habituation differs: SD = √0.05 ≈ 0.21 µV per 10 deviants; r = −0.45 with MMN size

Tokens: SD = √0.28 ≈ 0.52 µV. ICC = 0.02: most variance is trial-to-trial noise

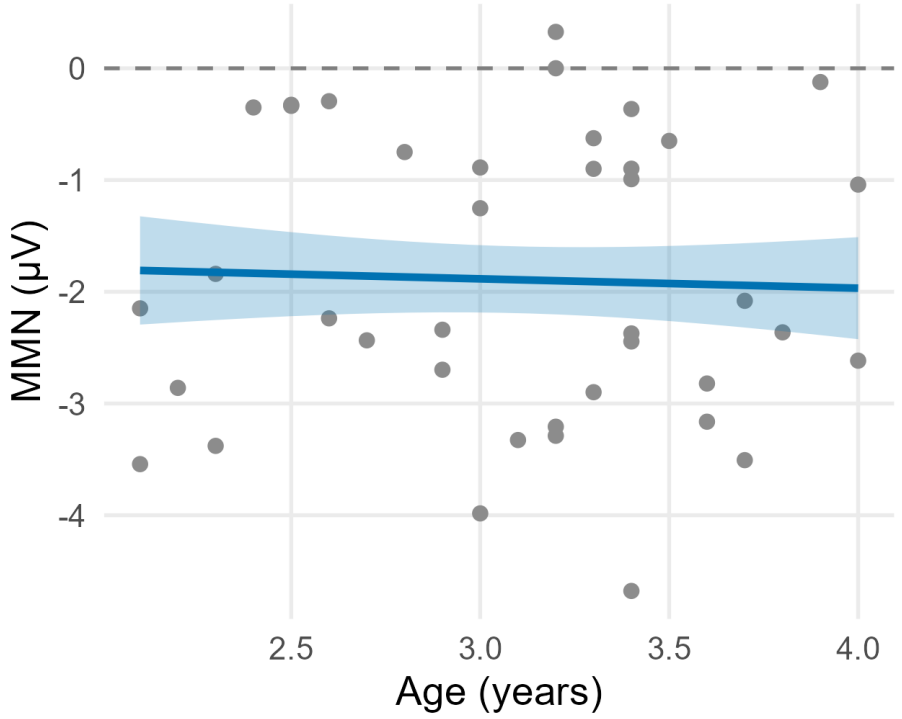
Research question: no evidence that the MMN changes with age in this sample (potentially because we were underpowered….!)

Simulation loosely based on Mon et al. (2024), Dev. Cogn. Neurosci., and Coy et al. (2024), Psychophysiology

Step 9: plotting the oddball results

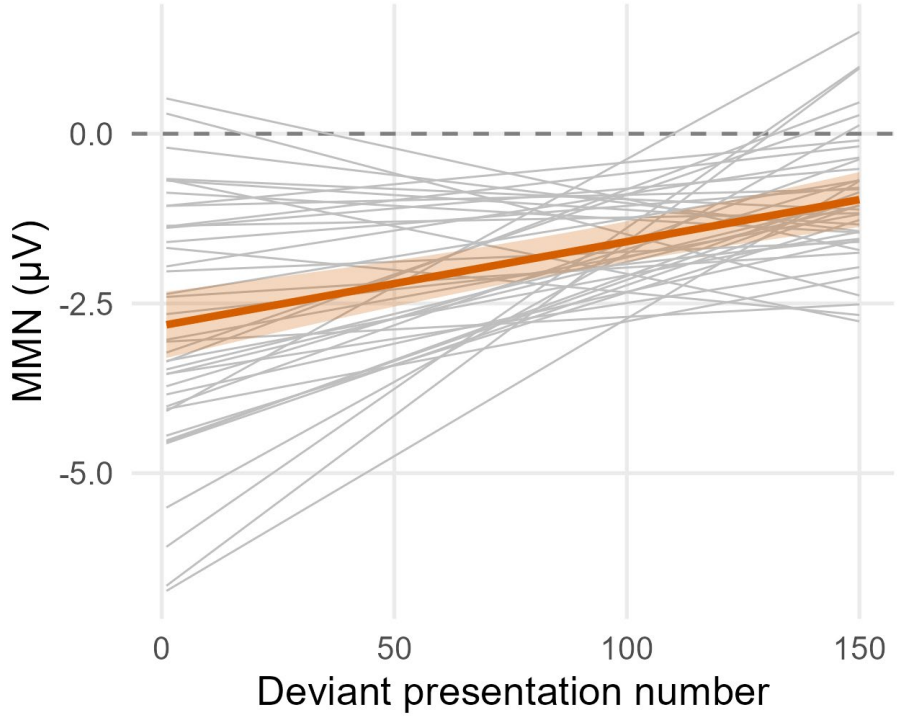
The MMN across age (our question)

EMM ± 1 SE (blue, mid-session); gray = each participant's mean MMN



The MMN habituates

EMM ± 1 SE (orange); gray = each participant's model-estimated line

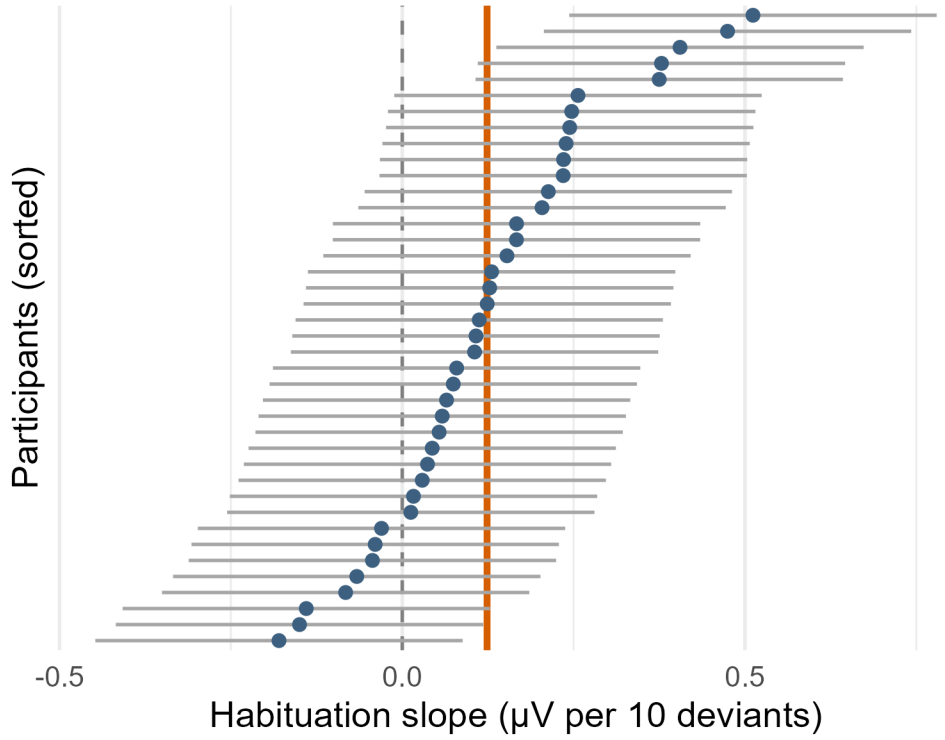


Simulation loosely based on Mon et al. (2024), Dev. Cogn. Neurosci., and Coy et al. (2024), Psychophysiology

Step 9: plotting the random slopes

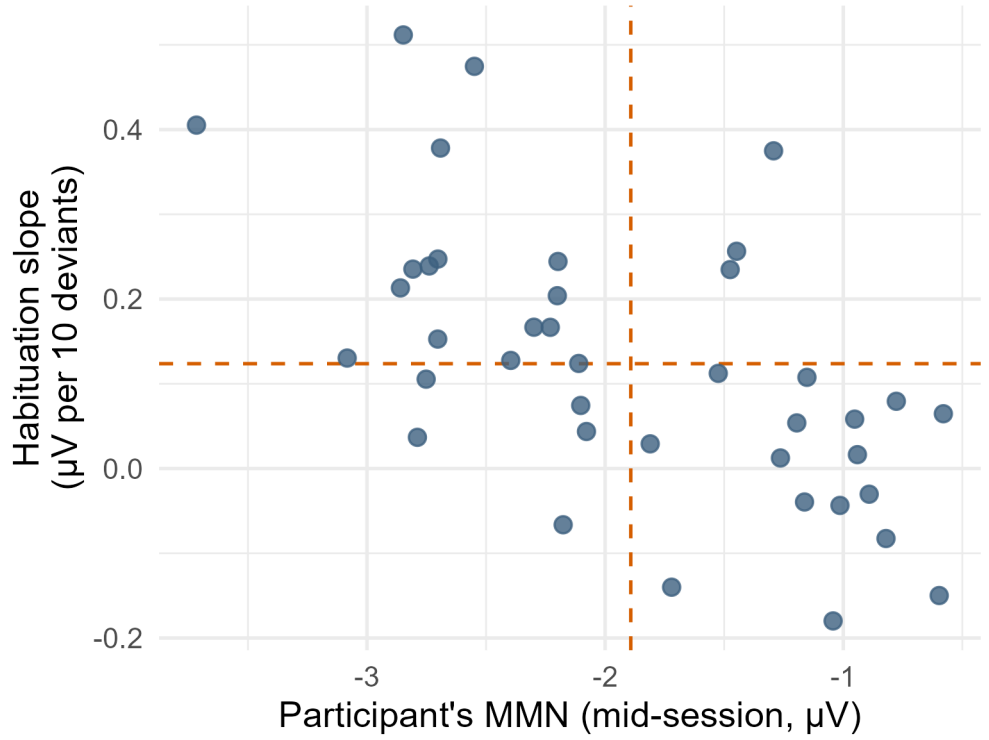
Each participant's habituation slope

Fixed + random slope, sorted; bars = 95% interval; orange = average; dashed = 0



Participants' MMN vs. their habituation

Each point = one participant; dashed orange = fixed effects



Most children's MMN habituates, but by different amounts: 32 of 40 slopes are above 0 (average 0.12, SD 0.21 µV per 10 deviants). Children with larger MMNs habituate faster (r = −0.45).

Simulation loosely based on Mon et al. (2024), Dev. Cogn. Neurosci., and Coy et al. (2024), Psychophysiology

07

Planning data collection for LME analysis

- ***Plan for trial-level covariates***
- *Unique event markers*
- *Pre-registration*
- *Power analyses*

Setting up experiments for LME

- Measures to plan to collect:
  - Trial-level covariates
    - E.g., reaction time, accuracy, looking/attention coding (video for infants), time since the last break
  - Session-level covariates
    - E.g., cap size, time of day, research assistant (RA), fussiness ratings
  - Other ideas?

07

Planning data collection for LME analysis

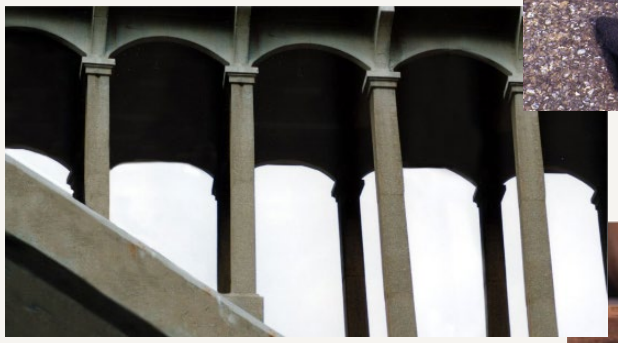
- *Plan for trial-level covariates*
- ***Unique event markers***
- *Pre-registration*
- *Power analyses*

Setting up experiments for LME

Neutral valence









Positive valence

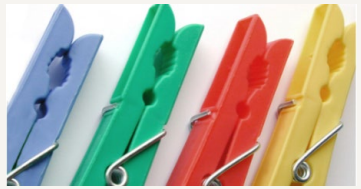
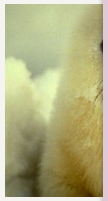
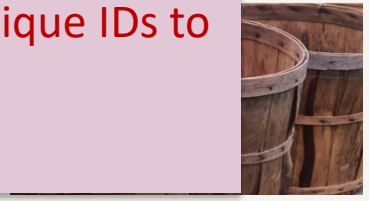
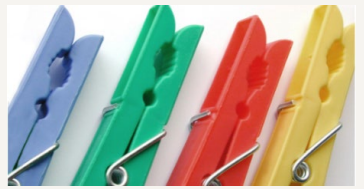








Setting up experiments for LME

	Participant 1	Participant 2	Participant 3
Trial 1	 1	 1	 2
Trial 2	 1	 1	 1
Trial 3			 1
Trial 4	 1	 2	 2
Trial 5	 2	 1	 1

If we present our stimuli in a randomized order, we don't have unique IDs to characterize our stimuli (e.g., for our random effects)

Setting up experiments for LME

Trial 1



11



14



24

Trial 2



12

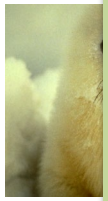


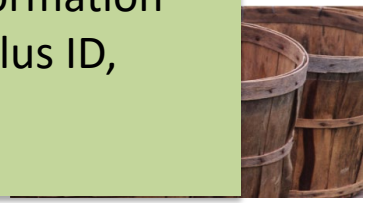
11



14

Trial 3





13

If each stimulus has a unique event marker, we can recover information needed for our model (e.g., trial presentation number, stimulus ID, condition)

Trial 4



13



21



23

Trial 5



22



13



11

Unique trial markers

- Give every trial a unique event-marker code that captures condition, stimulus identity, and presentation number
- Unique trial-level codes let you identify emotion, actor, and presentation number for every single trial — essential for LME, and also useful for flagging systematically outlying stimuli during pre-processing



Unique event markers



Stimulus computer
(e.g., PsychoPy, E-Prime, Presentation, Psychtoolbox)







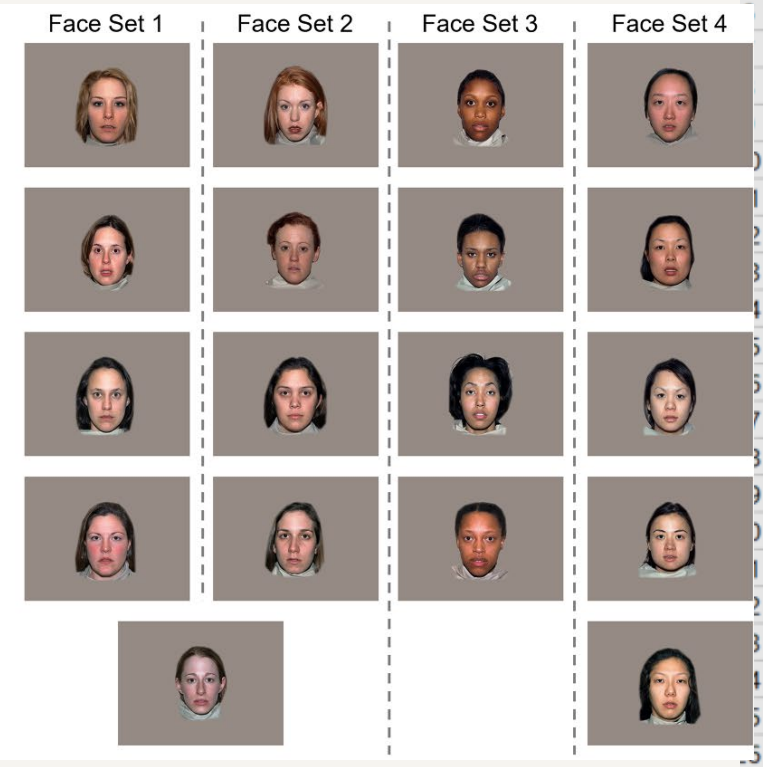
EEG computer
(e.g., BrainVision, Net Station)

The unique markers/flags that are sent to the EEG computer are limited by bits available in the Trigger Box
(8 bits = 0-255)

Unique event markers

- Think about what information can be recovered in post-processing (e.g., trial presentation number, stimulus set)
- I would suggest some combination of condition (e.g., first digit) and stimulus

Unique event markers



	A	B	C	D	E	F	G	H
1	Stim Image	Value Label	Numerical Value	Emotion	EmotionCode	Model	NewPreceedingCode	TrialNumber
2	[st1]A40.jpg	Angry4	40	Angry40	2	21	221	01-10
3	[st2]A40.jpg	Angry4	41		2	22	222	
4	[st3]A40.jpg	Angry4	42		2	23	223	
5	[st4]A40.jpg	Angry4	43		2	24	224	
	[st5]A40.jpg	Angry4	44		2	25	225	
	[st1]A100.jpg	Angry	50	Angry100	3	21	321	
	[st2]A100.jpg	Angry	51		3	22	322	
	[st3]A100.jpg	Angry	52		3	23	323	
	[st4]A100.jpg	Angry	53		3	24	324	
	[st5]A100.jpg	Angry	54		3	25	325	
	[st1]F40.jpg	Fear4	80	Fear40	4	21	421	
	[st2]F40.jpg	Fear4	81		4	22	422	
	[st3]F40.jpg	Fear4	82		4	23	423	
	[st4]F40.jpg	Fear4	83		4	24	424	
	[st5]F40.jpg	Fear4	84		4	25	425	
	[st1]F100.jpg	Fear	90	Fear100	5	21	521	
	[st2]F100.jpg	Fear	91		5	22	522	
	[st3]F100.jpg	Fear	92		5	23	523	
	[st4]F100.jpg	Fear	93		5	24	524	
	[st5]F100.jpg	Fear	94					
	[st1]H100.jpg	Happy	10	Happy100				
	[st2]H100.jpg	Happy	11					
	[st3]H100.jpg	Happy	12					
	[st4]H100.jpg	Happy	13					
	[st5]H100.jpg	Happy	14					
27	[st1]N00.jpg	Neutral	20	Neutral10				
28	[st2]N00.jpg	Neutral	21					
29	[st3]N00.jpg	Neutral	22		8	23	823	
30	[st4]N00.jpg	Neutral	23		8	24	824	
31	[st5]N00.jpg	Neutral	24		8	25	825	

The unique event markers need to be inserted before artifact rejection

Setting up an experiment for LME

Pre-processing

Insert trial-level event markers

1. Add channel locations
2. Filter

Artifact rejection

1. Continuous rejection by eye
2. Channel interpolation
3. ICA
4. Blink correction
5. Re-reference
6. Epoch
7. Simple voltage threshold & sample-to-sample to reject individual trials

Mean averaging

1. Mean average each condition
2. Export participants' CONDITION-LEVEL mean amplitude for data analysis

LME analyses

1. Export participants' TRIAL-LEVEL mean amplitude for data analysis

Epoching

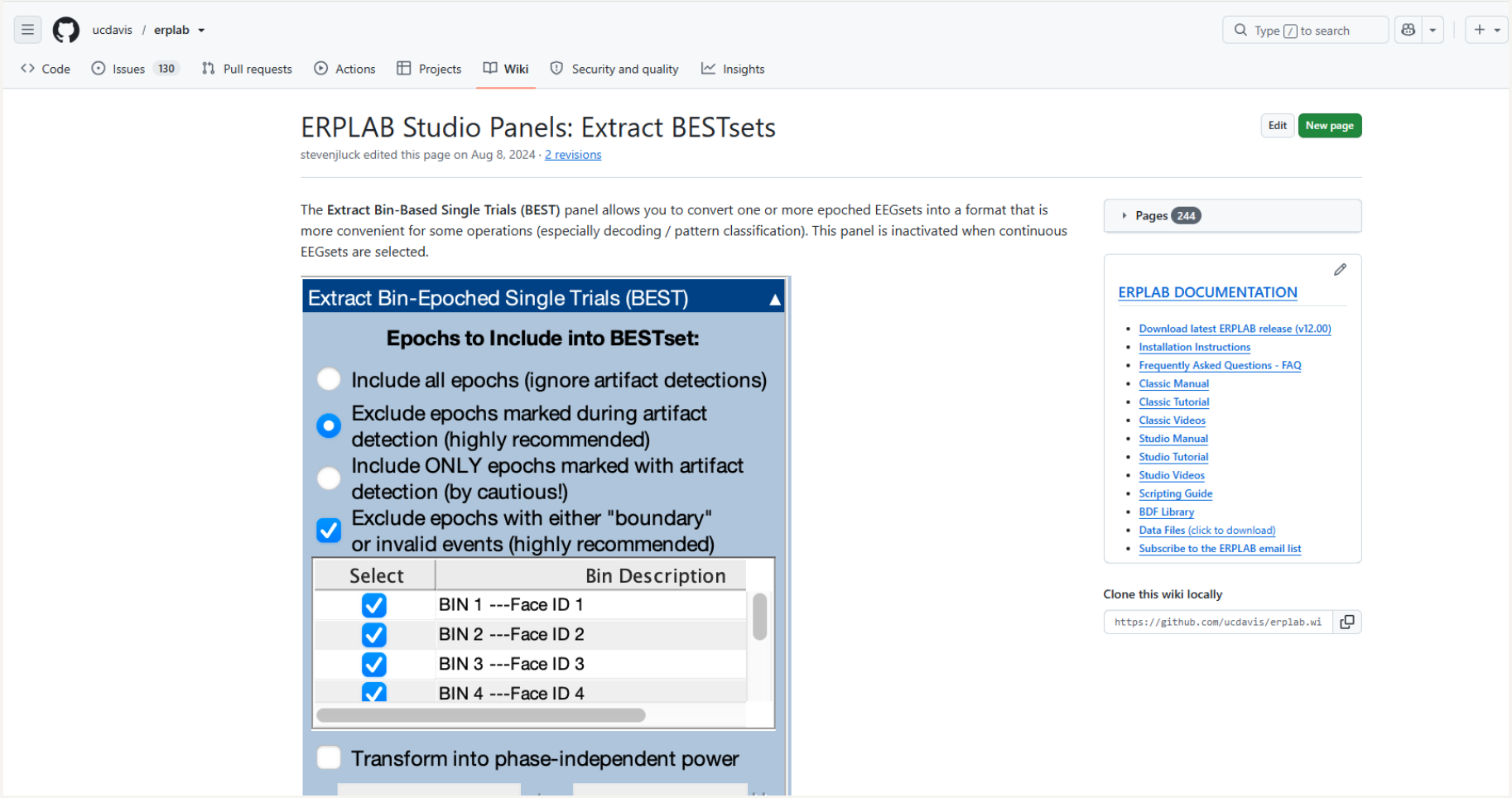
If you're epoching in ERPLAB you also need to assign unique bins to each of these unique flags

Or….

	A	B	C
1	binNumber	eventMarker	description
2	1	20101;20102;20103;20104;20105;20106;20107;20108;20109;20110;20201;202C	Angry40
3	2	30101;30102;30103;30104;30105;30106;30107;30108;30109;30110;30201;302C	Angry100
4	3	40101;40102;40103;40104;40105;40106;40107;40108;40109;40110;40201;402C	Fear40
5	4	50101;50102;50103;50104;50105;50106;50107;50108;50109;50110;50201;502C	Fear100
6	5	60101;60102;60103;60104;60105;60106;60107;60108;60109;60110;60201;602C	Happy100
7	6	80101;80102;80103;80104;80105;80106;80107;80108;80109;80110;80201;802C	Neutral100
8	7	20101;20102;20103;20104;20105;20106;20107;20108;20109;20110;20201;202C	AllEmotion
9	8	20101	20101
10	9	20102	20102
11	10	20103	20103
12	11	20104	20104
13	12	20105	20105
14	13	20106	20106
15	14	20107	20107
16	15	20108	20108
17	16	20109	20109
18	17	20110	20110
19	18	20201	20201
20	19	20202	20202
21	20	20203	20203
22	21	20204	20204

Epoching

The new version of ERPLab allows you to export trial-level data more easily, but you have to remember to assign trial presentation number before artifact rejection for it to work



Setting up an experiment for LME

Pre-processing

Insert trial-level event markers

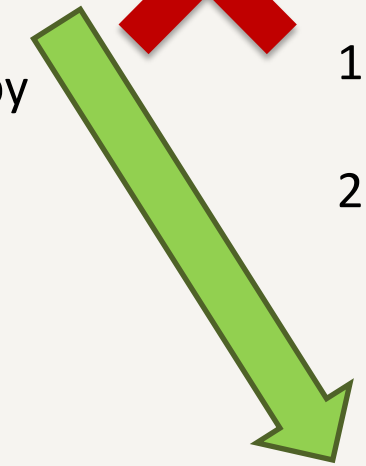
1. Add channel locations
2. Filter



Artifact rejection

1. Continuous rejection by eye
2. Channel interpolation
3. ICA
4. Blink correction
5. Re-reference
6. Epoch
7. Simple voltage threshold & sample-to-sample to reject individual trials





Mean averaging

1. Mean average each condition
2. Export participants' CONDITION-LEVEL mean amplitude for data analysis

LME analyses

1. Export participants' TRIAL-LEVEL mean amplitude for data analysis

07

Planning data collection for LME analysis

- *Plan for trial-level covariates*
- *Unique event markers*
- ***Pre-registration***
- *Power analyses*

What do you currently pre-register now, or what would you pre-register for a new experiment? What additional things would you pre-register for an LME analysis?

Nobody has responded yet.

Hang tight! Responses are coming in.

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Pre-register: The dependent variable

- **Measurement:** mean amplitude (or peak, or latency) and how it's computed
- **Time window(s):** if a window hasn't been selected yet, you can pre-register how it will be picked, e.g., from the grand-mean waveform across the full sample, agnostic to condition
- **Electrodes**

Pre-register: Missing data & trial inclusion

- **Artifact rejection thresholds**
- **How you will check for missingness patterns/ how you will model missing data:** does it relate to condition, trial number, or participant characteristics?

Pre-register: Model specification

- **Model formula**: Wilkinson notation may be especially useful here
- **Random effects:** the maximal structure the design supports and how you will simplify if the maximal model doesn't converge
- **Software:** package versions, optimizer

Pre-register: Inference

- Alpha level and directionality
- **p-values:** Satterthwaite or Kenward-Roger df, likelihood-ratio tests, or parametric bootstrap
- **Multiple comparisons:** correction across components, windows, or follow-up contrasts
- **Interactions:** how they'll be followed up (emmeans / emtrends, which contrasts, correction)
- **Effect sizes:** standardized betas?

Pre-register: Assumption checks & outliers

- **Checks to run:** e.g., performance::check_model()
- **Outliers:** definition at the trial and participant level, and handling (keep, winsorize, robust model)
- **Transformations:** whether you'll transform mean amplitude or predictors, and under what conditions

08

Power analyses for LMEs

The hard way & the easy way

How do you usually decide how many participants to recruit?

A power calculator (e.g., G*Power) 0%

A rule of thumb (e.g., match or double the N of published studies) 0%

Whatever we can afford or recruit (budget, time, and participant pool constraints) 0%

Simulating data 0%

I haven't had to plan a sample size yet 0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

When you run a power analysis, where does the effect size come from?

Effect sizes published in the literature

0%

My own pilot data

0%

The smallest effect I'd care about detecting

0%

A conventional benchmark (e.g., a "medium" effect, d = .5)

0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Which of these does your power analysis account for?

Number of trials per condition 0%

Trials lost to artifact rejection 0%

Number of stimuli 0%

Participants differing in the size of the effect 0%

None of the above 0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Power analyses the harder way: simulate ERP waveforms

Instead of simulating a single value for mean amplitude, simulate EEG: build single-trial waveforms, extract mean amplitude, and analyze the number of simulated datasets where you detect your effect of interest

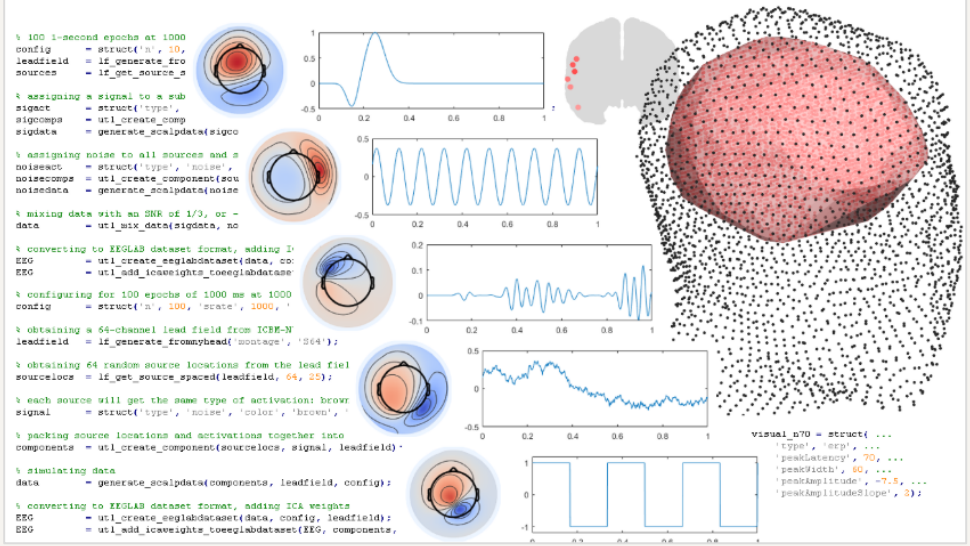
SEREEGA: Simulating Event-Related EEG Activity

SEREEGA is a modular, general-purpose open-source MATLAB-based toolbox to generate simulated toy data mimicking event-related electroencephalography (EEG) data.

Reference:

- Krol, L. R., Pawlitzki, J., Lotte, F., Gramann, K., & Zander, T. O. (2018). SEREEGA: Simulating Event-Related EEG Activity. *Journal of Neuroscience Methods, 309*, 13-24.

(See also this PDF mirror, or this bioRxiv preprint.)



1. **Simulate** single-trial ERP waveforms with known effects in SEREEGA, a free MATLAB toolbox (EEGLAB plug-in):
   head model → brain source → ERP signal (latency, width, amplitude) → project to scalp
2. **Add noise** real resting-state EEG from infants or preschoolers
3. **Measure** extract single-trial mean amplitude from the component window
4. **Fit** the LME
5. **Repeat** 1,000 simulated studies → power = % with a significant effect

Krol, Pawlitzki, Lotte, Gramann & Zander (2018), J. Neurosci. Methods, 309, 13–24; github.com/lrkrol/SEREEGA; Heise, Mon & Bowman (2022; 2025), Dev. Cogn. Neurosci.

How we simulated the Nc in SEREEGA

Ingredient	What we used (Heise et al., 2022; 2025)
Lead field	Pediatric Head Atlas, Atlas 1 (0–2 years; Song et al., 2013)
Source	Nc dipole at the prefrontal ICA cluster from Reynolds & Richards (2005)
Condition means	2022: Happy −10 vs. Angry −12 µV (SD 5) · 2025: Happy − Neutral = ±2 µV by group
Habituation	1.5 µV less negative per repeated presentation (Borgström et al., 2016)
Random effects	Subject intercepts (SD 10 µV) and actor intercepts (SD 5 µV)
Noise	2022: pink Gaussian noise · 2025: real infant resting-state EEG
Scale	2025: 48 subjects × 100 trials, 1,000 simulated datasets

SEREEGA's trial-level knobs

`peakAmplitude`
the ERP's size

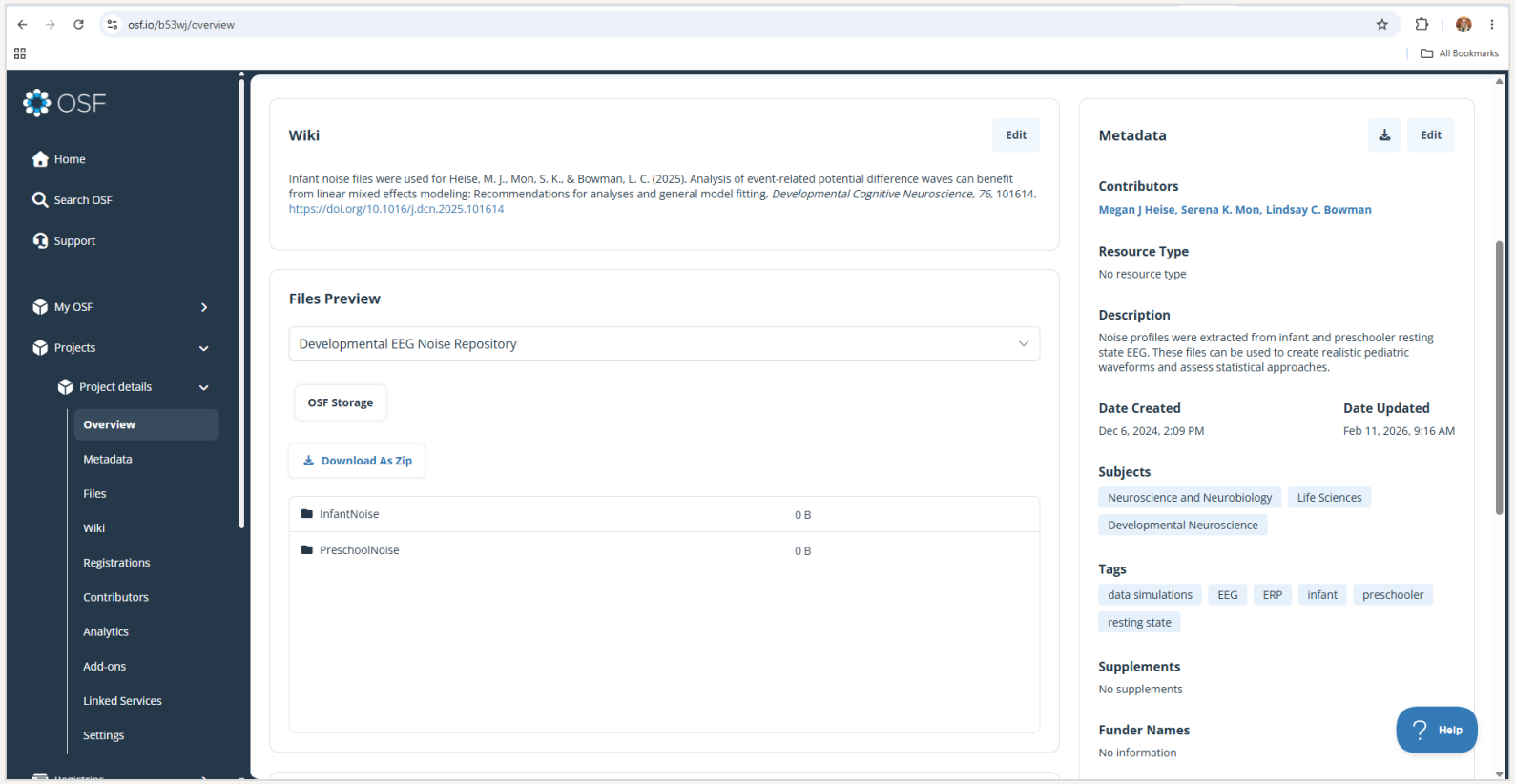
`peakAmplitudeDv`
trial-to-trial variability in size

`peakAmplitudeSlope`
linear change across trials, e.g., habituation

`peakLatency / peakLatencyDv`
timing and latency jitter

`noise class`
white, pink, brown… or your own recorded EEG

Realistic noise: a developmental EEG noise library



osf.io/b53wj

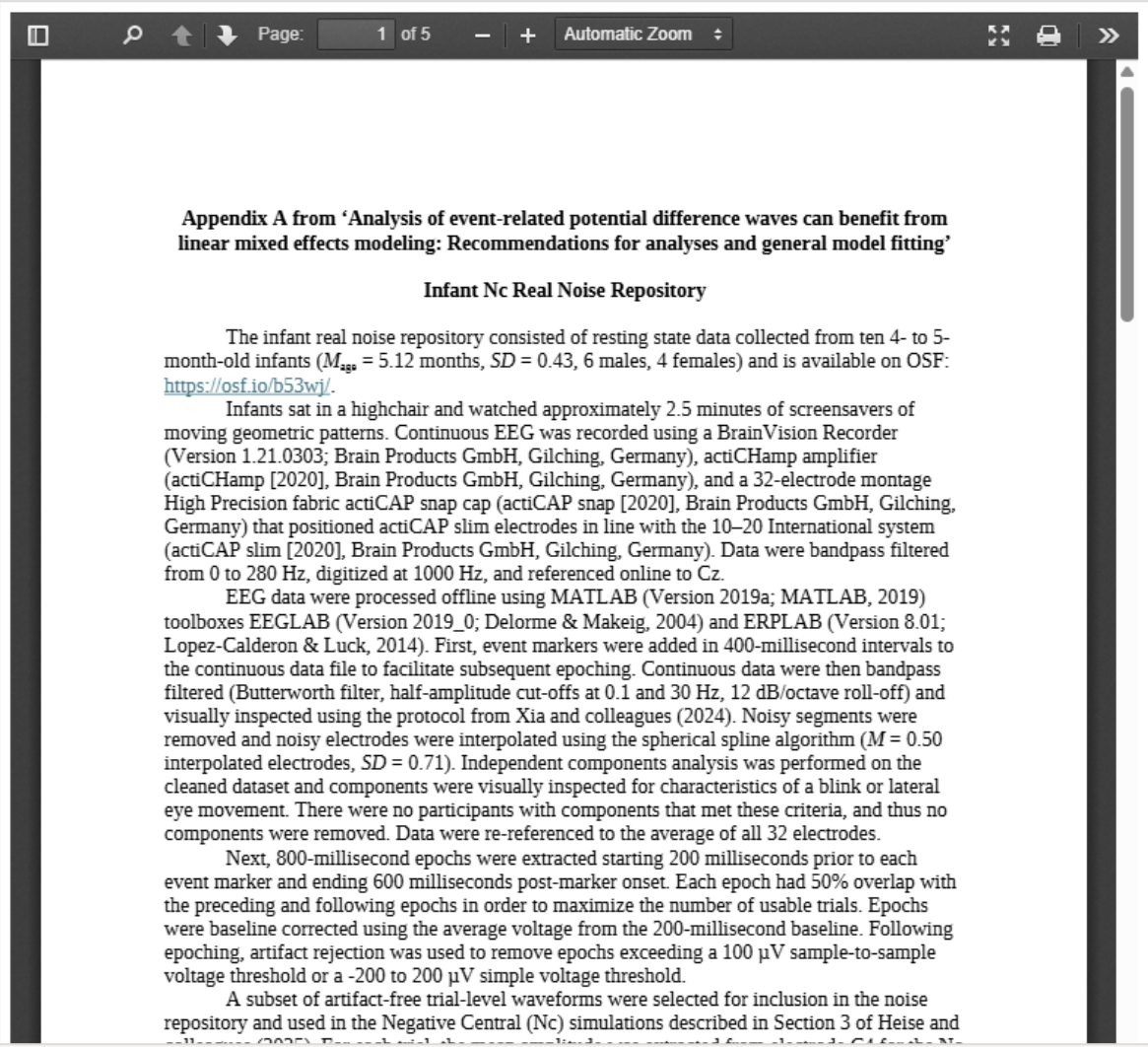
What: resting-state EEG from infants and preschoolers, cut into epochs you can add to simulated waveforms

Why: real infant EEG has the spectrum, spatial pattern, and artifacts that synthetic (e.g., pink) noise lacks

How we used it: each simulated subject was assigned 1 of 10 infant noise profiles, and its noise epochs were added to the simulated Nc trials

The Developmental EEG Noise Repository on OSF has infant and preschooler files for your own simulations.

Inside the infant noise files



Appendix A from 'Analysis of event-related potential difference waves can benefit from linear mixed effects modeling: Recommendations for analyses and general model fitting'

Infant Nc Real Noise Repository

The infant real noise repository consisted of resting state data collected from ten 4- to 5-month-old infants (M_{age} = 5.12 months, *SD* = 0.43, 6 males, 4 females) and is available on OSF: https://osf.io/b53wj/.

Infants sat in a highchair and watched approximately 2.5 minutes of screensavers of moving geometric patterns. Continuous EEG was recorded using a BrainVision Recorder (Version 1.21.0303; Brain Products GmbH, Gilching, Germany), actiCHamp amplifier (actiCHamp [2020], Brain Products GmbH, Gilching, Germany), and a 32-electrode montage High Precision fabric actiCAP snap cap (actiCAP snap [2020], Brain Products GmbH, Gilching, Germany) that positioned actiCAP slim electrodes in line with the 10–20 International system (actiCAP slim [2020], Brain Products GmbH, Gilching, Germany). Data were bandpass filtered from 0 to 280 Hz, digitized at 1000 Hz, and referenced online to Cz.

EEG data were processed offline using MATLAB (Version 2019a; MATLAB, 2019) toolboxes EEGLAB (Version 2019_0; Delorme & Makeig, 2004) and ERPLAB (Version 8.01; Lopez-Calderon & Luck, 2014). First, event markers were added in 400-millisecond intervals to the continuous data file to facilitate subsequent epoching. Continuous data were then bandpass filtered (Butterworth filter, half-amplitude cut-offs at 0.1 and 30 Hz, 12 dB/octave roll-off) and visually inspected using the protocol from Xia and colleagues (2024). Noisy segments were removed and noisy electrodes were interpolated using the spherical spline algorithm (*M* = 0.50 interpolated electrodes, *SD* = 0.71). Independent components analysis was performed on the cleaned dataset and components were visually inspected for characteristics of a blink or lateral eye movement. There were no participants with components that met these criteria, and thus no components were removed. Data were re-referenced to the average of all 32 electrodes.

Next, 800-millisecond epochs were extracted starting 200 milliseconds prior to each event marker and ending 600 milliseconds post-marker onset. Each epoch had 50% overlap with the preceding and following epochs in order to maximize the number of usable trials. Epochs were baseline corrected using the average voltage from the 200-millisecond baseline. Following epoching, artifact rejection was used to remove epochs exceeding a 100 µV sample-to-sample voltage threshold or a -200 to 200 µV simple voltage threshold.

A subset of artifact-free trial-level waveforms were selected for inclusion in the noise repository and used in the Negative Central (Nc) simulations described in Section 3 of Heise and

Who	10 infants, 4–5 months old (M = 5.12 months)
Task	~2.5 min of resting state: watching screensavers
Recording	32-channel actiCAP, BrainVision actiCHamp, 1000 Hz
Cleaning	0.1–30 Hz filter, bad channels interpolated, ICA for blinks and eye movements, average reference
Epochs	800 ms (−200 to 600 ms), 50% overlap, ±100 µV artifact rejection
Result	artifact-free noise trials to add to simulated waveforms

Today: power analysis the easier way

SEREEGA for power analyses

Simulate whole waveforms

- Benefits are that you can include real EEG "noise" and look at other ERP processing decisions (e.g., impact of measurement window)
- Many parameters; slow; coding intensive
- Maybe best for methods papers and testing new pipelines

Simulations in R for power analyses

Simulate mean amplitudes

- Benefits are that it's fast; easy to vary N, trials, and stimuli
- Models effects, random effects, and trial-level noise as numbers
- Easier for planning your next study

What is statistical power?

Power = 1 – β: the probability your study detects an effect when there is an effect in the population

	True effect	No effect
Significant	**Power** 1 – β	**Type I error** α (false positive)
Not significant	**Type II error** β (false negative)	**True negative** 1 – α

probability of a Type II error

Power depends on…

- **Effect size**
  e.g., the Happy vs. Angry amplitude difference
- **Number of participants**
- **Number of trials / stimuli**
- **Variability**
  in an LME, every variance component: participants, stimuli, random slopes, trial noise
- **Alpha and # of comparisons**
  stricter thresholds lower power

Formula-based power tool for LMEs

Westfall, Kenny & Judd (2014) app: https://jakewestfall.shinyapps.io/crossedpower/

- Analytic power for designs where a sample of participants responds to a sample of stimuli
- Inputs: effect size, numbers of participants and stimuli, and the **VPCs**
- Solve for any one unknown by entering "X" (e.g., how many stimuli for 80% power?)
- Designs: fully crossed, counterbalanced, stimuli or participants (or both) within condition

VPC = variance partitioning coefficient

The share of the total variance that comes from each source: participants, stimuli, their random slopes for condition, and the residual

With standardized input, the VPCs must sum to 1

Same idea as the ICC: e.g., our Tutorial 1 model put ≈ 54% of the variance between children

Useful when your design is simple [e.g., one fixed effect with two conditions (e.g., Angry vs. Happy), participants and stimuli as random factors]

How a simulation-based power analysis works

1 Simulate

a new dataset from a model with known effects and variances

2 Fit & test

your planned LME. Is the effect significant (|t| > 2)? [approx. *p*<.05, see Kumle et al. 2021]

3 Repeat

1,000+ times. Power = % of simulations that were significant

repeat

Then do it again for each candidate design

N = 10 | N = 20 | N = 30 | N = 40 → **power curve**

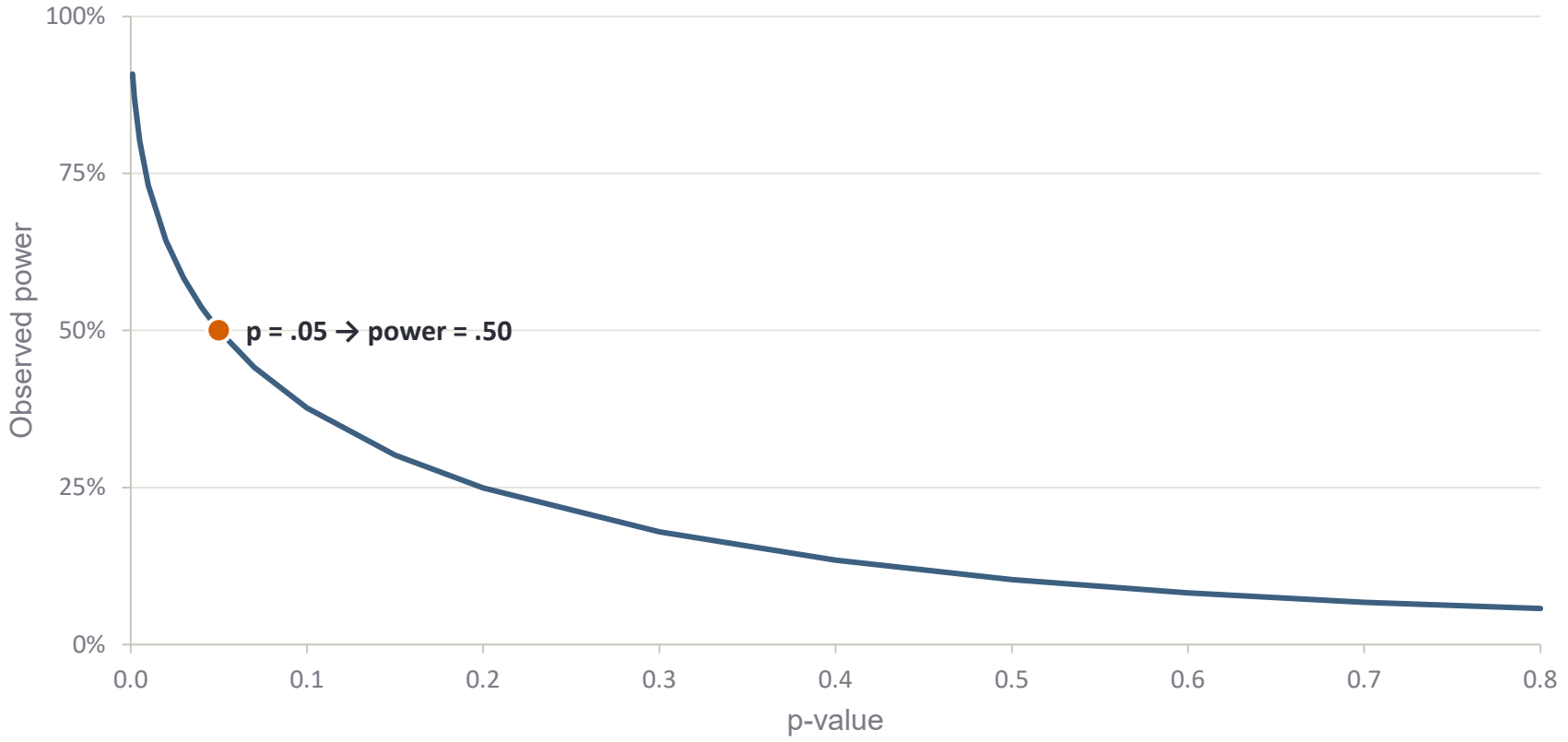
You can vary participants, trials, stimuli, effect size, or variance components



Why post hoc power analyses are bad

Post hoc power is the p-value

"Observed" power for a two-sided test (α = .05), computed from the p-value alone



Tutorial 3: Power analysis in R

Open: `LME_tutorial3_power.html`

Design 1

The N400 in dyslexic and typical adults

Starting from published results (ERP CORE, Kappenman et al., 2021, and a meta-analysis, Basma et al., 2024): how many participants for the N400 effect, and for a group difference between dyslexic and non-dyslexic participants?

Design 2

Planning a new study accounting for artifact rejection and number of stimuli

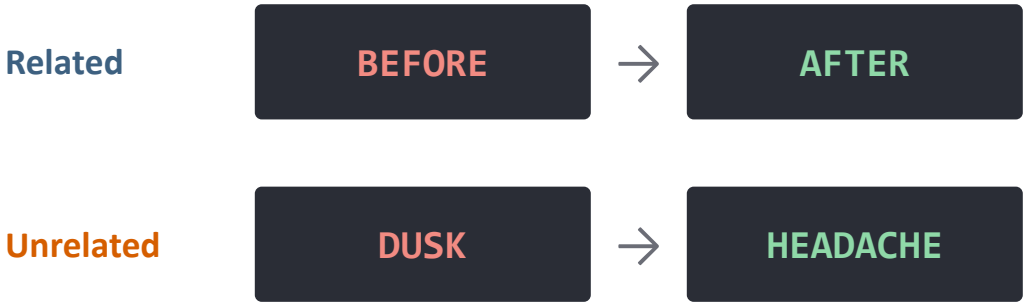
Trials per condition × artifact rejection (simulation loop with faux); how many stimuli (mixedpower over actors, R2power())

Our walkthrough today is based on the mixedpower notebooks (which also includes more example tutorials!): **lkumle.github.io/power_notebooks**

Design 1: the N400 in dyslexic and typical adults

Modeled after ERP CORE (Kappenman et al., 2021), with the group difference from a meta-analysis (Basma et al., 2024)

The task: ERP CORE word pair judgment



Prime (red) then target (green): "are these words related?"

60 related + 60 unrelated pairs per participant; each target word appears in both conditions

N400: mean amplitude at CPz, 300–500 ms

Example pairs from the ERP CORE stimulus lists

ERP CORE (Kappenman et al., 2021)

39 typical adults: unrelated − related = **−7.61 µV** (SD 3.27; d_z = 2.33)

Meta-analysis (Basma et al., 2024)

20 studies: the N400 differs between typical and dyslexic readers, **g = 0.66**

Kappenman et al. (2021), NeuroImage; Basma, Savage & Bertone (2024), Int. J. Psychophysiol.

Design 1: Research questions

1 The N400 effect

Unrelated vs. related targets: a within-participant effect

SESOI: −5.5 µV

ERP CORE estimate (−6.5 µV averaged over groups), 15% smaller

2 Condition × Group

Is the N400 effect smaller in dyslexic adults?

SESOI: 1.9 µV

g = 0.66 × 3.27 µV ≈ 2.2 µV, 15% smaller

```
MeanAmp ~ Condition_e * Group_e +
          (1 + Condition_e | SubjectNum) + (1 | ItemNum)
```

Condition_e: related = −0.5, unrelated = +0.5: the N400 effect, averaged over groups

Group_e: typical = −0.5, dyslexic = +0.5

Condition_e:Group_e: dyslexic minus typical N400 effect (positive = smaller in dyslexic readers)

Random effects: participants differ in overall amplitude and in their N400 effect; some target words elicit a more negative or less negative N400 than others overall

Pilot: 40 adults (20 per group, like ERP CORE) × 60 target words × 2 conditions = 120 trials each

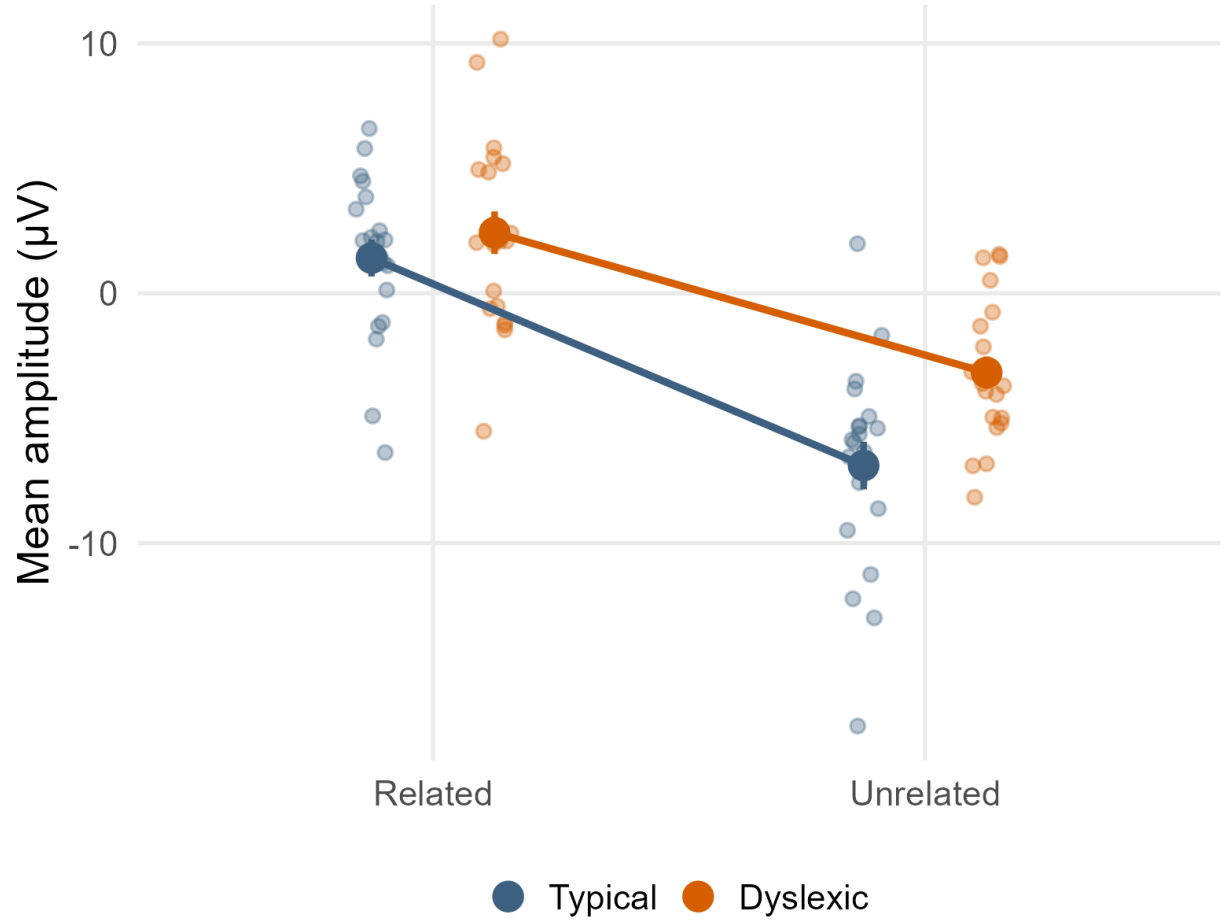
Kappenman et al. (2021), NeuroImage; Basma, Savage & Bertone (2024), Int. J. Psychophysiol.

Power analysis: Design 1

Parameter	Value	Reference
N400 effect, typical readers	−7.61 μV	Kappenman et al. (2021)
Group difference in the N400 effect	2.2 μV	g = 0.66 (Basma et al., 2024) × 3.27 μV (SD reported by Kappenman et al.)
Trials per participant	60 + 60	Kappenman et al. (2021)
Single-trial noise SD	11.5 μV	ERP CORE's averaged noise is 1.5 μV; Averaging 60 trials divides noise by √60, so single trials: 1.5 × √60 ≈ 11.5 μV
Participants' N400 effect (random slope SD)	2.5 μV	Derived from ERP CORE after subtracting single-trial noise SD: N400 effects vary with SD 3.27 μV. Averaging noise adds $11.5^2 \times (1/60 + 1/60) \approx 4.4\ \mu V^2$; $\sqrt{(3.27^2 - 4.4)} \approx 2.5$ μV
Participants' overall amplitude (intercept SD)	3 μV	Estimate (not based on any study)
Target words (intercept SD)	1.5 μV	Estimate (not based on any study)
Related-target amplitude	2 μV	Estimate (not based on any study)

Kappenman et al. (2021), NeuroImage; Basma, Savage & Bertone (2024), Int. J. Psychophysiol.

Design 1: the simulated pilot data



Group means (µV)

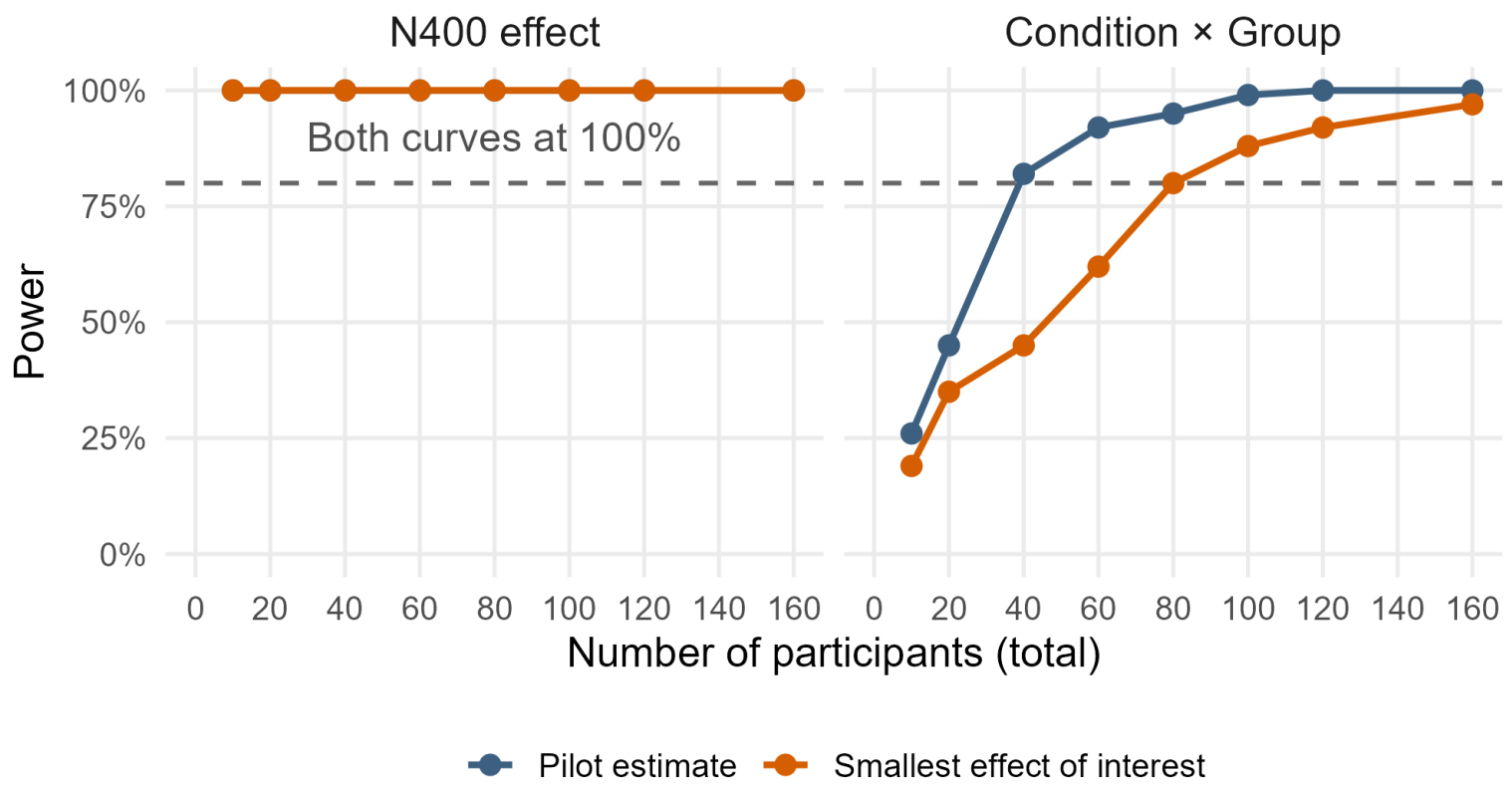
	Related	Unrelated	N400 effect
Typical	1.4	−6.9	−8.3
Dyslexic	2.4	−3.2	−5.6

N400 effect = unrelated − related, from 40 simulated adults (20 per group)

Q1: Both groups show an N400 effect (more negative for unrelated targets).

Q2: The line is less steep for dyslexic readers: their effect is 2.7 µV smaller.

Sample size planning for interactions



N ≤ 10

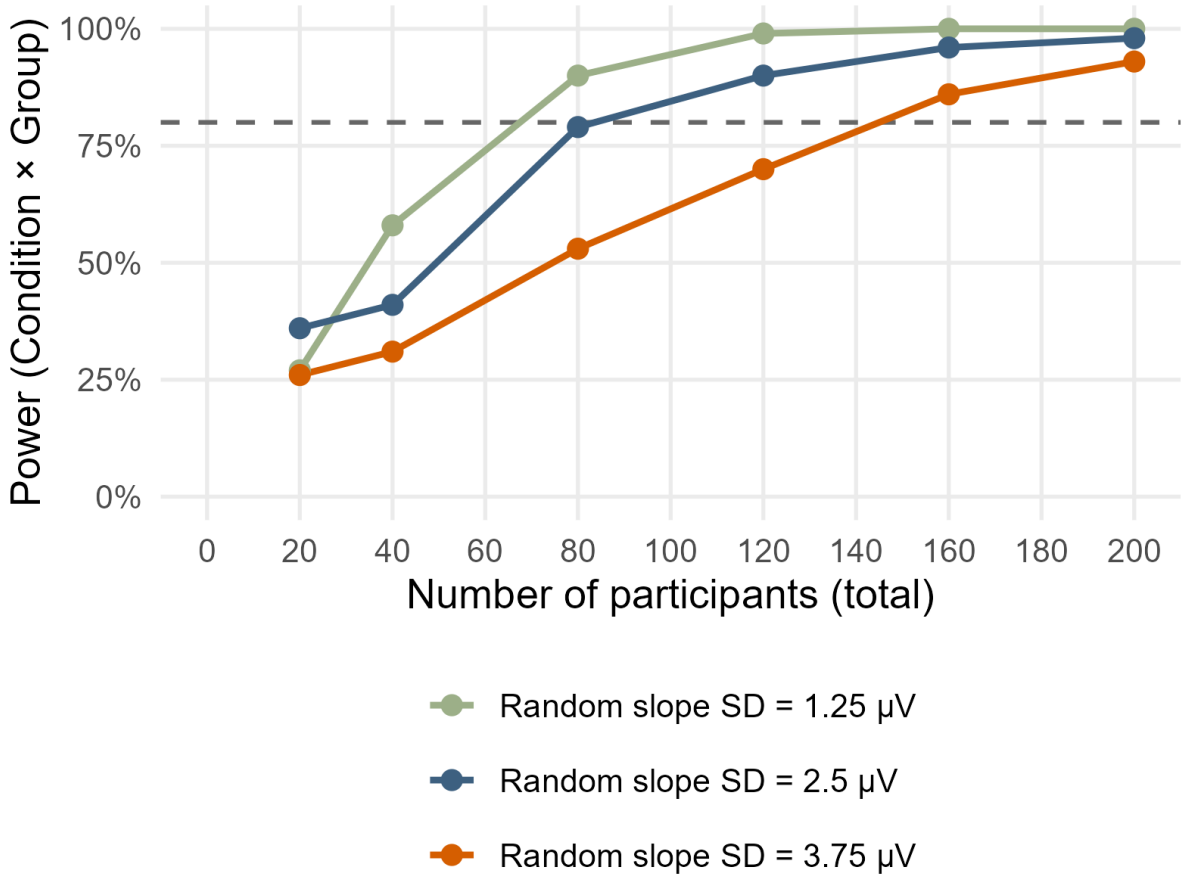
for 80% power for the N400 effect (SESOI = −5.5 µV): 100% at every sample size

N ≈ 80

for Condition × Group (SESOI = 1.9 µV): 80% at N = 80 (total, 40 in each group)

SESOI = smallest effect size of interest: the smallest effect you'd still care about detecting

Sample size planning for random slopes



Held constant across the three curves

- Group difference: 1.9 µV
- 60 trials per condition
- Trial-to-trial noise within a person: SD 11.5 µV
- Overall amplitude (SD 3) and target words (SD 1.5)

Only the random slope SD changes: 1.25, 2.5, or 3.75 µV

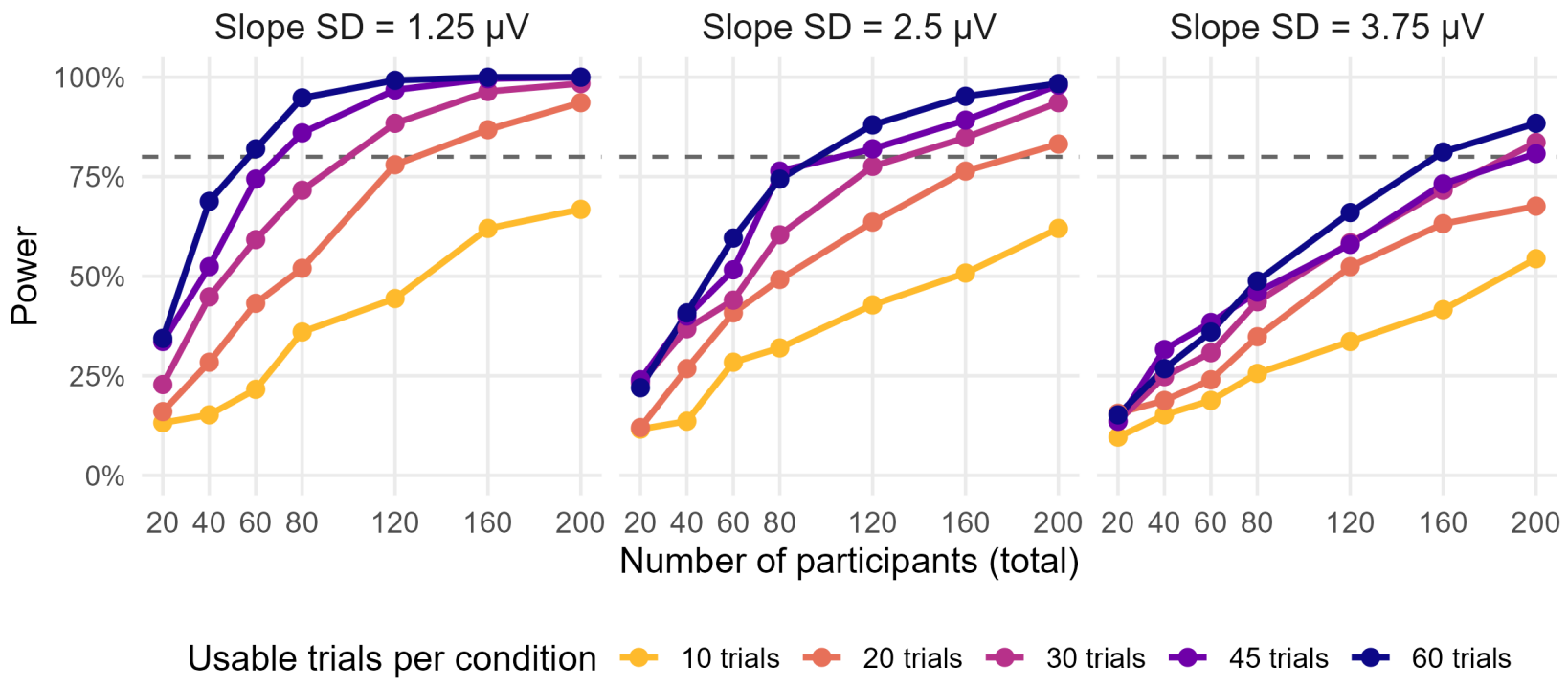
What is a random slope?

Each person's own N400 effect. With SD = 2.5 µV, 95% of typical readers have an effect between about −12.5 and −2.7 µV: nearly everyone shows an N400 effect, but by different amounts.

Why does it take more participants?

The ability to detect a 1.9 µV group difference is influenced by how much people differ **within** each group. More trials measure each person more precisely but don't make people more alike: more people are needed to estimate each group's average.

Your turn: how many usable trials do you need?



Think · pair · share

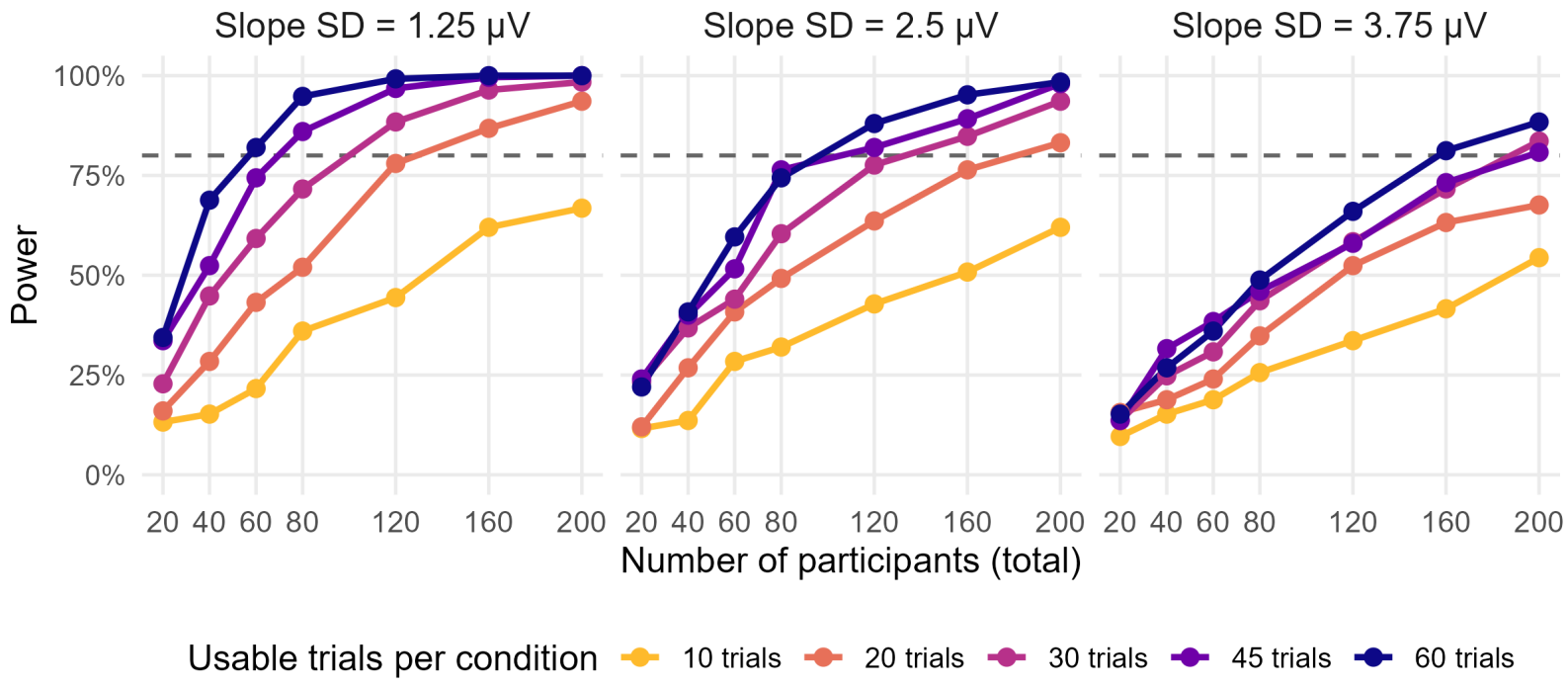
1 min on your own, then talk with a neighbor

1. How many more participants do you need if each keeps 10 vs. 60 trials?

2. What do you notice as the slope SD increases?

3. How many usable trials do participants in your studies retain?

Usable trials vs. participants



Our N400 simulation (Design 1): 250 simulated studies per point

DeBolt, Rhemtulla & Oakes (2020):

In infant looking-time data, infants were noisy from trial to trial but barely differed from each other (ICC ≈ 0). So trials and infants were interchangeable: doubling trials per infant (8 → 16) cut the infants needed from ~56 to ~32. Like our left panel: more trials help most when people differ little.

Our simulation: N for 80%	10 trials	20	30	45	60 trials
Slope SD = 1.25 µV	> 200	130	101	70	57
Slope SD = 2.5 µV	> 200	182	134	106	97
Slope SD = 3.75 µV	> 200	> 200	189	196	157

Design 2: the Nc to emotional faces in preschoolers

Research question: is the Nc more negative for angry than for happy faces?

Participants: 3- to 6-year-olds

Task: passively viewing faces of several actors showing different emotions

Component: Nc, central electrodes (C3/Cz/C4), about 300–600 ms

Trials: each actor showed each emotion 10 times

Data loss: children kept about 27 of 50 trials per condition (SD ≈ 11), so roughly 45% were rejected

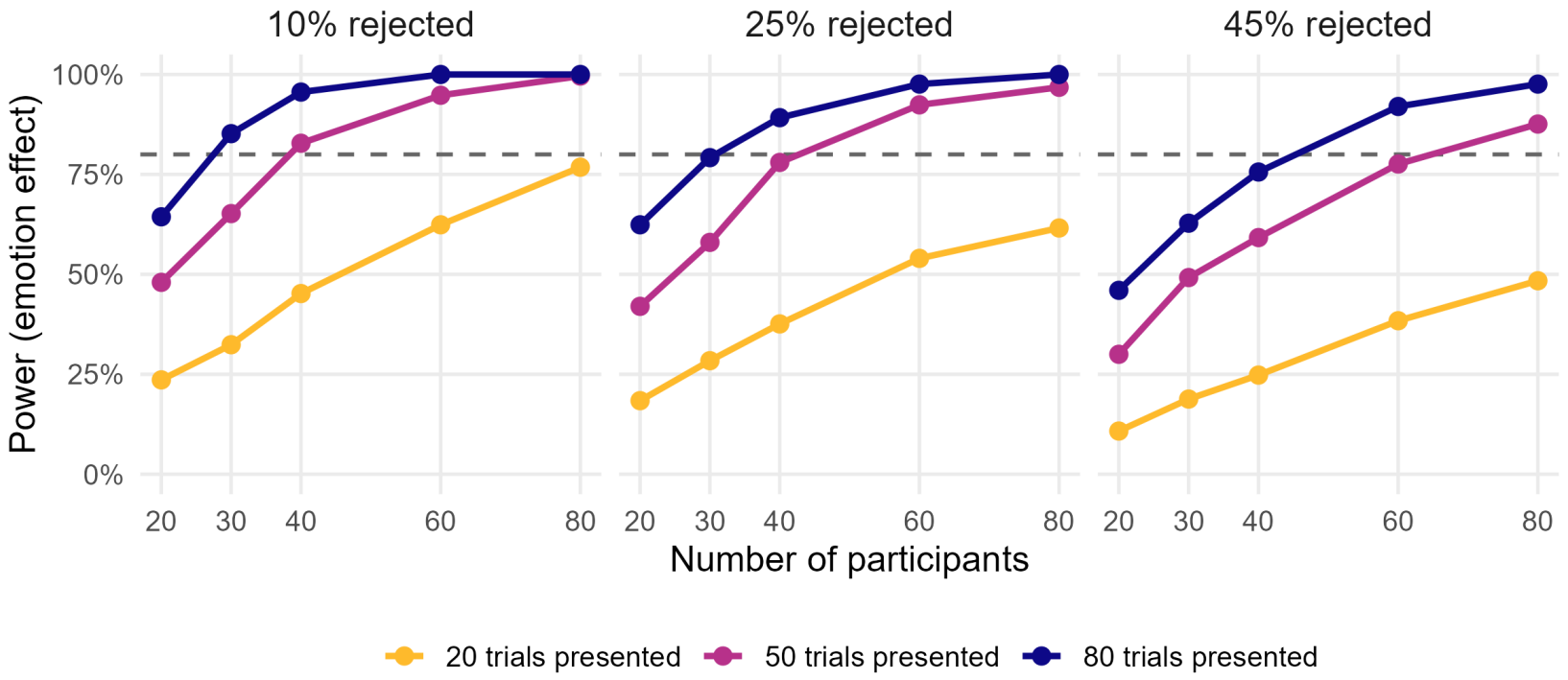
Simulation

Parameter	Value
Emotion effect (angry – happy)	−2 µV
Actors × presentations	5 × 10 = 50 / condition
Habituation	+0.5 µV per presentation
Trial-to-trial noise (SD)	20 µV (large)
Children differ in the effect (SD)	1 µV (small)
Trials rejected	45%
Pilot sample	30 children

Heise, Mon & Bowman (2022), Developmental Cognitive Neuroscience; DeBolt, Rhemtulla & Oakes (2020), Infancy

Participants, trials, and data loss

Design 2: is the Nc more negative for angry than for happy faces?



Think · pair · share

1 min on your own, then talk with a neighbor

1. Compare the panels: what is the impact of having fewer trials?
2. Is it better to present participants with more trials, or to recruit more children?
3. Which panel looks most like your data?

Recommendations for LME power analyses

1 Run 1,000+ simulations

Fewer is fine for a demo or while working out simulation parameters

2 Plan for the SESOI

The smallest effect you'd still care about

3 Power the weakest effect

With several effects, base N on the one with the lowest power (e.g., interactions)

4 Explore your assumptions

Rerun across plausible effects, variances, slopes, and trial counts

Packages

`mixedpower` fast, parallel, all fixed effects at once (Kumle et al., 2018)

`simr` slower but more flexible: build models from scratch, CIs for power (Green & MacLeod, 2016)

`faux / lmem_sim` simulate mixed-model data from scratch (DeBruine & Barr, 2021)

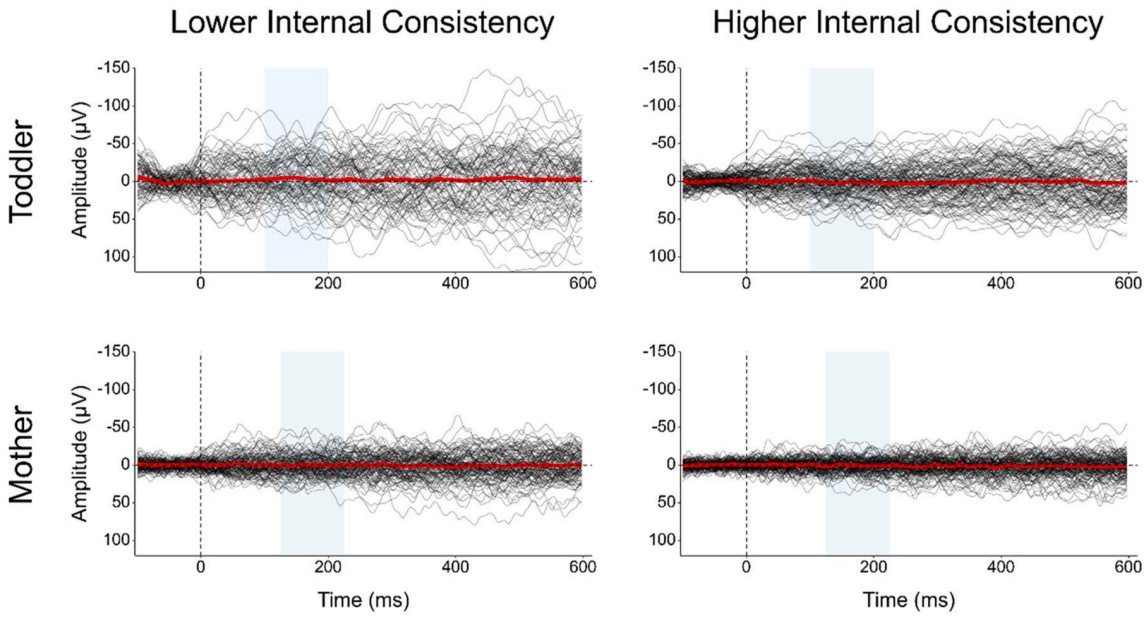
09

Special cases: Location scale models

When trial-to-trial variability is the signal

LMEs assume everyone's trial-to-trial variability is the same

- An LME estimates **one** residual variance (σ^2) shared by every participant and every trial
- But trial-by-trial variability can differ by participant, age, group, etc.
- **Unequal variability (heteroscedasticity)** lowers power and inflates Type I error (DeShon & Alexander, 1996)
- Summary measures (e.g., each person's SD) mix neural variability with artifact noise and are unreliable with few trials



Mon et al., 2024. Single-trial MMN waveforms (black) and their average (red) for 2 toddlers and 2 mothers

Mon, Manning, Wakschlag & Norton (2024), Developmental Cognitive Neuroscience

When homogeneity of variance isn't met

1 Robust standard errors

Same model and estimates; SEs and p-values that don't assume everyone is equally noisy

- Easy approach if you're not interested in interpreting variability

```
parameters::model_parameters
(
  fit, vcov = "CR2")
# cluster-robust
(clubSandwich)
```

2 Model the variance (nlme)

nlme::lme() takes the same fixed and random effects, plus **weights** that let the residual SD differ by group or change with a predictor

```
fit_h <- nlme::lme(MeanAmp ~
Condition * Group,
  random  = ~ 1 +
Condition|SubjectID,
  weights = varIdent(form = ~
1|Group),
  data = dat)
anova(fit_h0, fit_h) # LRT
```

3 Fit a MELSM

Fixed and random scale effects: each person gets their own variability, estimated with the mean

- Separable predictors/random effects for means and variability

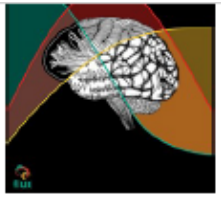
```
brms: sigma ~ x + (1 | id)
glmmTMB: dispformula =
  ~ x + (1 | id)
```

Mixed effects location scale models (MELSM)

ELSEVIER

Developmental Cognitive Neuroscience

journal homepage: www.elsevier.com/locate/dcn



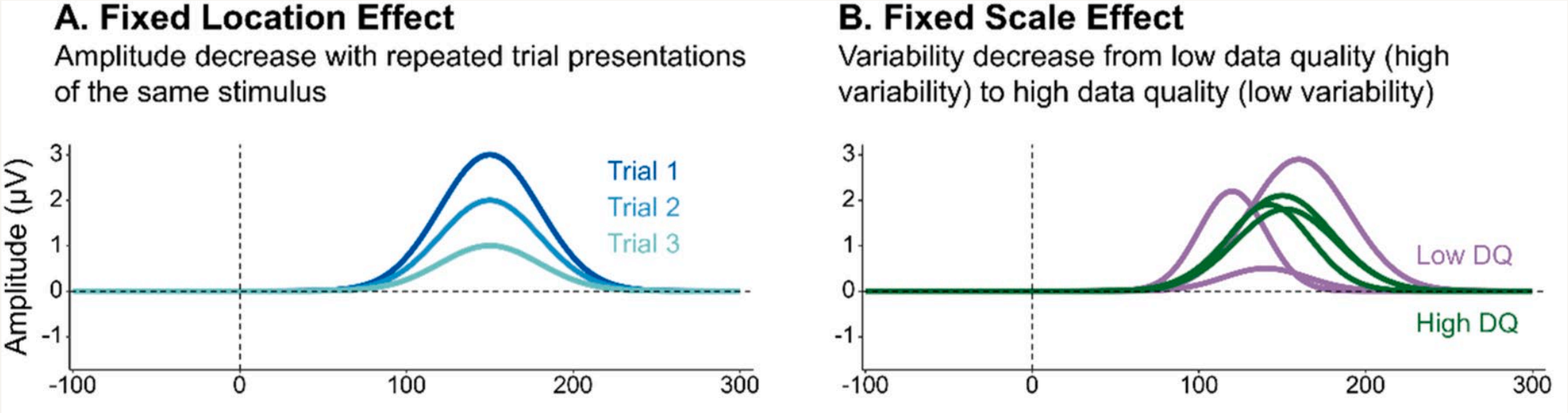
Check for updates

Leveraging mixed-effects location scale models to assess the ERP mismatch negativity's psychometric properties and trial-by-trial neural variability in toddler-mother dyads

Serena K. Mon [a], Brittany L. Manning [b,c], Lauren S. Wakschlag [b,c], Elizabeth S. Norton [a,b,c,*]

[a] Roxelyn and Richard Pepper Department of Communication Sciences and Disorders, Northwestern University, Evanston, IL, USA
[b] Department of Medical Social Sciences, Feinberg School of Medicine, Northwestern University, Chicago, IL, USA
[c] Northwestern University Institute for Innovations in Developmental Sciences, Chicago, IL, USA

Mixed-effects location scale models (MELSMs)



Mon et al. (2024), Fig. 1: simulated trial-level ERPs. DQ = EEG data quality

Location sub-model

Predicts the mean of single-trial amplitude (this is the LME you already know)

$$amp_{ij} = \beta_0 + \beta_1 \cdot trial_{ij} + u_{0j} + \varepsilon_{ij}$$

Scale sub-model

Predicts the variability (log SD) of single-trial amplitude. This part is new

$$\log(\sigma_{ij}) = \gamma_0 + \gamma_1 \cdot DQ_j + v_{0j}$$

LMEs vs. MELSMs: what changes

	LME (mean only)	MELSM (mean + variability)
What's modeled	The mean of single-trial amplitude	The mean **and the SD** of single-trial amplitude
Residual variance	One σ shared by every person and trial	σ predicted by covariates (log scale) and allowed to differ by person
Random effects	Location only: people differ in their mean	Location and scale: people differ in mean and variability, and the two can correlate
Software	lme4 / lmerTest (frequentist)	Usually brms (Bayesian): priors, chains, R-hat
Comparing models	Likelihood-ratio tests, AIC/BIC	Leave-one-out cross-validation (ELPD difference)
Fitting time	Seconds to minutes	Hours: ~20 h per toddler model on 4 cores in Mon et al.

Mon, Manning, Wakschlag & Norton (2024), Developmental Cognitive Neuroscience; Bürkner (2017), brms

Extending the questions we can ask of ERP data

	LME	MELSM
Individual differences	Do participants differ in their average MMN amplitude?	Do participants differ in how consistent their MMN is from trial to trial (participant-level internal consistency)?
Change across the session	Does amplitude habituate across trials?	Does trial-to-trial variability change across the session?
Predictors & group differences	Does amplitude differ by hemisphere, age, sex, or group?	Is the response more variable in one hemisphere, age, or group, after accounting for EEG data quality?
Brain–behavior relations	Does average MMN relate to language or psychopathology?	Does neural variability relate to language or psychopathology (e.g., excessive noise as a risk marker)?
Brain-behavior relations in dyads	Are toddlers' and mothers' MMN amplitudes similar?	Are toddlers' and mothers' trial-by-trial variability similar?

Mon, Manning, Wakschlag & Norton (2024), Developmental Cognitive Neuroscience

Practical challenges: data, convergence & power

	Challenges	Possible solutions
Trials per person	Each person's variability is estimated from their own trials; toddlers in Mon et al. had 5–195	Partial pooling shrinks sparse participants toward the group, so no minimum trial cutoff is needed, but more trials improve the accuracy of scale estimates
Convergence	Many more parameters.	Build up from simple to complex; drop random scale slopes first (Mon et al. removed the random slopes of trial number)
Compute time	~20 hours on 4 cores per toddler model (4 chains × 10,000 iterations)	Computing cluster; debug with fewer iterations first
Sample size & power	Brain–behavior effects are small; dyadic models need data from both members, so more data are lost	Run power simulations using published estimates or SESOI before collecting data

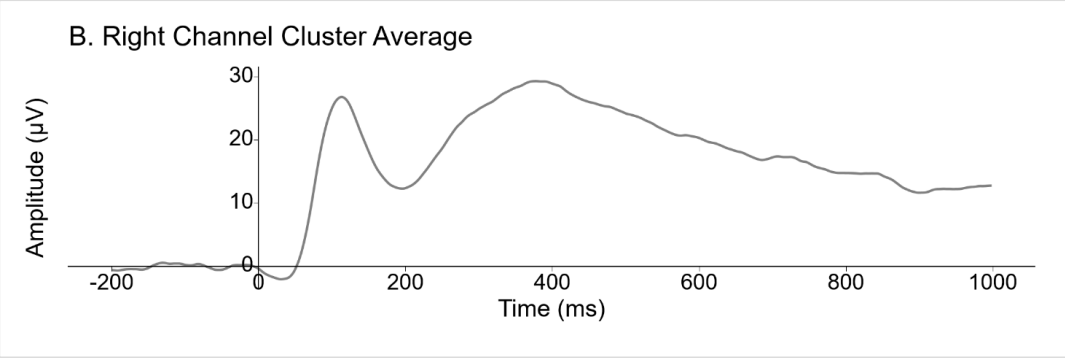
Mon, Manning, Wakschlag & Norton (2024), Developmental Cognitive Neuroscience

10

Special cases: Latency

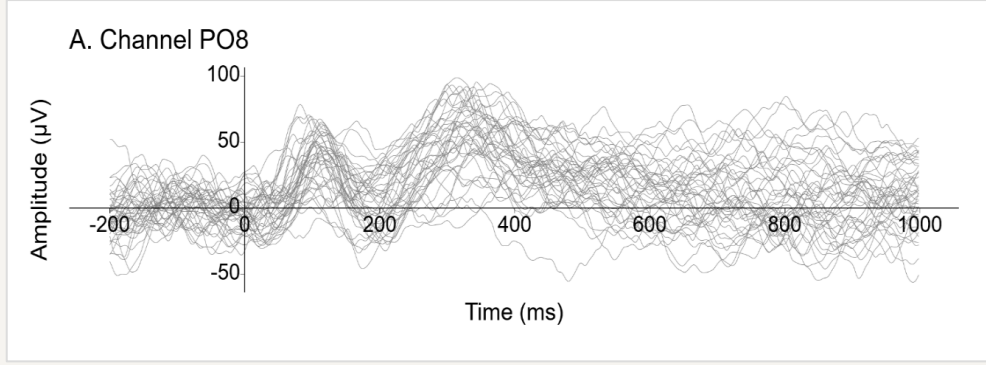
Uncharted territory! What do we know so far?

Can I use LMEs for questions about ERP latency?



Peak latency of the averaged waveform

≠



Average of single-trial peak latencies

Latency extracted from the averaged waveform isn't equal to the mean of single trial's latencies

- Single-trial ERPs are noisy and vulnerable to artifactual noise spikes, which would be identified as the peak in ERP measurement
- Averaging single-trial peaks pulls values toward the center of the time window

Preliminary analyses

- Simulated P1 (SEREEGA): 48 children × 50 trials, 1,000 datasets each
- 10 or 20 μV peak; infant or preschooler noise
- 3 missingness patterns (2 MAR and 1 MCAR) × 0–80% low-trial-count children

Can I use LMEs for questions about ERP latency?

Research question	Example	LME	ANOVA	What we found
Single-group estimate	Average P1 latency of 6-month-olds	✓	✓	Both within ±10% bias; LME had lower error, ANOVA less bias (LME off by 1–3 ms)
Group difference	6- vs. 12-month-olds	✗	✓	LME underestimated a 10 ms difference by 30–56% (3–6 ms)
Individual differences	Latency and a behavioral score	✗	✓	Real data: all models significant, but LME β = −0.24 vs. −0.67 ms per point (ANCOVA)

Where LME still helps

- **Many children with few trials:** at 80% low-trial-count children, power was .66 (LME) vs. .10 (ANOVA)
- **Systematic missingness** (more loss late in the session and in younger children): in the worst missing data scenarios LME finally performed better than ANOVA

Where LME struggles

- **Small peaks (10 µV):** single trials are more vulnerable to noise spikes
- **Differences between groups or individuals:** biased toward 0 at every level of missingness

Mon, Heise & Bowman (2025, preprint), ssrn.com/abstract=6233097

Key takeaways

LMEs allow you to retain all clean ERP trials.

LMEs use every clean trial from every participant: no arbitrary trial-count cutoffs.

LMEs are unbiased under realistic patterns of missing data. Analyses that use casewise deletion (regression, ANOVA) are biased unless data are missing completely at random.

LMEs allow you to model more nuisance variability and extend results to new stimuli and participants. Random effects are assumed to be sampled from the population, Barr et al. show that findings transfer best to new participants and stimuli when the maximal model is fit.

Follow the 9-step roadmap. Start maximal and simplify one step at a time to identify the LME that your data can support.

Centering and coding can help make models more interpretable.

Plan with simulation. Power depends on participants, trials, and how much people differ, and simulating realistic values ahead of time helps in planning both aspects of your design (# trials, # stimuli) and sample size.

Ask new questions. Trial-level data let you study habituation and trial-level covariates (LMEs), and trial-to-trial variability (MELSMs).

Thank you!



Scan to open the survey

tinyurl.com/spr2026lme

Anonymous · about 3 minutes · every question is optional

1. What did you like best? (anything: a topic, an example, the tutorials, the polls…)
2. What do you wish we had covered in more detail?
3. What did you like least, or what was most confusing?
4. Any other feedback?

Key R packages for LME/ERP analysis

Package	Version used	What it does
lme4	1.1-36	Fit linear mixed effects models
lmerTest	3.1-3	p-values with Satterthwaite df
emmeans	1.11.0	Marginal means, contrasts, and slopes (emtrends)
performance + see	0.18.2 / 0.14.2	check_model(), ICC, R^2, convergence and singularity checks
sjPlot	2.8.17	tab_model() tables; random-effect plots
parameters	0.29.3	Standardized and robust (sandwich) estimates
mixedpower	0.1.0	Simulation-based power for mixed models
faux	1.2.4	Simulate crossed designs (participants × stimuli)
simr	—	Power by simulation (an alternative to mixedpower)
nlme	3.1-167	Model residual variance (weights = varIdent)
glmmTMB	1.1.11	Dispersion models and MELSMs (frequentist)
brms	2.22.0	Bayesian mixed models and MELSMs
tidyverse	2.0.0	Data wrangling (dplyr) and plots (ggplot2)

Versions are the ones the tutorials were run with (R 4.4.3)

Resources

Textbooks

Raudenbush, S. W., & Bryk, A. S. (2002). *Hierarchical linear models: Applications and data analysis methods* (2nd ed.). Sage.

Snijders, T. A. B., & Bosker, R. J. (2012). *Multilevel analysis: An introduction to basic and advanced multilevel modeling* (2nd ed.). Sage.

Power & study planning

DeBruine, L. M., & Barr, D. J. (2021). Understanding mixed-effects models through data simulation. *Advances in Methods and Practices in Psychological Science, 4*(1).

Kappenman, E. S., Farrens, J. L., Zhang, W., Stewart, A. X., & Luck, S. J. (2021). ERP CORE: An open resource for human event-related potential research. *NeuroImage, 225*, 117465.

Kumle, L., Võ, M. L.-H., & Draschkow, D. (2021). Estimating power in (generalized) linear mixed models: An open introduction and tutorial in R. *Behavior Research Methods, 53*, 2528–2543.

Westfall, J., Kenny, D. A., & Judd, C. M. (2014). Statistical power and optimal design in experiments in which samples of participants respond to samples of stimuli. *Journal of Experimental Psychology: General, 143*(5), 2020–2045.

Code & materials: our tutorials (Rmd + HTML) · github.com/basclab/LME_MixedEffectsERPTutorial · lkumle.github.io/power_notebooks · erpinfo.org/erp-core

LME methods for ERP research

Barr, D. J., Levy, R., Scheepers, C., & Tily, H. J. (2013). Random effects structure for confirmatory hypothesis testing: Keep it maximal. *Journal of Memory and Language, 68*(3), 255–278.

Ferron, J. M., Hogarty, K. Y., Dedrick, R. F., Hess, M. R., Niles, J. D., & Kromrey, J. D. (2008). Reporting results from multilevel analyses. In A. A. O'Connell & D. B. McCoach (Eds.), *Multilevel modeling of educational data* (pp. 391–426). Information Age.

Heise, M. J., Mon, S. K., & Bowman, L. C. (2022). Utility of linear mixed effects models for event-related potential research with infants and children. *Developmental Cognitive Neuroscience, 54*, 101070.

Heise, M. J., Mon, S. K., & Bowman, L. C. (2025). Analysis of event-related potential difference waves can benefit from linear mixed effects modeling: Recommendations for analyses and general model fitting. *Developmental Cognitive Neuroscience, 76*, 101614.

Matuschek, H., Kliegl, R., Vasishth, S., Baayen, H., & Bates, D. (2017). Balancing Type I error and power in linear mixed models. *Journal of Memory and Language, 94*, 305–315.

Mon, S. K., Manning, B. L., Wakschlag, L. S., & Norton, E. S. (2024). Leveraging mixed-effects location scale models to assess the ERP mismatch negativity's psychometric properties and trial-by-trial neural variability in toddler-mother dyads. *Developmental Cognitive Neuroscience, 70*, 101459.

Schielzeth, H., Dingemanse, N. J., Nakagawa, S., Westneat, D. F., Allegue, H., Teplitsky, C., Réale, D., Dochtermann, N. A., Garamszegi, L. Z., & Araya-Ajoy, Y. G. (2020). Robustness of linear mixed-effects models to violations of distributional assumptions. *Methods in Ecology and Evolution, 11*(9), 1141–1152.

Volpert-Esmond, H. I., Page-Gould, E., & Bartholow, B. D. (2021). Using multilevel models for the analysis of event-related potentials. *International Journal of Psychophysiology, 162*, 145–156.